beatbreaker / REV_A

UNPOWERED MECHANICAL TRIAL ONLY — SCOPED DOCUMENTATION RELEASE

This package retains drawing revision B26-CONSOLIDATED-02 and approved trial model B26-build123d03.

Engineer approved the drawing for control reconstruction and accepted the bounded C03/C05 trial exception.

The included drawing pages retain their historical DRAFT / NOT APPROVED labels; this cover records the later decisions.

The original 69-page drawing is reproduced unaltered as document pages 2-70; add 1 to its internal page index.

Read this PDF with the companion shell/, panel/, jog/ and reference_geometry_DRAFT.json data in this directory.

Nine cap definitions are exact faceted mesh data, not parametric feature histories. All lengths are in mm.

VALIDATION LIMITS

C03/C05: reverse Boolean cuts returned invalid shapes and intersection results depended on operand order.

Sampled tests agree, but strict equivalence remains unresolved; the accepted exception is trial-only, not a pass.

Both P01 STEP strict-volume comparisons failed by about 0.641091522 mm3; a separate trial exception was accepted.

No exact surface equivalence, physical fit, strength, electrical safety, thermal safety or production readiness is claimed.

Purchased parts and wiring are incomplete reference envelopes. Unspecified PCB mounts are omitted.

Material for jogs/caps, print orientation, usable printer margins, supports and fit calibration remain TBD.

The concept PNG is illustrative only, not a dimensioned source or actual model view.

Frozen approvals and evidence: docs/APPROVALS.md and the package REV_A_handoff.md / VALIDATION evidence.

Scope: unpowered prototype; existing accepted model and geometry unchanged; no new STL or STEP generated.

Units: mm. STL is unitless. Third-angle projection on drawing sheets. Revision date: see recorded approvals.

Assembly, drawing scope and construction datum

Existing trial basis: B26-build123d03; drawing revision B26-CONSOLIDATED-02 is not approved.

Table datum: X to right, Y toward operator, Z up; model CAD coordinates are (X,-Y,Z).

Envelope nominal XY 600 x 320 mm; individual six-piece shell boundaries X=220/380, Y=140.

Two non-pressing jogs: local O at floor top, CAD origins (110,-165,2) and (490,-165,2).

Each local frame is rotated +60 degrees around CAD Z. Print orientation is TBD.

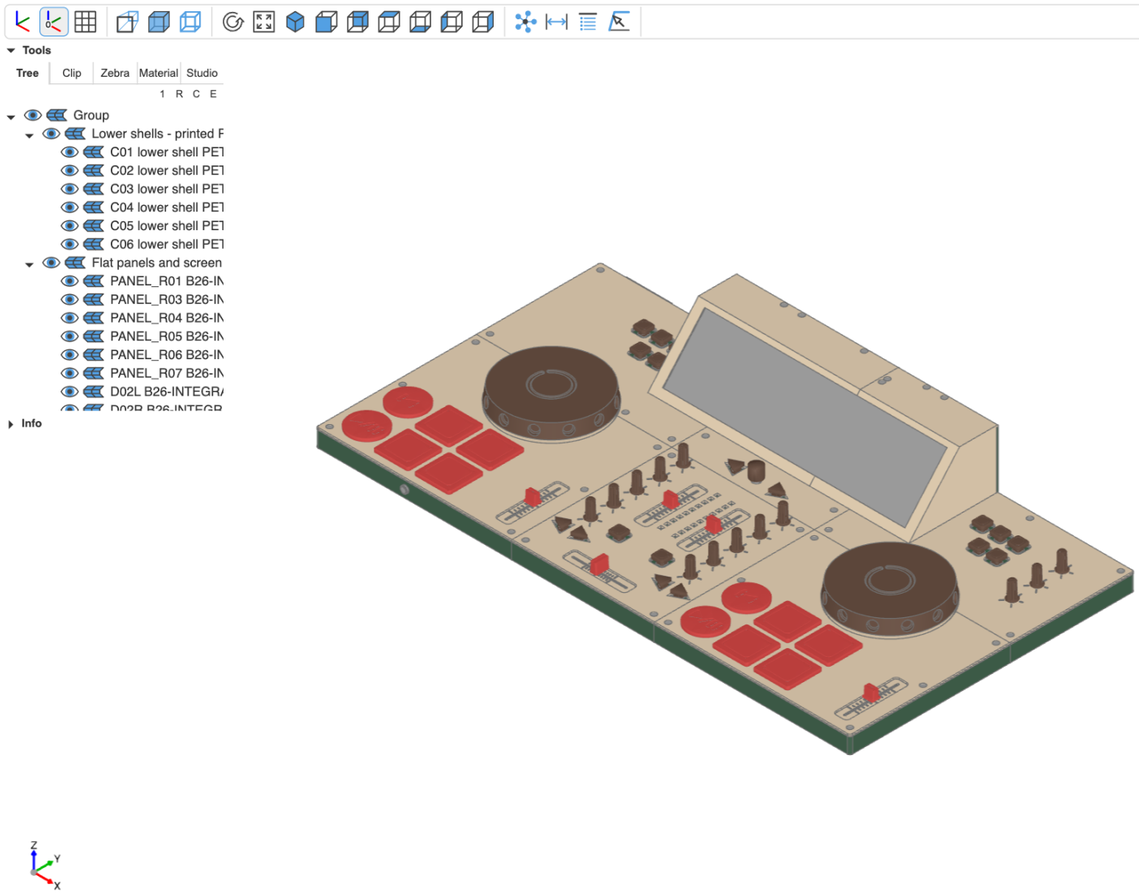


Image: actual previously approved trial model, not a dimensioned view or proof of the new documentation.

Read the enclosure, front-panel and jog sheets with their companion data; do not scale graphics to obtain dimensions.

Purchased parts are incomplete reference envelopes; absent PCB mounts remain absent. No physical or electrical approval.

Printed BOM / quantities per complete controller

36 definitions, 155 occurrences. Quantities and material copied from the released STL manifest.

PART ID	QTY	MATERIAL	PART ID	QTY	MATERIAL
C01	1	PETG	encoder_retainer	1	PETG
C02	1	PETG	display_toe	4	PETG
C03	1	PETG	jog_P01_rotor	2	TBD
C04	1	PETG	jog_P02_carrier	2	TBD
C05	1	PETG	jog_P04_collar	2	TBD
C06	1	PETG	jog_P07_collar	2	TBD
PANEL_R01	1	PETG	jog_cover	4	TBD
PANEL_R03	1	PETG	jog_sensor_plate	2	TBD
PANEL_R04	1	PETG	jog_toe	4	TBD
PANEL_R05	1	PETG	CAP_B	12	TBD
PANEL_R06	1	PETG	CAP_T	6	TBD
PANEL_R07	1	PETG	CAP_P	13	TBD
D02L	1	PETG	CAP_F	4	TBD
D02R	1	PETG	CAP_C	1	TBD
C07_bridge	16	PETG	CAP_E	1	TBD
button_retainer	30	PETG	CAP-CUE40	2	TBD
pot_retainer	13	PETG	CAP-PP40	2	TBD
fader_retainer	10	PETG	CAP-HC40	8	TBD

Enclosure parts PETG; jogs and cap material TBD. Orientations, supports and fit calibration TBD.

Nine inherited cap definitions are exact faceted meshes, not parametric feature histories.

Reference fasteners, bearings, electronics and components are not included in the printed BOM.

P01 STEP strict volume comparison failed; only the documented bounded trial exception was accepted.

Drawing index and companion data

This single document contains all listed sheets; no older PDF needs to be opened separately.

Pages 1-3: assembly datum, printed BOM and this index.

Pages 4-28: enclosure, retained B26-CONSOLIDATED-01 sheets.

Pages 29-32: enclosure section and construction details, B26-CONSOLIDATED-02.

Pages 33-52: front panel and caps, retained B26-CONSOLIDATED-01 sheets.

Pages 53-55: front panel operation and cap-data schedule, B26-CONSOLIDATED-02.

Pages 56-67: jog and its operation schedule, B26-CONSOLIDATED-02.

Pages 68-69: incomplete purchased-equipment and control reference geometry.

Retained sheets carry their original 01 revision and DRAFT labels; the three original 01 PDFs were accepted as presented.

Where 02 schedules provide more complete construction detail, use those schedules with the same nominal trial geometry.

Companion data in the package:

shell/shell_dataset_DRAFT.json: complete shell numeric schedules and construction operations.

panel/panel_addendum.json and panel/frontpanel_data.json: panel construction and positions.

panel/B26_CONSOLIDATED_01_caps_mesh_DRAFT.json: nine exact inherited facet meshes.

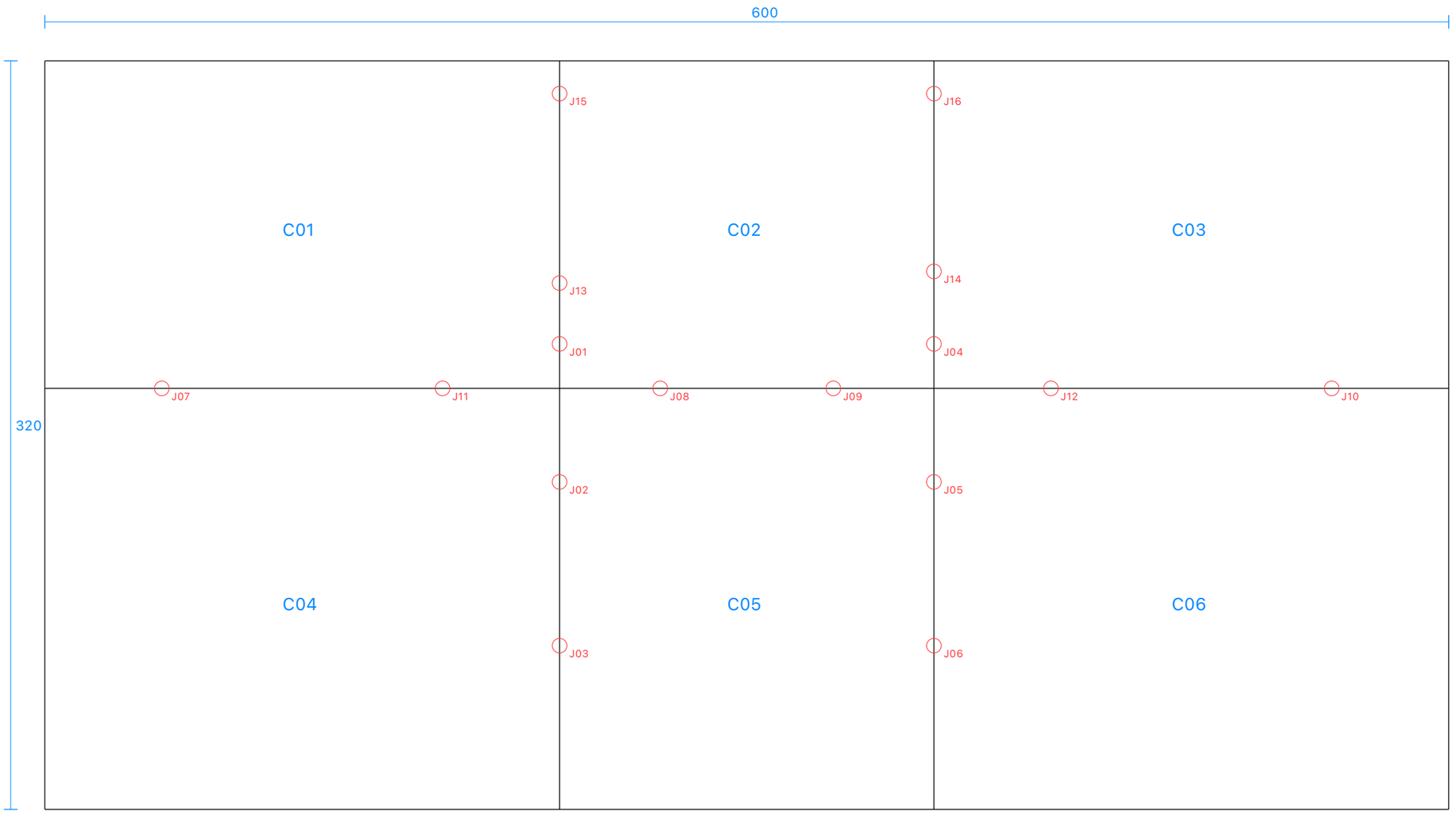
panel/B26_CONSOLIDATED_01_caps_poses_DRAFT.json: 49 local-to-CAD placements.

jog/jog_data.json: nominal jog operations, source cross-check and limitations.

reference_geometry_DRAFT.json: nominal purchased-part reference bodies and reference planes.

02 documentation is pending engineering review; transcription and reconstructibility remain to be independently checked.

1 / Six-shell layout and 16 bridge positions



J01 X 220,121

J02 X 220,180

J03 X 220,250

J04 X 380,121

J05 X 380,180

J06 X 380,250

J07 Y 50,140

J08 Y 263,140

J09 Y 337,140

J10 Y 550,140

J11 Y 170,140

J12 Y 430,140

J13 X 220,95

J14 X 380,90

J15 X 220,14

J16 X 380,14

X boundaries 0 / 220 / 380 / 600; Y boundaries 0 / 140 / 320.

Floor Z0..2; main external wall top Z18. Roof rear-wall height follows sheet 9.

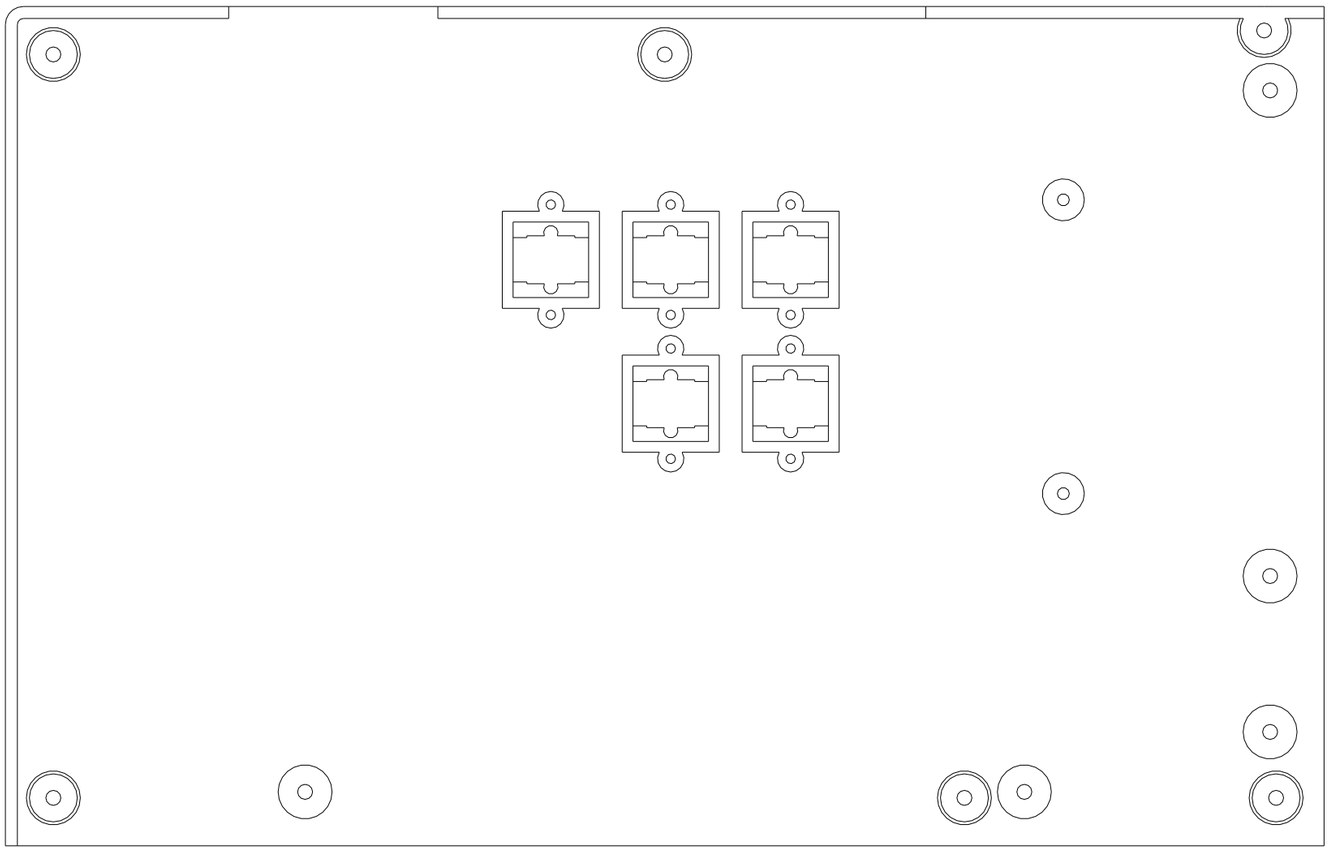
Red J labels: bridge centres. Two pad axes per bridge at +/-9 along its normal.

Exact supports, cuts and 98 control CSG groups: enclosure_data.json and shell_operations_DRAFT.json.

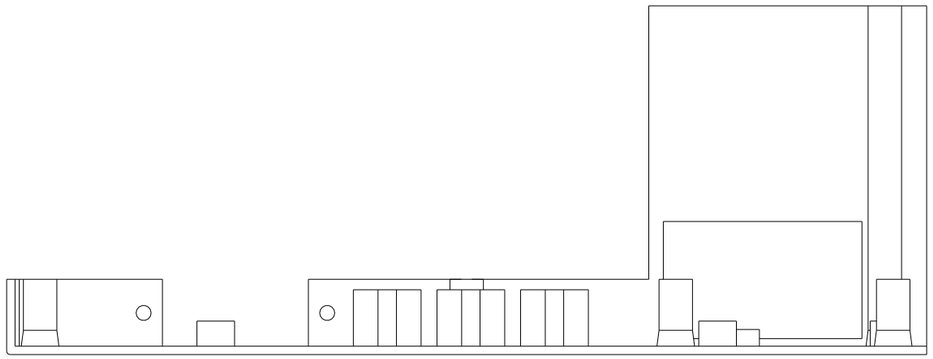
Schedules and graphics are draft; full independent reconstruction and physical checks remain open.

C01 / orthographic exterior views / quantity 1

TOP / visible edges



FRONT / visible edges



STEP bounds: 220.00 x 140.00 x 83.377 mm

Nominal XY: X 0..220, Y 0..140.

Z0..2 floor; exterior wall Z2..18; see rear roof-wall formula when applicable.

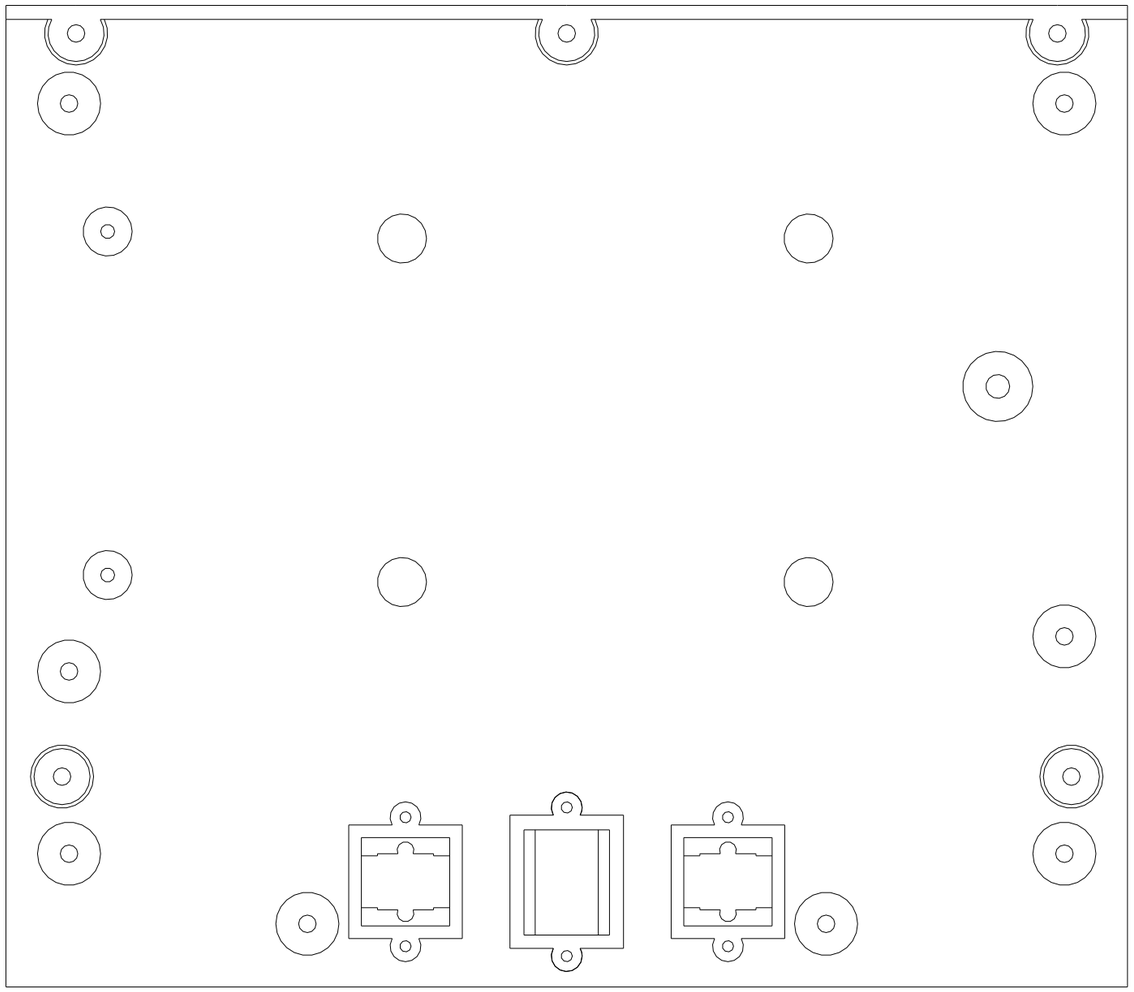
Views use visible STEP edges sampled to 0.05 mm; hidden edges are omitted.

Do not measure views. Reconstruction requires both companion numeric JSON schedules.

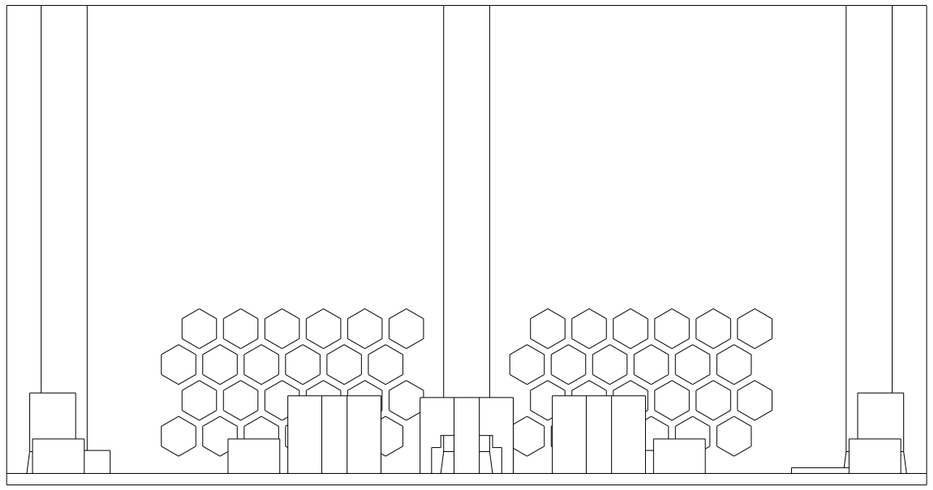
Detailed sections and source-data verification remain incomplete.

C02 / orthographic exterior views / quantity 1

TOP / visible edges



FRONT / visible edges



STEP bounds: 160.00 x 140.00 x 83.377 mm

Nominal XY: X 220..380, Y 0..140.

Z0..2 floor; exterior wall Z2..18; see rear roof-wall formula when applicable.

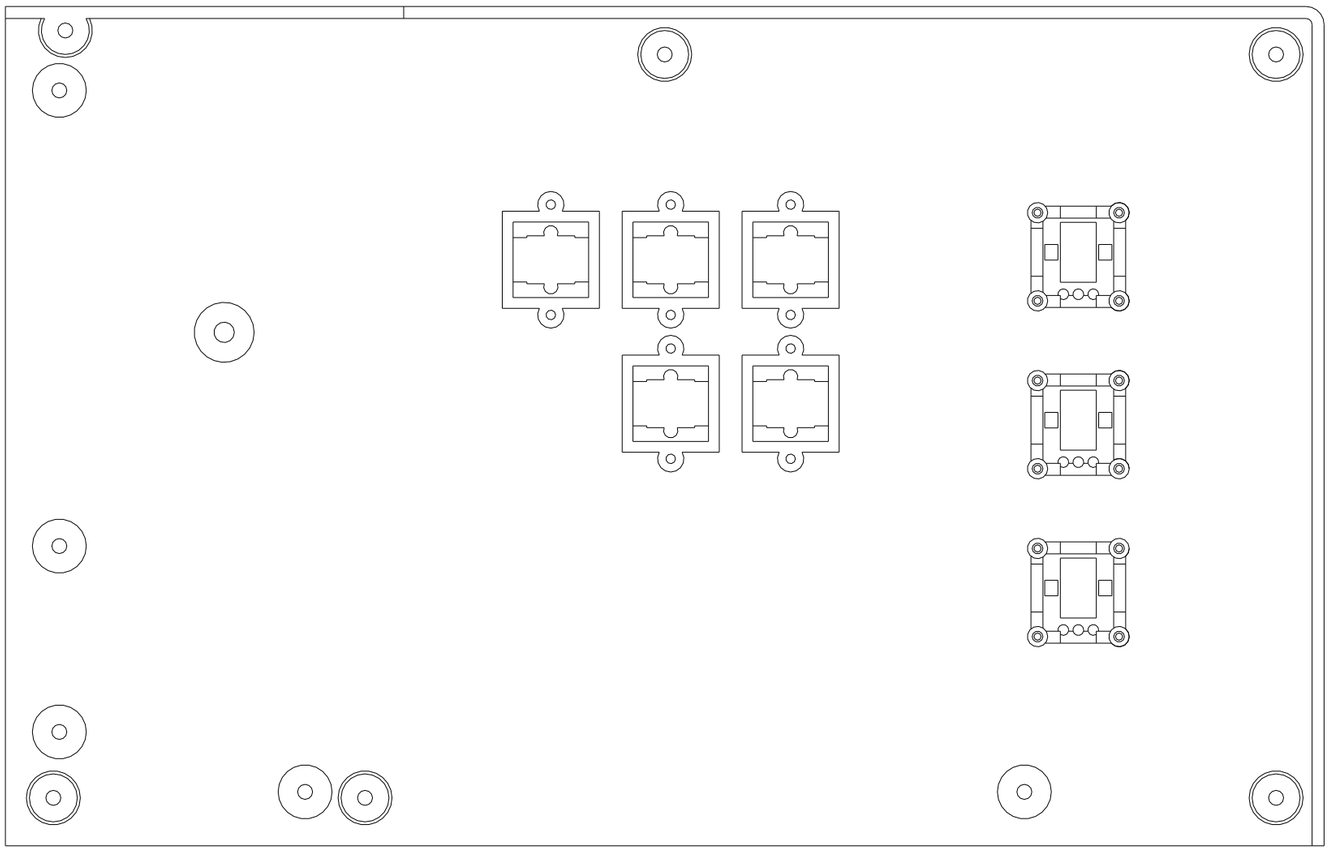
Views use visible STEP edges sampled to 0.05 mm; hidden edges are omitted.

Do not measure views. Reconstruction requires both companion numeric JSON schedules.

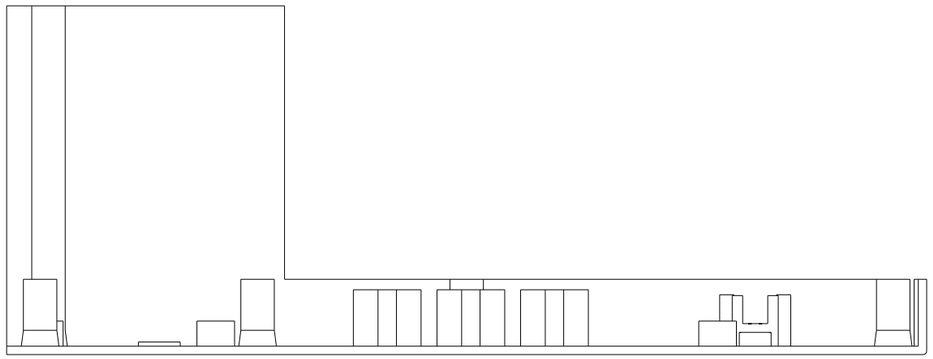
Detailed sections and source-data verification remain incomplete.

C03 / orthographic exterior views / quantity 1

TOP / visible edges



FRONT / visible edges



STEP bounds: 220.00 x 140.00 x 83.377 mm

Nominal XY: X 380..600, Y 0..140.

Z0..2 floor; exterior wall Z2..18; see rear roof-wall formula when applicable.

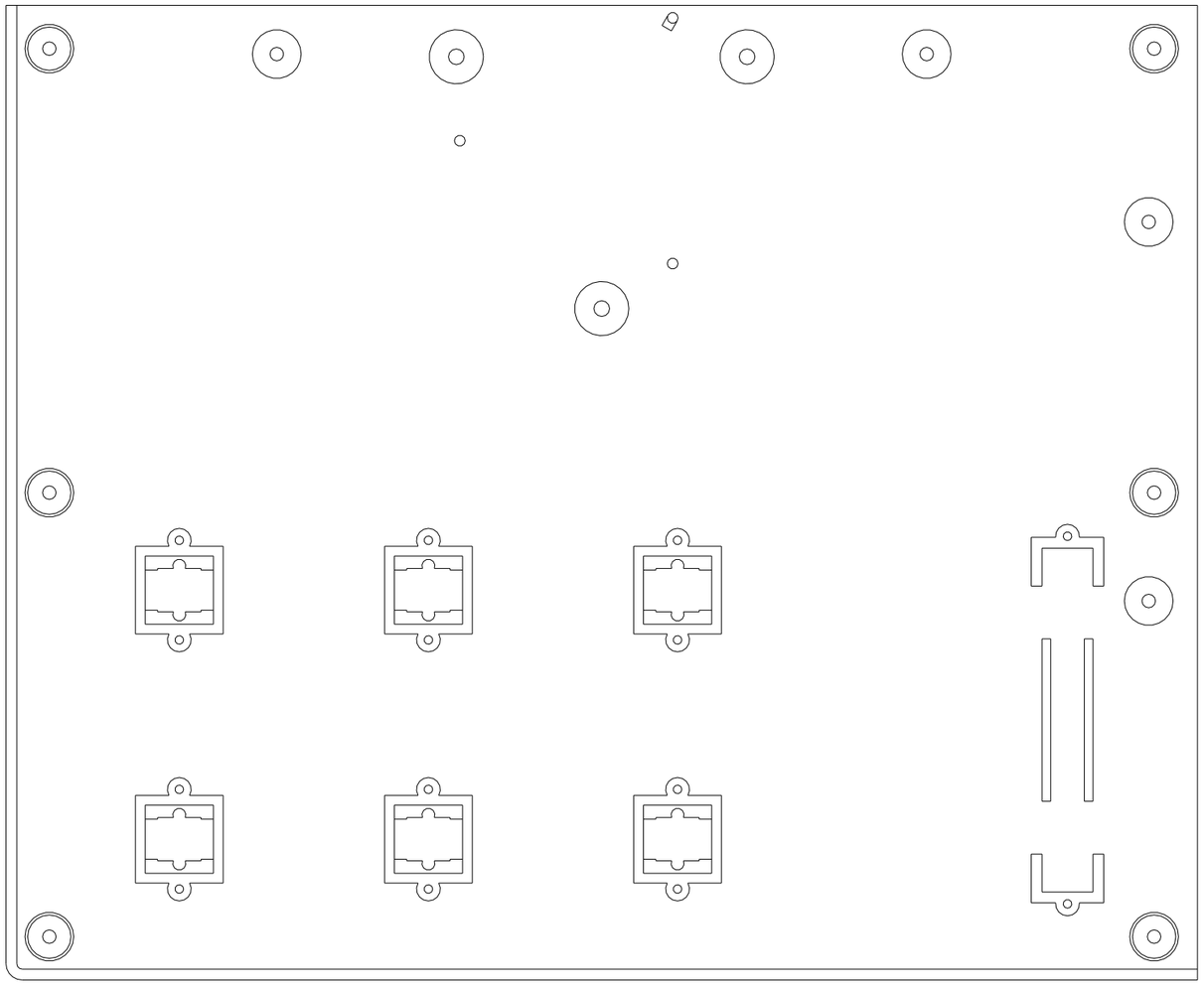
Views use visible STEP edges sampled to 0.05 mm; hidden edges are omitted.

Do not measure views. Reconstruction requires both companion numeric JSON schedules.

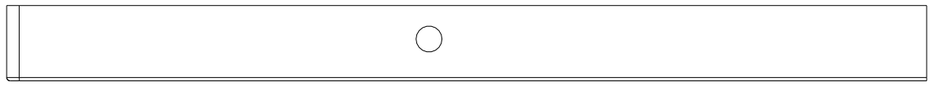
Detailed sections and source-data verification remain incomplete.

C04 / orthographic exterior views / quantity 1

TOP / visible edges



FRONT / visible edges



STEP bounds: 220.00 x 180.00 x 18.000 mm

Nominal XY: X 0..220, Y 140..320.

Z0..2 floor; exterior wall Z2..18; see rear roof-wall formula when applicable.

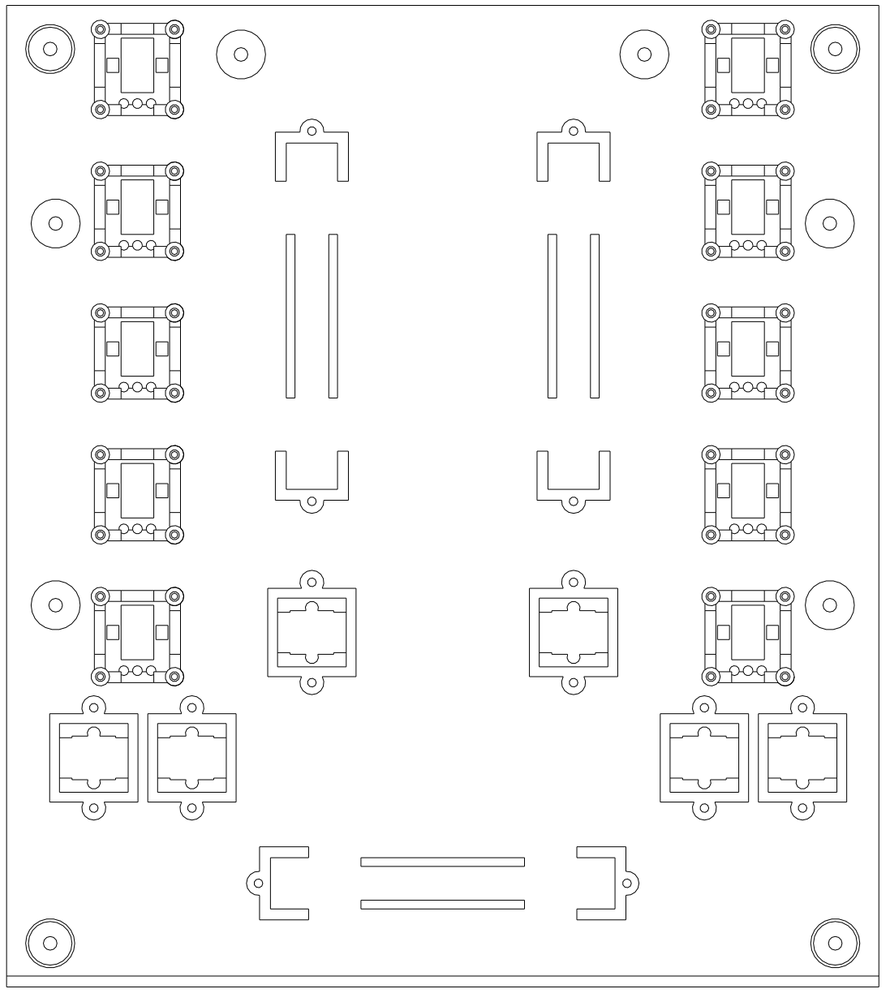
Views use visible STEP edges sampled to 0.05 mm; hidden edges are omitted.

Do not measure views. Reconstruction requires both companion numeric JSON schedules.

Detailed sections and source-data verification remain incomplete.

C05 / orthographic exterior views / quantity 1

TOP / visible edges



FRONT / visible edges



STEP bounds: 160.00 x 180.00 x 18.000 mm

Nominal XY: X 220..380, Y 140..320.

Z0..2 floor; exterior wall Z2..18; see rear roof-wall formula when applicable.

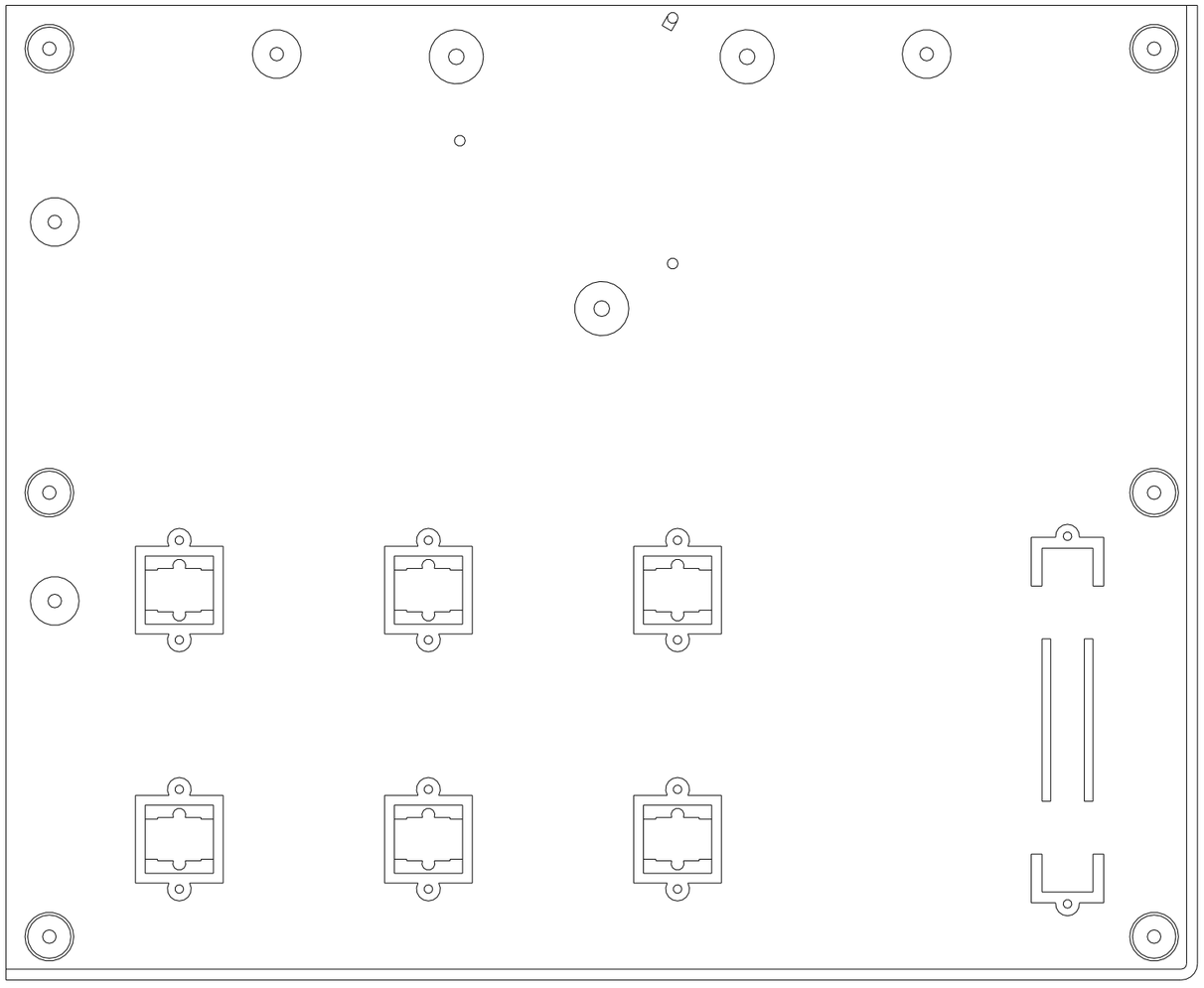
Views use visible STEP edges sampled to 0.05 mm; hidden edges are omitted.

Do not measure views. Reconstruction requires both companion numeric JSON schedules.

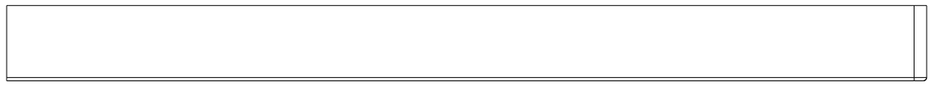
Detailed sections and source-data verification remain incomplete.

C06 / orthographic exterior views / quantity 1

TOP / visible edges



FRONT / visible edges



STEP bounds: 220.00 x 180.00 x 18.000 mm

Nominal XY: X 380..600, Y 140..320.

Z0..2 floor; exterior wall Z2..18; see rear roof-wall formula when applicable.

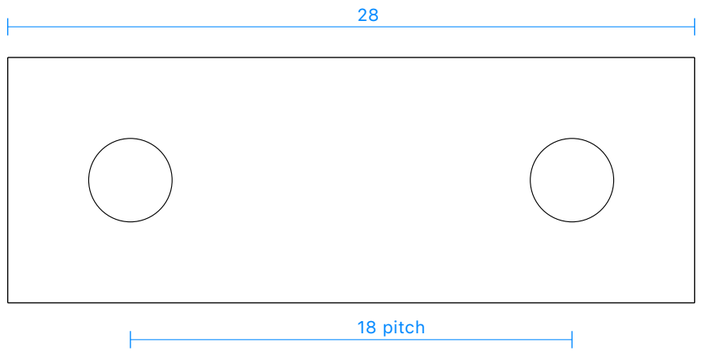
Views use visible STEP edges sampled to 0.05 mm; hidden edges are omitted.

Do not measure views. Reconstruction requires both companion numeric JSON schedules.

Detailed sections and source-data verification remain incomplete.

8 / C07 bridge, pad and fader-end detail

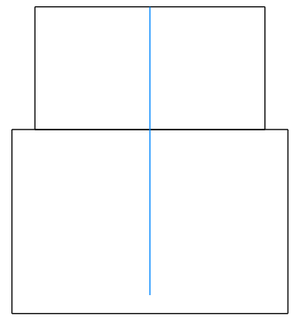
C07 / PLAN / quantity 16



Plan shown 4:1. Plate 28 x 10 x 3; two D3.4 through holes.

Local datum at plate bottom centre; installed plate Z8..11.

PAD / SECTION / schematic



Floor Z0..2; pad D9, Z2..8; blind pilot D2.5, Z2.5..8.

M3x8 head and shank are nominal references, not thread or strength proof.

FADER END / plan dimensions, all five

F01-F04 inner wall |u|=31.75; boss/pilot |u|=33.95. F05: 31.60 / 33.80.

Boss R2.2, pilot D1.6. Ten retainers 8.4 x 13.4 x 1.5, D2.2, countersink D4.4/90 deg.

Retainer inner edges |u|=28.75 / 28.60; outer |u|=37.15 / 37.00 respectively.

Source coordinates and exact ordered cuts are in the companion CSG schedules.

Parts are not qualified for physical fit, thread formation or strength.

PRINTED BOM / FULL CONTROLLER

C01-C06 x1 each; six flat panels x1 each; D02L/D02R x1 each; C07 x16; display toes x4.

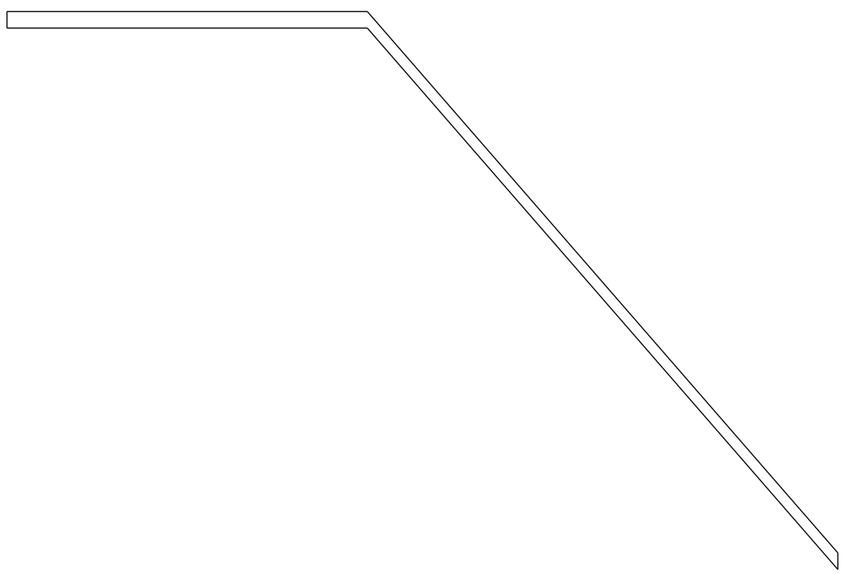
Button retainers x30; pot retainers x13; fader retainers x10; encoder retainer x1.

Cap occurrences x49; two static jogs, 18 printed occurrences total. See companion sheets.

Purchased fasteners, controls, bearing and electronics are nominal or incomplete references.

9 / Roof and rear wall / nominal sections

ROOF SECTION / table Y,Z / schematic scale



Slope 49 deg; top Z=85.376766; breakpoint Y=43.515278.

Front junction Y=100.346434; main panel Z18..20.

D02L X153.55..335; D02R X335..446.45; opening 272 x 72.

Rear wall Y0..2, raised to Z=83.376766.

Roof section, return, lap, 4 display supports and transformed features are specified numerically in shell_operations_DRAFT.json.

FLOOR / WALL AND JOINTS

Wall 2 thick; outer corner R3 / inner R1; bottom outer edge R0.8.

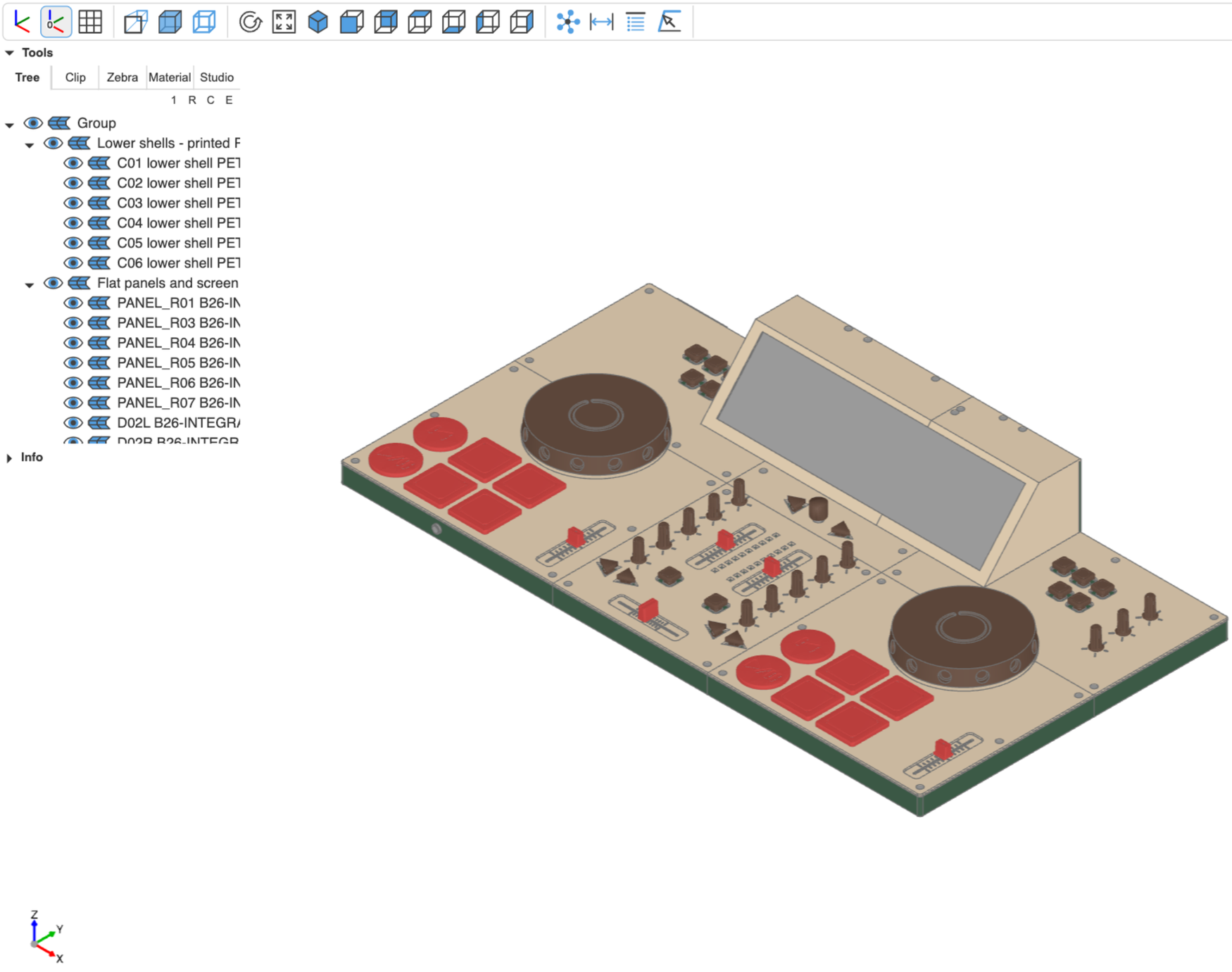
Two rear J15/J16 bridges use the same C07, at (220,14)/(380,14).

Display rear, cable and connector bodies are incomplete references; PCB mounts are omitted.

No physical fit, tool access, strength, thermal or electrical certification.

Reference / B26-build123d03 assembled model

Actual-model review image from the approved trial snapshot; not a dimensioned view.



Numerical construction schedule / continued

PRIMITIVES, DATUM AND OPERATION ORDER

Numerical schedule display is rounded to 0.000000001 mm; the companion JSON retains the original source numbers. This is not a manufacturing tolerance.

Table X right, Y toward operator, Z up; CAD (X,-Y,Z). All regions are closed nominal bounds.

CSG=[identifier,4x4 local-to-table transform, additive primitives, subtractive primitives]. Primitive=[kind,origin,size]; kind 0 is box with min corner and xyz size; kind 1 is vertical cone/frustum with bottom centre and [bottom radius,top radius,height]; kind 2 is vertical rounded rectangle with centre and [width,height,corner radius,extrusion height]. Union additive primitives, then subtract cuts in listed order; transform with local-to-table matrix; map table Y to negative CAD Y. The 48 groups named FIXED and E01-fixed_cage are attached to lower shells, clipped to shell XY boundaries. The 48 RETAINER groups and E01-body_retainer are separate parts; each fader retainer group contains two separate parts.

box: box(x0,x1,y0,y1,z0,z1) is the table-coordinate axis-aligned solid over the listed intervals.

cylinder: cylinder(x,y,z0,z1,r0,r1) is vertical at table (x,y), with bottom radius r0 and top radius r1. If r1 is absent it equals r0.

section: section(x0,width,[(Y,Z),...]) is a closed polygon in table Y/Z extruded from x0 to x0+width. Listed polygons are ordered and closed back to the first point.

For each domain [x0,x1,y0,y1], start box(x0,x1,y0,y1,0,18) and subtract box(max(x0,2),min(x1,598),max(y0,2),min(y1,318),2,102).

For each rear shell y0=0, union box(max(x0,153.55),min(x1,446.45),0,2,18,23+80*sin(49 degrees)) when the X interval is positive.

For C01/C03/C04/C06 only, fillet the outside vertical corner R3 and inside vertical corner R1, before adding other features. Fillet the bottom exterior perimeter edges R0.8; apply only to external X=0/600 or Y=0/320 straight lines and their R3 corner arcs. Internal seam edges are not filleted.

C01: subtract box(37.25,72.15,-0.01,2.01,2,18); subtract two horizontal circular cuts radius1.8 at X32.75 and 76.65, Z9.95, with axes along table Y from -0.01 to 2.01. Subtract rear AC01 box defined by enclosure_data.json inlet_definition centre and width/height, table Y0..2.

C02: for each polygon_XZ in enclosure_data.json ventilation_cells_original, subtract the polygon extruded along table Y from -0.01 to 2.01.

C04: subtract horizontal circular cut radius3.1 on X101/Z10, axis table Y from 317.99 to 320.01.

Attach 33 posts from enclosure_data.json posts: union their foot and core, subtract their pilot, intersect with box(0,600,0,320,0,100), then intersect each post with each shell domain and union its intersecting piece to that shell.

For each of the 16 joints in enclosure_data.json joints, at offsets -9 and +9 along the normal axis union a pad cylinder(px,py,2,8,4.5), subtract its pilot cylinder(px,py,2.5,8.01,1.25), clip to each owning shell and attach.

For the 48 FIXED rows and E01-fixed_cage in enclosure_data.json controls, evaluate listed additive and subtractive local CSG primitives and local-to-table transform; intersect with each shell domain and attach. The 48 RETAINER rows and E01-body_retainer are separate printed parts. Fader end bosses/pilots and retainers already contain the B26-FADER-01 translation.

Device mount axes are in enclosure_data.json device_mount_axes_original_table_XY, with Y translation +20. At PI and DAC axes union cylinder(x,y,2,6,3.5); at PSU axes union cylinder(x,y,2,3,5). PI axes subtract cylinder(x,y,1,6.01,1); PSU axes subtract cylinder(x,y,-0.01,3.01,1.7) and frustum cylinder(x,y,0,1.3,3,1.7). DAC pads have no modeled retention holes.

For each jog centre (110,165) and (490,165), for angles 150,270,30 degrees take x=cx+31*cos(a), y=cy-31*sin(a). Union cylinder(x,y,2,8,5); subtract cylinder(x,y,-0.01,2.5,2.4) and cylinder(x,y,2.5,8.01,1.45).

For each jog centre and angles 60,180,300 degrees take x=cx+26.2*cos(a), y=cy-26.2*sin(a); union cylinder(x,y,2,3.5,1). Union an anti-rotation key defined in local CAD by box X24..26, Y-1..1, Z0..1; rotate +60 degrees about CAD Z and translate to (cx,-165,2).

ROOF PARAMETERS AND ORDERED OPERATIONS

Roof X ranges and angles: {"angle_deg": 49, "roof_halves_X": {"D02L": [153.55, 335], "D02R": [335, 446.45]}}

Let F=96+5/tan(49 degrees), J=96-80*cos(49 degrees), U=25+80*sin(49 degrees), CY=96-40*cos(49 degrees), H=23+40*sin(49 degrees). Angles in degrees.

For each half X=x0..x1, form section(x0,x1-x0,[(0,U-2),(J,U-2),(F,18),(F,20),(J,U),(0,U)]). Union the end-return section at X153.55..155.55 for left or X444.45..446.45 for right: [(2,18),(F,18),(F,20),(J,U),(2,U)].

Place display-plane cutters/features using CAD Plane origin (300,-CY,H), x_dir=(side,0,0), z_dir=(0,-sin49,cos49); local y_dir follows the right-handed frame. When side*row>0, reflect local geometry through XZ before placement. For the opening subtract the placed local box X[-136,136], Y[-36,36], Z[-5,5] with side=1,row=1.

For each X in 210,230,300,370,390 strictly inside the half X interval, subtract cylinder(X,4,U-2.01,U+0.01,1.7) and countersink frustum cylinder(X,4,U-1.3,U,1.7,3).

Left only: union box(325,345,13,29,U-10,U-2), then subtract pilot cylinders at (340,18) and (340,24) from Z=U-9 to U-1.99 radius1.25. Right only: subtract through cylinders at these axes from U-2.01 to U+0.01 radius1.7 and countersink frusta U-1.3..U radii1.7..3.

At X=228 on left or 372 on right, union section(X-5,10,[(95,16),(116,16),(116,18),(F,18),(95,23+tan49)]); subtract cylinder(X,110,15.99,18.01,1.7).

For row=-1,+1, construct local display support: union boxes X138.5..144.2 Y18..30 Z-21.2..0; X138.25..140 Y20..39.797470349551496 Z-21..-0.25; and X130..140 Y38.297470349551496..39.797470349551496 Z-21..-0.25. Subtract box X136..146 Y19.9..28.1 Z-21.2..-18 and local cylinders at (140.7,24), radius1.25 Z-18.1..-11 and radius4 Z-60..-21. Place with side=-1 for left or +1 for right and current row, then union.

Where present, fillet only the exterior top edge at Z=U R0.8. Do not fillet internal seams.

FOUR PRINTED DISPLAY TOES

Four separate printed display toes, one for each side=-1,+1 and row=-1,+1. In the display-plane local frame union box X136.5..144.2, Y20..28, Z-21..-18 and box X136.5..137.5, Y20..28, Z-18..-17. Subtract cylinder at local (140.7,24), radius1.7, Z-21.1..-17.9.

Numerical construction schedule / continued

For each side,row, mirror the local toe through the local XZ plane when side*row>0; then place in the display frame specified in roof_parameters, origin (300,-CY,H), x_dir=(side,0,0), z_dir=(0,-sin49,cos49), H=23+40*sin49. This includes the B26 Z+5 displacement.

ENCLOSURE FASTENER REFERENCE GEOMETRY AND PLACEMENTS

The B26 model includes 197 nominal enclosure fastener references. They have no modeled threads or drive recesses; real purchased geometry and physical fit are not certified. Jog hardware is listed separately in the jog drawing. No U1-U4 mounting screws are specified.

Define local socket_screw(r,length,headR,headH) as a cylinder radius r over Z=-length..0 union a cylinder radius headR over Z=0..headH. Define local countersunk(r,length,headR) as a cylinder radius r over Z=-length..0 union a frustum from radius r at Z=-(headR-r) to headR at Z=0.

For each of the 16 C07 joints, at each pad axis +/-9 along the joint normal place socket_screw(1.5,8,2.75,3) at table (x,y), local Z0=global Z11. Quantity 32.

At each of the 28 flat-panel closure axes in the panel drawing place countersunk(1.5,8,3) with its top at global Z20. At five rear roof axes (X210/230/300/370/390,Y4) and two lap axes (340,18)/(340,24), use the same screw with top Z=25+80*sin(49 degrees). Quantity 35.

For the two PSU fasteners at X361.5/416.5, table Y54.35, place countersunk(1.5,6,3) in CAD frame origin (x,-54.35,0), x_dir=(1,0,0), z_dir=(0,0,-1). Quantity 2.

For each of the 49 removable control CSG groups in enclosure_data.json, inspect the listed subtractive primitives. Select every kind-1 frustum with unequal bottom and top radii and top radius 1.6 for P controls or 2.2 for all other controls. For local origin (u,v,w), size (r0,r1,h), and local-to-table transform M, place a screw at table x=M00*u+M01*v+M03, y=M10*u+M11*v+M13, global top z=M23+w+h. P controls use countersunk(0.8,5,1.6), four per control; B/T/F/E controls use countersunk(1,6,2.2), two per control. Quantity 124.

For each of four display toes, start with socket_screw(1.5,8,2.75,3) placed in local frame origin (140.7,24,-21), x_dir=(1,0,0), z_dir=(0,0,-1). Reflect this local shape through XZ if side*row>0, and place using the same display frame and side/row combinations defined in display_toes. Quantity 4.

Count: 32 joint + 35 closure + 2 PSU + 124 control + 4 display-toe = 197 enclosure fastener references.

JOINTS, POST GEOMETRY, DEVICE AXES AND INLET

C07: {"hole_centres": [[-9, 0], [9, 0]], "hole_diameter": 3.4, "local_origin": "centre of bottom face", "placement": "for each joint centre, bottom Z=8; long direction X for normal X, Y for normal Y", "size": [28, 10, 3]}

All posts: core radius 4 from Z2 to the listed top; foot bottom radius 4.5 / top radius 4, Z2..5.8. Pilot radius 1.25; post clipped to the case XY boundary.

S1-M1-POST: XY [8, 8]; core Z [2, 18]; pilot Z [8, 18.01]

S1-M2-POST: XY [110, 8]; core Z [2, 18]; pilot Z [8, 18.01]

Numerical construction schedule / continued

S1-M3-POST: XY [210, 4]; core Z [2, 83.376766418]; pilot Z [73.376766418, 83.386766418]

S1-M4-POST: XY [8, 132]; core Z [2, 18]; pilot Z [8, 18.01]

S1-M5-POST: XY [160, 132]; core Z [2, 18]; pilot Z [8, 18.01]

S1-M6-POST: XY [212, 132]; core Z [2, 18]; pilot Z [8, 18.01]

S2a-M1-POST: XY [230, 4]; core Z [2, 83.376766418]; pilot Z [73.376766418, 83.386766418]

S2a-M2-POST: XY [300, 4]; core Z [2, 83.376766418]; pilot Z [73.376766418, 83.386766418]

S2a-M3-POST: XY [370, 4]; core Z [2, 83.376766418]; pilot Z [73.376766418, 83.386766418]

S2b-M3-POST: XY [228, 110]; core Z [2, 16]; pilot Z [6, 16.01]

S2b-M4-POST: XY [372, 110]; core Z [2, 16]; pilot Z [6, 16.01]

S3-M1-POST: XY [390, 4]; core Z [2, 83.376766418]; pilot Z [73.376766418, 83.386766418]

S3-M2-POST: XY [490, 8]; core Z [2, 18]; pilot Z [8, 18.01]

S3-M3-POST: XY [592, 8]; core Z [2, 18]; pilot Z [8, 18.01]

S3-M4-POST: XY [388, 132]; core Z [2, 18]; pilot Z [8, 18.01]

S3-M5-POST: XY [440, 132]; core Z [2, 18]; pilot Z [8, 18.01]

S3-M6-POST: XY [592, 132]; core Z [2, 18]; pilot Z [8, 18.01]

S4-M1-POST: XY [8, 148]; core Z [2, 18]; pilot Z [8, 18.01]

S4-M2-POST: XY [212, 148]; core Z [2, 18]; pilot Z [8, 18.01]

S4-M3-POST: XY [8, 230]; core Z [2, 18]; pilot Z [8, 18.01]

Numerical construction schedule / continued

S4-M4-POST: XY [212, 230]; core Z [2, 18]; pilot Z [8, 18.01]

S4-M5-POST: XY [8, 312]; core Z [2, 18]; pilot Z [8, 18.01]

S4-M6-POST: XY [212, 312]; core Z [2, 18]; pilot Z [8, 18.01]

S5-M1-POST: XY [228, 148]; core Z [2, 18]; pilot Z [8, 18.01]

S5-M2-POST: XY [372, 148]; core Z [2, 18]; pilot Z [8, 18.01]

S5-M3-POST: XY [228, 312]; core Z [2, 18]; pilot Z [8, 18.01]

S5-M4-POST: XY [372, 312]; core Z [2, 18]; pilot Z [8, 18.01]

S6-M1-POST: XY [388, 148]; core Z [2, 18]; pilot Z [8, 18.01]

S6-M2-POST: XY [592, 148]; core Z [2, 18]; pilot Z [8, 18.01]

S6-M3-POST: XY [388, 230]; core Z [2, 18]; pilot Z [8, 18.01]

S6-M4-POST: XY [592, 230]; core Z [2, 18]; pilot Z [8, 18.01]

S6-M5-POST: XY [388, 312]; core Z [2, 18]; pilot Z [8, 18.01]

S6-M6-POST: XY [592, 312]; core Z [2, 18]; pilot Z [8, 18.01]

{"axes": [[176.505, 12.25], [176.505, 61.25], [234.505, 12.25], [234.505, 61.25]], "id": "PI", "translation_Y": 20}

{"axes": [[276.5, 13.25], [276.5, 62.25], [334.5, 13.25], [334.5, 62.25]], "id": "DAC", "translation_Y": 20}

{"axes": [[361.5, 34.35], [416.5, 34.35]], "id": "PSU", "translation_Y": 20}

AC inlet centre X,Z: [180.775, 17.817175976]; cutout width,height: [47.5, 28]

Numerical construction schedule / continued

48 VENTILATION CELL POLYGONS — X,Z MM

{"bank": "LEFT", "column": 0, "polygon_XZ": [[252.9, 10.196152423], [249.9, 11.92820323], [246.9, 10.196152423], [246.9, 6.732050808], [249.9, 5], [252.9, 6.732050808]], "row": 0}

{"bank": "LEFT", "column": 1, "polygon_XZ": [[260.1, 10.196152423], [257.1, 11.92820323], [254.1, 10.196152423], [254.1, 6.732050808], [257.1, 5], [260.1, 6.732050808]], "row": 0}

{"bank": "LEFT", "column": 2, "polygon_XZ": [[267.3, 10.196152423], [264.3, 11.92820323], [261.3, 10.196152423], [261.3, 6.732050808], [264.3, 5], [267.3, 6.732050808]], "row": 0}

{"bank": "LEFT", "column": 3, "polygon_XZ": [[274.5, 10.196152423], [271.5, 11.92820323], [268.5, 10.196152423], [268.5, 6.732050808], [271.5, 5], [274.5, 6.732050808]], "row": 0}

{"bank": "LEFT", "column": 4, "polygon_XZ": [[281.7, 10.196152423], [278.7, 11.92820323], [275.7, 10.196152423], [275.7, 6.732050808], [278.7, 5], [281.7, 6.732050808]], "row": 0}

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{"bank": "LEFT", "column": 0, "polygon_XZ": [[256.5, 16.43153533], [253.5, 18.163586138], [250.5, 16.43153533], [250.5, 12.967433715], [253.5, 11.235382907], [256.5, 12.967433715]], "row": 1}

{"bank": "LEFT", "column": 1, "polygon_XZ": [[263.7, 16.43153533], [260.7, 18.163586138], [257.7, 16.43153533], [257.7, 12.967433715], [260.7, 11.235382907], [263.7, 12.967433715]], "row": 1}

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{"bank": "LEFT", "column": 4, "polygon_XZ": [[285.3, 16.43153533], [282.3, 18.163586138], [279.3, 16.43153533], [279.3, 12.967433715], [282.3, 11.235382907], [285.3, 12.967433715]], "row": 1}

{"bank": "LEFT", "column": 5, "polygon_XZ": [[292.5, 16.43153533], [289.5, 18.163586138], [286.5, 16.43153533], [286.5, 12.967433715], [289.5, 11.235382907], [292.5, 12.967433715]], "row": 1}

{"bank": "LEFT", "column": 0, "polygon_XZ": [[252.9, 22.666918237], [249.9, 24.398969045], [246.9, 22.666918237], [246.9, 19.202816622], [249.9, 17.470765814], [252.9, 19.202816622]], "row": 2}

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{"bank": "LEFT", "column": 4, "polygon_XZ": [[281.7, 22.666918237], [278.7, 24.398969045], [275.7, 22.666918237], [275.7, 19.202816622], [278.7, 17.470765814], [281.7, 19.202816622]], "row": 2}

Numerical construction schedule / continued

{"bank": "LEFT", "column": 5, "polygon_XZ": [[288.9, 22.666918237], [285.9, 24.398969045], [282.9, 22.666918237], [282.9, 19.202816622], [285.9, 17.470765814], [288.9, 19.202816622]], "row": 2}

{"bank": "LEFT", "column": 0, "polygon_XZ": [[256.5, 28.902301144], [253.5, 30.634351952], [250.5, 28.902301144], [250.5, 25.438199529], [253.5, 23.706148722], [256.5, 25.438199529]], "row": 3}

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{"bank": "LEFT", "column": 4, "polygon_XZ": [[285.3, 28.902301144], [282.3, 30.634351952], [279.3, 28.902301144], [279.3, 25.438199529], [282.3, 23.706148722], [285.3, 25.438199529]], "row": 3}

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{"bank": "RIGHT", "column": 2, "polygon_XZ": [[327.9, 10.196152423], [324.9, 11.92820323], [321.9, 10.196152423], [321.9, 6.732050808], [324.9, 5], [327.9, 6.732050808]], "row": 0}

{"bank": "RIGHT", "column": 3, "polygon_XZ": [[335.1, 10.196152423], [332.1, 11.92820323], [329.1, 10.196152423], [329.1, 6.732050808], [332.1, 5], [335.1, 6.732050808]], "row": 0}

{"bank": "RIGHT", "column": 4, "polygon_XZ": [[342.3, 10.196152423], [339.3, 11.92820323], [336.3, 10.196152423], [336.3, 6.732050808], [339.3, 5], [342.3, 6.732050808]], "row": 0}

{"bank": "RIGHT", "column": 5, "polygon_XZ": [[349.5, 10.196152423], [346.5, 11.92820323], [343.5, 10.196152423], [343.5, 6.732050808], [346.5, 5], [349.5, 6.732050808]], "row": 0}

{"bank": "RIGHT", "column": 0, "polygon_XZ": [[317.1, 16.43153533], [314.1, 18.163586138], [311.1, 16.43153533], [311.1, 12.967433715], [314.1, 11.235382907], [317.1, 12.967433715]], "row": 1}

{"bank": "RIGHT", "column": 1, "polygon_XZ": [[324.3, 16.43153533], [321.3, 18.163586138], [318.3, 16.43153533], [318.3, 12.967433715], [321.3, 11.235382907], [324.3, 12.967433715]], "row": 1}

{"bank": "RIGHT", "column": 2, "polygon_XZ": [[331.5, 16.43153533], [328.5, 18.163586138], [325.5, 16.43153533], [325.5, 12.967433715], [328.5, 11.235382907], [331.5, 12.967433715]], "row": 1}

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{"bank": "RIGHT", "column": 4, "polygon_XZ": [[345.9, 16.43153533], [342.9, 18.163586138], [339.9, 16.43153533], [339.9, 12.967433715], [342.9, 11.235382907], [345.9, 12.967433715]], "row": 1}

Numerical construction schedule / continued

{"bank": "RIGHT", "column": 5, "polygon_XZ": [[353.1, 16.43153533], [350.1, 18.163586138], [347.1, 16.43153533], [347.1, 12.967433715], [350.1, 11.235382907], [353.1, 12.967433715]], "row": 1}

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{"bank": "RIGHT", "column": 0, "polygon_XZ": [[317.1, 28.902301144], [314.1, 30.634351952], [311.1, 28.902301144], [311.1, 25.438199529], [314.1, 23.706148722], [317.1, 25.438199529]], "row": 3}

{"bank": "RIGHT", "column": 1, "polygon_XZ": [[324.3, 28.902301144], [321.3, 30.634351952], [318.3, 28.902301144], [318.3, 25.438199529], [321.3, 23.706148722], [324.3, 25.438199529]], "row": 3}

{"bank": "RIGHT", "column": 2, "polygon_XZ": [[331.5, 28.902301144], [328.5, 30.634351952], [325.5, 28.902301144], [325.5, 25.438199529], [328.5, 23.706148722], [331.5, 25.438199529]], "row": 3}

{"bank": "RIGHT", "column": 3, "polygon_XZ": [[338.7, 28.902301144], [335.7, 30.634351952], [332.7, 28.902301144], [332.7, 25.438199529], [335.7, 23.706148722], [338.7, 25.438199529]], "row": 3}

{"bank": "RIGHT", "column": 4, "polygon_XZ": [[345.9, 28.902301144], [342.9, 30.634351952], [339.9, 28.902301144], [339.9, 25.438199529], [342.9, 23.706148722], [345.9, 25.438199529]], "row": 3}

{"bank": "RIGHT", "column": 5, "polygon_XZ": [[353.1, 28.902301144], [350.1, 30.634351952], [347.1, 28.902301144], [347.1, 25.438199529], [350.1, 23.706148722], [353.1, 25.438199529]], "row": 3}

CONTROL CSG RECIPES — UNION ADDITIONS, THEN SUBTRACT CUTS

RECIPE 1: [additions, cuts] = [[[0, [-8.1, -8.1, -2], [16.2, 16.2, 6.7]], [1, [0, -9.2, -2], [2.2, 2.2, 6.7]], [1, [0, 9.2, -2], [2.2, 2.2, 6.7]], [0, [-8.1, -8.1, -8.8], [1.8, 16.2, 6.8]], [0, [6.3, -8.1, -8.8], [1.8, 16.2, 6.8]], [0, [-6.3, -8.1, -8.8], [12.6, 1.8, 6.8]], [0, [-6.3, 6.3, -8.8], [12.6, 1.8, 6.8]], [1, [0, -9.2, -8.8], [2.2, 2.2, 6.8]], [1, [0, 9.2, -8.8], [2.2, 2.2, 6.8]]], [[0, [-6.3, -6.3, 0], [12.6, 12.6, 4.8]], [0, [-4, -4, -2.1], [8, 8, 2.2]], [0, [-8.2, -3.7, -6], [16.4, 7.4, 6.1]], [1, [-6.25, -2.5, -2.1], [1.2, 1.2, 2.2]], [1, [-6.25, 2.5, -2.1], [1.2, 1.2, 2.2]], [1, [6.25, -2.5, -2.1], [1.2, 1.2, 2.2]], [1, [6.25, 2.5, -2.1], [1.2, 1.2, 2.2]], [1, [0, -4.5, -2.1], [1.2, 1.2, 2.2]], [1, [0, 4.5, -2.1], [1.2, 1.2, 2.2]], [1, [0, -9.2, -0.8], [0.8, 0.8, 5.6]], [1, [0, 9.2, -0.8], [0.8, 0.8, 5.6]]]]

RECIPE 2: [additions, cuts] = [[[0, [-7.9, -7.7, -2], [15.8, 16.9, 8.8]], [0, [-7.9, -7.7, -5.3], [4.1, 3.3, 3.3]], [0, [3.8, -7.7, -5.3], [4.1, 3.3, 3.3]], [0, [-7.9, 5.4, -5.3], [4.1, 3.8, 3.3]], [0, [3.8, 5.4, -5.3], [4.1, 3.8, 3.3]], [1, [-6.8, -6.6, -5.3], [1.7, 1.7, 12.1]], [1, [-6.8, 8.1, -5.3], [1.7, 1.7, 12.1]], [1, [6.8, -6.6, -5.3], [1.7, 1.7, 12.1]], [1, [6.8, 8.1, -5.3], [1.7, 1.7, 12.1]], [1, [-6.8, -6.6, 6.8], [1.7, 1.7, 0.2]], [1, [-6.8, 8.1, 6.8], [1.7, 1.7, 0.2]], [1, [6.8, -6.6, 6.8], [1.7, 1.7, 0.2]], [1, [6.8, 8.1, 6.8], [1.7, 1.7, 0.2]]], [[0, [-5.9, -5.7, 0], [11.8, 12.9, 6.9]], [0, [-3, -5, -2.1], [6, 10, 2.2]], [0, [-8, -4, 0.1], [16, 8, 6.8]], [0, [-3, -7.8, 0.1], [6, 17.1, 6.8]], [0, [-5.6, -1.3, -2.1], [2.2, 2.6, 2.2]], [0, [3.4, -1.3, -2.1], [2.2, 2.6, 2.2]], [1, [-2.5, 7, -2.1], [0.9, 0.9, 2.2]], [1, [0, 7, -2.1], [0.9, 0.9, 2.2]], [1, [2.5, 7,

Numerical construction schedule / continued

-2.1], [0.9, 0.9, 2.2]], [1, [-6.8, -6.6, 2.3], [0.6, 0.6, 4.6]], [1, [-6.8, 8.1, 2.3], [0.6, 0.6, 4.6]], [1, [6.8, -6.6, 2.3], [0.6, 0.6, 4.6]], [1, [6.8, 8.1, 2.3], [0.6, 0.6, 4.6]], [1, [-6.8, -6.6, 6.8], [0.9, 0.9, 0.3]], [1, [-6.8, 8.1, 6.8], [0.9, 0.9, 0.3]], [1, [6.8, -6.6, 6.8], [0.9, 0.9, 0.3]], [1, [6.8, 8.1, 6.8], [0.9, 0.9, 0.3]]]]

RECIPE 3: [additions, cuts] = [[[0, [-15, -4.7, -7.3], [30, 1.6, 7.3]], [0, [-15, 3.1, -7.3], [30, 1.6, 7.3]], [0, [-33.75, -6.7, -7.3], [2, 13.4, 13.5]], [0, [31.75, -6.7, -7.3], [2, 13.4, 13.5]], [0, [-31.75, -6.7, -7.3], [7, 2, 13.5]], [0, [-31.75, 4.7, -7.3], [7, 2, 13.5]], [0, [24.75, -6.7, -7.3], [7, 2, 13.5]], [0, [24.75, 4.7, -7.3], [7, 2, 13.5]], [1, [-33.95, 0, -7.3], [2.2, 2.2, 13.5]], [1, [33.95, 0, -7.3], [2.2, 2.2, 13.5]]], [[1, [-33.95, 0, 0.7], [0.8, 0.8, 5.6]], [1, [33.95, 0, 0.7], [0.8, 0.8, 5.6]]]]

RECIPE 4: [additions, cuts] = [[[0, [-15, -4.7, -6.8], [30, 1.6, 6.8]], [0, [-15, 3.1, -6.8], [30, 1.6, 6.8]], [0, [-33.6, -6.7, -6.8], [2, 13.4, 13]], [0, [31.6, -6.7, -6.8], [2, 13.4, 13]], [0, [-31.6, -6.7, -6.8], [7, 2, 13]], [0, [-31.6, 4.7, -6.8], [7, 2, 13]], [0, [24.6, -6.7, -6.8], [7, 2, 13]], [0, [24.6, 4.7, -6.8], [7, 2, 13]], [1, [-33.8, 0, -6.8], [2.2, 2.2, 13]], [1, [33.8, 0, -6.8], [2.2, 2.2, 13]]], [[1, [-33.8, 0, 0.7], [0.8, 0.8, 5.6]], [1, [33.8, 0, 0.7], [0.8, 0.8, 5.6]]]]

RECIPE 5: [additions, cuts] = [[[0, [-8.1, -9.5, -2], [16.2, 19, 8.7]], [0, [-8.1, -9.5, -6.5], [2, 19, 4.5]], [0, [6.1, -9.5, -6.5], [2, 19, 4.5]], [1, [0, -10.6, -6.5], [2.2, 2.2, 13.2]], [1, [0, 10.6, -6.5], [2.2, 2.2, 13.2]]], [[0, [-6.1, -7.4, 0], [12.2, 15, 6.8]], [0, [-4.5, -9.6, -6], [9, 19.2, 6.1]], [0, [-4.5, -6, -2.1], [9, 12, 2.2]], [1, [0, -10.6, 1.2], [0.8, 0.8, 5.6]], [1, [0, 10.6, 1.2], [0.8, 0.8, 5.6]]]]

RECIPE 6: [additions, cuts] = [[[0, [-8.1, -8.1, 4.7], [16.2, 16.2, 1.5]], [1, [0, -9.2, 4.7], [2.2, 2.2, 1.5]], [1, [0, 9.2, 4.7], [2.2, 2.2, 1.5]]], [[2, [0, 0, 4.6], [8, 8, 1.4, 1.7]], [1, [0, -9.2, 4.6], [1.1, 1.1, 1.7]], [1, [0, 9.2, 4.6], [1.1, 1.1, 1.7]], [1, [0, -9.2, 5.1], [1.1, 2.2, 1.1]], [1, [0, 9.2, 5.1], [1.1, 2.2, 1.1]]]]

RECIPE 7: [additions, cuts] = [[[0, [-7.9, -7.7, 7], [15.8, 16.9, 1.5]], [1, [-6.8, -6.6, 7], [1.7, 1.7, 1.5]], [1, [-6.8, 8.1, 7], [1.7, 1.7, 1.5]], [1, [6.8, -6.6, 7], [1.7, 1.7, 1.5]], [1, [6.8, 8.1, 7], [1.7, 1.7, 1.5]]], [[1, [0, 0, 6.9], [4.25, 4.25, 1.7]], [1, [-6.8, -6.6, 6.9], [0.9, 0.9, 1.7]], [1, [-6.8, 8.1, 6.9], [0.9, 0.9, 1.7]], [1, [6.8, -6.6, 6.9], [0.9, 0.9, 1.7]], [1, [6.8, 8.1, 6.9], [0.9, 0.9, 1.7]], [1, [-6.8, -6.6, 7.8], [0.9, 1.6, 0.7]], [1, [-6.8, 8.1, 7.8], [0.9, 1.6, 0.7]], [1, [6.8, -6.6, 7.8], [0.9, 1.6, 0.7]], [1, [6.8, 8.1, 7.8], [0.9, 1.6, 0.7]]]]

RECIPE 8: [additions, cuts] = [[[0, [-37.15, -6.7, 6.2], [8.4, 13.4, 1.5]], [0, [28.75, -6.7, 6.2], [8.4, 13.4, 1.5]], [1, [-33.95, 0, 6.2], [2.2, 2.2, 1.5]], [1, [33.95, 0, 6.2], [2.2, 2.2, 1.5]]], [[1, [-33.95, 0, 6.1], [1.1, 1.1, 1.7]], [1, [33.95, 0, 6.1], [1.1, 1.1, 1.7]], [1, [-33.95, 0, 6.6], [1.1, 2.2, 1.1]], [1, [33.95, 0, 6.6], [1.1, 2.2, 1.1]]]]

RECIPE 9: [additions, cuts] = [[[0, [-37, -6.7, 6.2], [8.4, 13.4, 1.5]], [0, [28.6, -6.7, 6.2], [8.4, 13.4, 1.5]], [1, [-33.8, 0, 6.2], [2.2, 2.2, 1.5]], [1, [33.8, 0, 6.2], [2.2, 2.2, 1.5]]], [[1, [-33.8, 0, 6.1], [1.1, 1.1, 1.7]], [1, [33.8, 0, 6.1], [1.1, 1.1, 1.7]], [1, [-33.8, 0, 6.6], [1.1, 2.2, 1.1]], [1, [33.8, 0, 6.6], [1.1, 2.2, 1.1]]]]

RECIPE 10: [additions, cuts] = [[[0, [-8.1, -9.5, 6.7], [16.2, 19, 1.5]], [1, [0, -10.6, 6.7], [2.2, 2.2, 1.5]], [1, [0, 10.6, 6.7], [2.2, 2.2, 1.5]]], [[1, [0, 0, 6.6], [3.7, 3.7, 1.7]], [1, [0, -10.6, 6.6], [1.1, 1.1, 1.7]], [1, [0, 10.6, 6.6], [1.1, 1.1, 1.7]], [1, [0, -10.6, 7.1], [1.1, 2.2, 1.1]], [1, [0, 10.6, 7.1], [1.1, 2.2, 1.1]]]]

49 FIXED AND 49 RETAINER GROUPS — OWNER, RECIPE, TRANSFORM

B01-FIXED / owner B01 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 91], [0, 1, 0, 42.25], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B02-FIXED / owner B02 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 111], [0, 1, 0, 42.25], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B03-FIXED / owner B03 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 131], [0, 1, 0, 42.25], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B04-FIXED / owner B04 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 111], [0, 1, 0, 66.25], [0, 0, 1, 10.8], [0, 0, 0, 1]]

Numerical construction schedule / continued

B05-FIXED / owner B05 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 131], [0, 1, 0, 66.25], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B06-FIXED / owner B06 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 471], [0, 1, 0, 42.25], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B07-FIXED / owner B07 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 491], [0, 1, 0, 42.25], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B08-FIXED / owner B08 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 511], [0, 1, 0, 42.25], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B09-FIXED / owner B09 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 491], [0, 1, 0, 66.25], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B10-FIXED / owner B10 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 511], [0, 1, 0, 66.25], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B11-FIXED / owner B11 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 32], [0, 1, 0, 248], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B12-FIXED / owner B12 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 32], [0, 1, 0, 294], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B13-FIXED / owner B13 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 78], [0, 1, 0, 248], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B14-FIXED / owner B14 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 124], [0, 1, 0, 248], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B15-FIXED / owner B15 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 78], [0, 1, 0, 294], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B16-FIXED / owner B16 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 124], [0, 1, 0, 294], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B17-FIXED / owner B17 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 412], [0, 1, 0, 248], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B18-FIXED / owner B18 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 412], [0, 1, 0, 294], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B19-FIXED / owner B19 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 458], [0, 1, 0, 248], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B20-FIXED / owner B20 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 504], [0, 1, 0, 248], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B21-FIXED / owner B21 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 458], [0, 1, 0, 294], [0, 0, 1, 10.8], [0, 0, 0, 1]]

Numerical construction schedule / continued

B22-FIXED / owner B22 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 504], [0, 1, 0, 294], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B23-FIXED / owner B23 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 276], [0, 1, 0, 255], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B24-FIXED / owner B24 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 324], [0, 1, 0, 255], [0, 0, 1, 10.8], [0, 0, 0, 1]]

T01-FIXED / owner T01 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 277], [0, 1, 0, 125], [0, 0, 1, 10.8], [0, 0, 0, 1]]

T02-FIXED / owner T02 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 323], [0, 1, 0, 125], [0, 0, 1, 10.8], [0, 0, 0, 1]]

T03-FIXED / owner T03 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 236], [0, 1, 0, 278], [0, 0, 1, 10.8], [0, 0, 0, 1]]

T04-FIXED / owner T04 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 254], [0, 1, 0, 278], [0, 0, 1, 10.8], [0, 0, 0, 1]]

T05-FIXED / owner T05 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 348], [0, 1, 0, 278], [0, 0, 1, 10.8], [0, 0, 0, 1]]

T06-FIXED / owner T06 / recipe 1 / local-to-table 4x4 [[1, 0, 0, 366], [0, 1, 0, 278], [0, 0, 1, 10.8], [0, 0, 0, 1]]

P01-FIXED / owner P01 / recipe 2 / local-to-table 4x4 [[1, 0, 0, 559], [0, 1, 0, 41], [0, 0, 1, 7.3], [0, 0, 0, 1]]

P02-FIXED / owner P02 / recipe 2 / local-to-table 4x4 [[1, 0, 0, 559], [0, 1, 0, 69], [0, 0, 1, 7.3], [0, 0, 0, 1]]

P03-FIXED / owner P03 / recipe 2 / local-to-table 4x4 [[1, 0, 0, 559], [0, 1, 0, 97], [0, 0, 1, 7.3], [0, 0, 0, 1]]

P04-FIXED / owner P04 / recipe 2 / local-to-table 4x4 [[1, 0, 0, 244], [0, 1, 0, 151], [0, 0, 1, 7.3], [0, 0, 0, 1]]

P05-FIXED / owner P05 / recipe 2 / local-to-table 4x4 [[1, 0, 0, 244], [0, 1, 0, 177], [0, 0, 1, 7.3], [0, 0, 0, 1]]

P06-FIXED / owner P06 / recipe 2 / local-to-table 4x4 [[1, 0, 0, 244], [0, 1, 0, 203], [0, 0, 1, 7.3], [0, 0, 0, 1]]

P07-FIXED / owner P07 / recipe 2 / local-to-table 4x4 [[1, 0, 0, 244], [0, 1, 0, 229], [0, 0, 1, 7.3], [0, 0, 0, 1]]

P08-FIXED / owner P08 / recipe 2 / local-to-table 4x4 [[1, 0, 0, 244], [0, 1, 0, 255], [0, 0, 1, 7.3], [0, 0, 0, 1]]

P09-FIXED / owner P09 / recipe 2 / local-to-table 4x4 [[1, 0, 0, 356], [0, 1, 0, 151], [0, 0, 1, 7.3], [0, 0, 0, 1]]

Numerical construction schedule / continued

P10-FIXED / owner P10 / recipe 2 / local-to-table 4x4 [[1, 0, 0, 356], [0, 1, 0, 177], [0, 0, 1, 7.3], [0, 0, 0, 1]]

P11-FIXED / owner P11 / recipe 2 / local-to-table 4x4 [[1, 0, 0, 356], [0, 1, 0, 203], [0, 0, 1, 7.3], [0, 0, 0, 1]]

P12-FIXED / owner P12 / recipe 2 / local-to-table 4x4 [[1, 0, 0, 356], [0, 1, 0, 229], [0, 0, 1, 7.3], [0, 0, 0, 1]]

P13-FIXED / owner P13 / recipe 2 / local-to-table 4x4 [[1, 0, 0, 356], [0, 1, 0, 255], [0, 0, 1, 7.3], [0, 0, 0, 1]]

F01-FIXED / owner F01 / recipe 3 / local-to-table 4x4 [[0, 1, 0, 196], [1, 0, 0, 272], [0, 0, 1, 9.3], [0, 0, 0, 1]]

F02-FIXED / owner F02 / recipe 3 / local-to-table 4x4 [[0, 1, 0, 576], [1, 0, 0, 272], [0, 0, 1, 9.3], [0, 0, 0, 1]]

F03-FIXED / owner F03 / recipe 3 / local-to-table 4x4 [[0, 1, 0, 276], [1, 0, 0, 197], [0, 0, 1, 9.3], [0, 0, 0, 1]]

F04-FIXED / owner F04 / recipe 3 / local-to-table 4x4 [[0, 1, 0, 324], [1, 0, 0, 197], [0, 0, 1, 9.3], [0, 0, 0, 1]]

F05-FIXED / owner F05 / recipe 4 / local-to-table 4x4 [[1, 0, 0, 300], [0, -1, 0, 301], [0, 0, 1, 8.8], [0, 0, 0, 1]]

E01-fixed_cage / owner E01 / recipe 5 / local-to-table 4x4 [[1, 0, 0, 300], [0, 1, 0, 125], [0, 0, 1, 8.5], [0, 0, 0, 1]]

B01-RETAINER / owner B01 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 91], [0, 1, 0, 42.25], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B02-RETAINER / owner B02 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 111], [0, 1, 0, 42.25], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B03-RETAINER / owner B03 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 131], [0, 1, 0, 42.25], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B04-RETAINER / owner B04 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 111], [0, 1, 0, 66.25], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B05-RETAINER / owner B05 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 131], [0, 1, 0, 66.25], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B06-RETAINER / owner B06 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 471], [0, 1, 0, 42.25], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B07-RETAINER / owner B07 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 491], [0, 1, 0, 42.25], [0, 0, 1, 10.8], [0, 0, 0, 1]]

Numerical construction schedule / continued

B08-RETAINER / owner B08 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 511], [0, 1, 0, 42.25], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B09-RETAINER / owner B09 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 491], [0, 1, 0, 66.25], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B10-RETAINER / owner B10 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 511], [0, 1, 0, 66.25], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B11-RETAINER / owner B11 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 32], [0, 1, 0, 248], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B12-RETAINER / owner B12 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 32], [0, 1, 0, 294], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B13-RETAINER / owner B13 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 78], [0, 1, 0, 248], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B14-RETAINER / owner B14 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 124], [0, 1, 0, 248], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B15-RETAINER / owner B15 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 78], [0, 1, 0, 294], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B16-RETAINER / owner B16 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 124], [0, 1, 0, 294], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B17-RETAINER / owner B17 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 412], [0, 1, 0, 248], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B18-RETAINER / owner B18 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 412], [0, 1, 0, 294], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B19-RETAINER / owner B19 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 458], [0, 1, 0, 248], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B20-RETAINER / owner B20 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 504], [0, 1, 0, 248], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B21-RETAINER / owner B21 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 458], [0, 1, 0, 294], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B22-RETAINER / owner B22 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 504], [0, 1, 0, 294], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B23-RETAINER / owner B23 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 276], [0, 1, 0, 255], [0, 0, 1, 10.8], [0, 0, 0, 1]]

B24-RETAINER / owner B24 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 324], [0, 1, 0, 255], [0, 0, 1, 10.8], [0, 0, 0, 1]]

T01-RETAINER / owner T01 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 277], [0, 1, 0, 125], [0, 0, 1, 10.8], [0, 0, 0, 1]]

Numerical construction schedule / continued

T02-RETAINER / owner T02 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 323], [0, 1, 0, 125], [0, 0, 1, 10.8], [0, 0, 0, 1]]

T03-RETAINER / owner T03 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 236], [0, 1, 0, 278], [0, 0, 1, 10.8], [0, 0, 0, 1]]

T04-RETAINER / owner T04 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 254], [0, 1, 0, 278], [0, 0, 1, 10.8], [0, 0, 0, 1]]

T05-RETAINER / owner T05 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 348], [0, 1, 0, 278], [0, 0, 1, 10.8], [0, 0, 0, 1]]

T06-RETAINER / owner T06 / recipe 6 / local-to-table 4x4 [[1, 0, 0, 366], [0, 1, 0, 278], [0, 0, 1, 10.8], [0, 0, 0, 1]]

P01-RETAINER / owner P01 / recipe 7 / local-to-table 4x4 [[1, 0, 0, 559], [0, 1, 0, 41], [0, 0, 1, 7.3], [0, 0, 0, 1]]

P02-RETAINER / owner P02 / recipe 7 / local-to-table 4x4 [[1, 0, 0, 559], [0, 1, 0, 69], [0, 0, 1, 7.3], [0, 0, 0, 1]]

P03-RETAINER / owner P03 / recipe 7 / local-to-table 4x4 [[1, 0, 0, 559], [0, 1, 0, 97], [0, 0, 1, 7.3], [0, 0, 0, 1]]

P04-RETAINER / owner P04 / recipe 7 / local-to-table 4x4 [[1, 0, 0, 244], [0, 1, 0, 151], [0, 0, 1, 7.3], [0, 0, 0, 1]]

P05-RETAINER / owner P05 / recipe 7 / local-to-table 4x4 [[1, 0, 0, 244], [0, 1, 0, 177], [0, 0, 1, 7.3], [0, 0, 0, 1]]

P06-RETAINER / owner P06 / recipe 7 / local-to-table 4x4 [[1, 0, 0, 244], [0, 1, 0, 203], [0, 0, 1, 7.3], [0, 0, 0, 1]]

P07-RETAINER / owner P07 / recipe 7 / local-to-table 4x4 [[1, 0, 0, 244], [0, 1, 0, 229], [0, 0, 1, 7.3], [0, 0, 0, 1]]

P08-RETAINER / owner P08 / recipe 7 / local-to-table 4x4 [[1, 0, 0, 244], [0, 1, 0, 255], [0, 0, 1, 7.3], [0, 0, 0, 1]]

P09-RETAINER / owner P09 / recipe 7 / local-to-table 4x4 [[1, 0, 0, 356], [0, 1, 0, 151], [0, 0, 1, 7.3], [0, 0, 0, 1]]

P10-RETAINER / owner P10 / recipe 7 / local-to-table 4x4 [[1, 0, 0, 356], [0, 1, 0, 177], [0, 0, 1, 7.3], [0, 0, 0, 1]]

P11-RETAINER / owner P11 / recipe 7 / local-to-table 4x4 [[1, 0, 0, 356], [0, 1, 0, 203], [0, 0, 1, 7.3], [0, 0, 0, 1]]

P12-RETAINER / owner P12 / recipe 7 / local-to-table 4x4 [[1, 0, 0, 356], [0, 1, 0, 229], [0, 0, 1, 7.3], [0, 0, 0, 1]]

Numerical construction schedule / continued

P13-RETAINER / owner P13 / recipe 7 / local-to-table 4x4 [[1, 0, 0, 356], [0, 1, 0, 255], [0, 0, 1, 7.3], [0, 0, 0, 1]]

F01-RETAINER / owner F01 / recipe 8 / local-to-table 4x4 [[0, 1, 0, 196], [1, 0, 0, 272], [0, 0, 1, 9.3], [0, 0, 0, 1]]

F02-RETAINER / owner F02 / recipe 8 / local-to-table 4x4 [[0, 1, 0, 576], [1, 0, 0, 272], [0, 0, 1, 9.3], [0, 0, 0, 1]]

F03-RETAINER / owner F03 / recipe 8 / local-to-table 4x4 [[0, 1, 0, 276], [1, 0, 0, 197], [0, 0, 1, 9.3], [0, 0, 0, 1]]

F04-RETAINER / owner F04 / recipe 8 / local-to-table 4x4 [[0, 1, 0, 324], [1, 0, 0, 197], [0, 0, 1, 9.3], [0, 0, 0, 1]]

F05-RETAINER / owner F05 / recipe 9 / local-to-table 4x4 [[1, 0, 0, 300], [0, -1, 0, 301], [0, 0, 1, 8.8], [0, 0, 0, 1]]

E01-body_retainer / owner E01 / recipe 10 / local-to-table 4x4 [[1, 0, 0, 300], [0, 1, 0, 125], [0, 0, 1, 8.5], [0, 0, 0, 1]]

SOURCE AND UNVERIFIED LIMITS

This is a transcription for nominal trial geometry and has not yet undergone complete independent source or drawing validation.

No verified print orientation, tolerance, calibrated fit, physical strength, full purchased-part geometry, connected cable access, electrical or thermal safety is implied.

1 / Reconstruction contract and scope

Use shell_dataset_DRAFT.json with this addendum: the embedded operations and 98 CSG rows are dimensional authority.

C01-C06: one each; C07: 16; D02L/D02R: one each; display toes: four.

49 fixed control groups are clipped and attached to the shells; 49 removable groups form 54 printed solids.

The fixed and removable CSG entries supply individual seat, pilot, ligament, cut and placement dimensions.

Local control matrices map (u,v,w,1) to table (X,Y,Z,1); CAD is (X,-Y,Z).

Primitive 0: box min corner and size; 1: frustum bottom centre, bottom/top radius and height;

2: centred rounded rectangle, width/height/radius and extrusion. Union additions, then subtract cuts.

Apply fillets and boolean operations in listed source order. Do not scale or measure these schematics.

Earlier enclosure PDF visible edges omit hidden features; acceptance did not establish reconstructibility.

Source transcription and numerical formulas are not an independent Boolean equivalence check.

Purchased geometry, cable access, fit, print strength, tolerances and safety are not established.

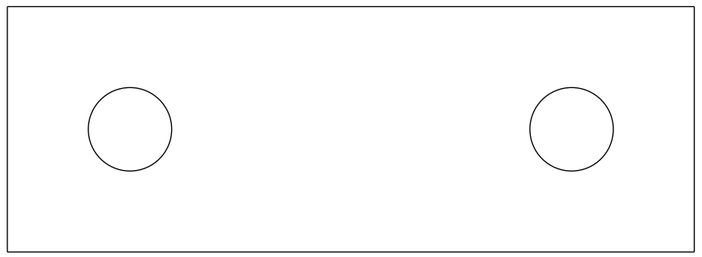
P01 STEP strict volume comparison failed; bounded exception applies only to the mechanical trial;

it is not proof of surface equivalence. No new model, release or approval is implied.

Provenance and exact hash references, ordered operations and verification actions are embedded in JSON.

2 / C07 bridges, posts and hidden pilot seats

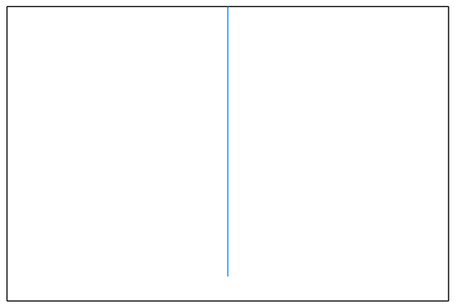
C07 PLAN / schematic



28 x 10 x 3; two D3.4 THRU, pitch 18; local bottom Z0; installed bottom Z8.

J01-J16 centres and normals: embedded schedules.joints. Place long direction along normal.

SEAM PAD SECTION / schematic



Each joint: pads at +/-9 normal; D9, Z2..8; pilot D2.5, Z2.5..8.01.

Floor Z0..2. Pilot depth from top 5.5 nominal; no modeled thread.

PANEL POSTS: 33 source rows. Foot frustum R4.5 at Z2 to R4 at Z5.8; core R4, Z2..top.

Pilot D2.5 from top-10 to top+0.01. Clip to global bounds then each owning shell before union.

Top and centre for each post: embedded schedules.posts. Clip at shell seams; no new fillet is implied.

PI: R3.5 Z2..6, pilot R1 Z1..6.01. DAC: same pad, no hole. PSU: R5 Z2..3,

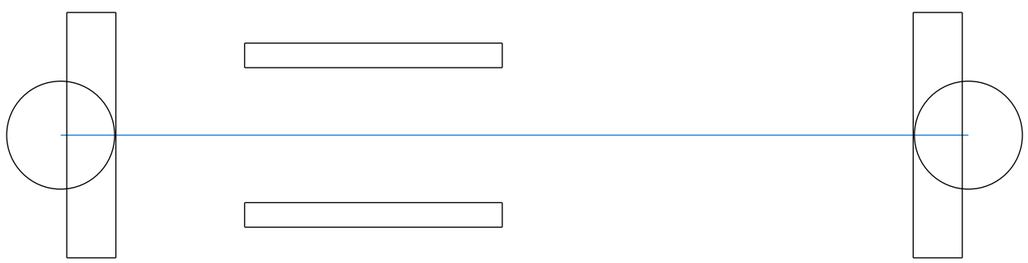
through R1.7 Z-0.01..3.01 and bottom frustum R3 to R1.7 over Z0..1.3.

Device axes: schedules.device_mount_axes_original_table_XY; add +20 to table Y.

These dimensions represent modeled features, not specification of purchased hardware or fit.

3 / Fader ends and internal ligaments

F01-F04 / local plan schematic, symmetric about u=0



Inner end wall |u|=31.75; end boss/pilot |u|=33.95; boss R2.2; pilot R0.8.

F05 uses |u|=31.60 and 33.80 respectively; its orientation is in its CSG matrix.

F01-F04: end walls u=31.75..33.75 and -33.75..-31.75; v=-6.7..6.7.

End ligaments positive u=24.75..31.75, negative u=-31.75..-24.75;

v=-6.7..-4.7 or 4.7..6.7. Rail u=-15..15; v=-4.7..-3.1 or 3.1..4.7.

F01-F04 end walls and ligaments w=-7.3..6.2, rails w=-7.3..0;

bosses w=-7.3..6.2, pilot w=0.7..6.3. Exact per-owner values are in JSON.

RETAINER / each of 10

F01-F04: boxes u=28.75..37.15 and -37.15..-28.75; F05: |u|=28.60..37.00.

Each box 8.4 x 13.4 x 1.5; v=-6.7..6.7; local w=6.2..7.7 for F01.

Add R2.2 boss, cut R1.1 through and countersink R1.1 to R2.2 over 1.1 height.

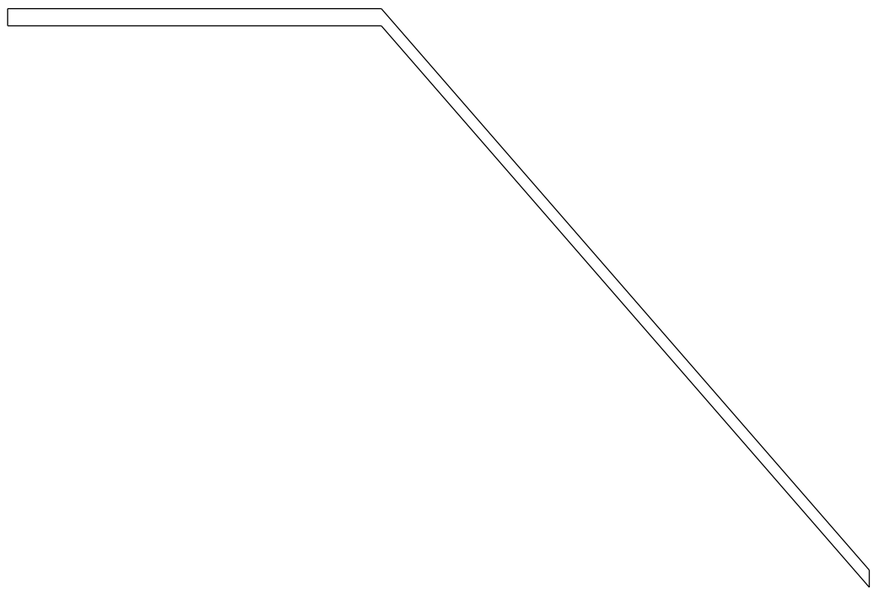
Per-row CSG is authoritative for exact w and all other controls, including E01.

B26 detail moved end primitives +/-1.5; B26-FADER-01 moved bosses/pilots another +/-1.2;

rails were not moved by the second change. The data already contain both changes: do not reapply.

4 / Roof and display support / local sections

ROOF SECTION in table Y/Z, not an evaluated projection



F=96+5/tan49 = 100.346434; J=96-80cos49 = 43.515278; U=25+80sin49 = 85.376766

D02L X153.55..335; D02R X335..446.45. Section is 2 thick in Z, not normal to slope.

End returns: X153.55..155.55 and X444.45..446.45. Rear seats: X210/230/300/370/390, Y4.

Lap: left box X325..345 Y13..29 Z(U-10)..(U-2), pilot axes (340,18)/(340,24).

Right lap holes: R1.7 through with R3 countersink. Front feet X228/372, Y110, R1.7.

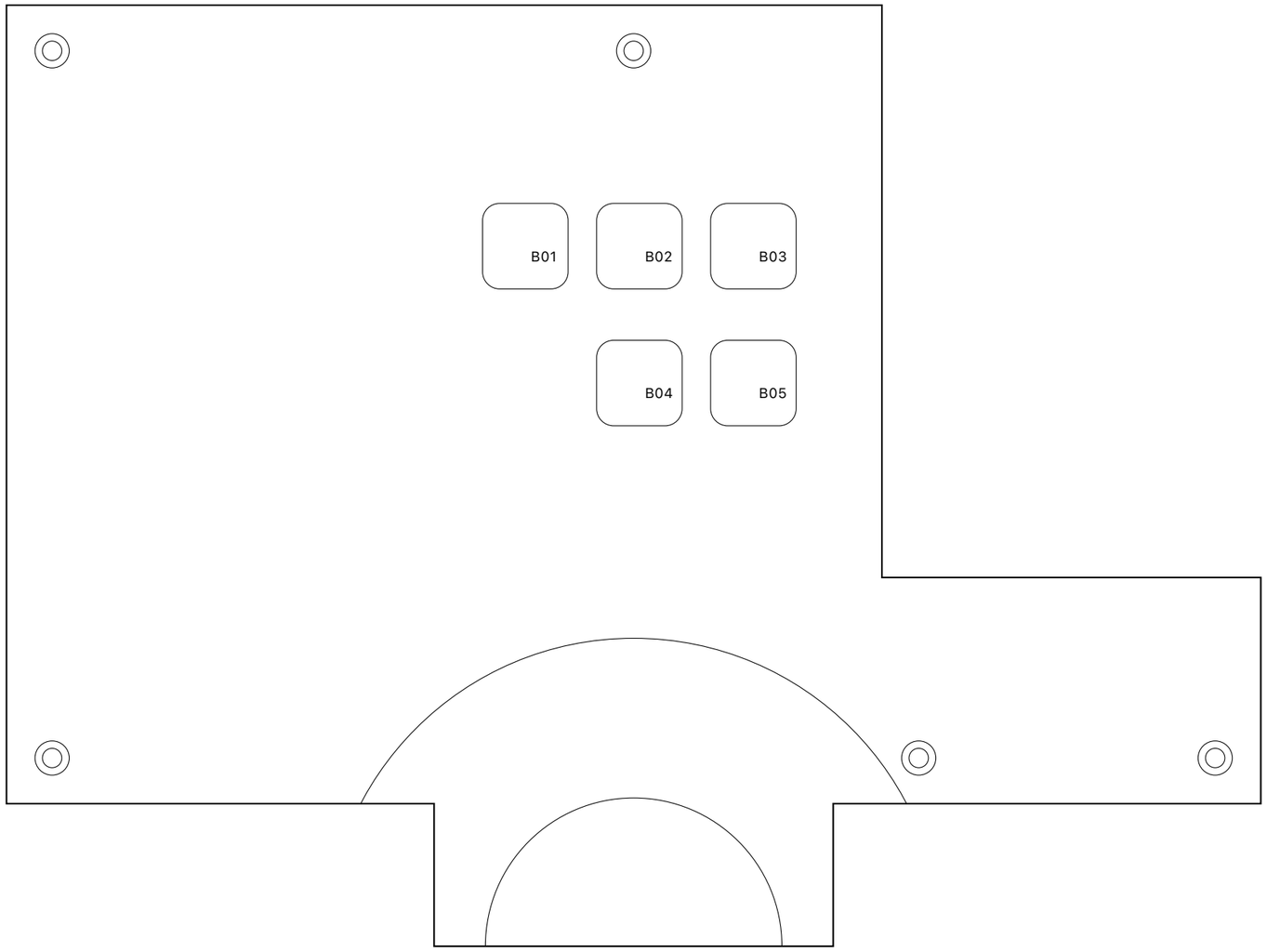
Display frame: CAD origin (300,-CY,H), CY=96-40cos49, H=23+40sin49;

x=(side,0,0), z=(0,-sin49,cos49), y=z cross x; mirror local y if side*row>0.

Support solids, lip, blind pilot, gallery and toe cuts: exact ordered local boxes/cylinders in JSON.

Toe base 7.7 x 8 x 3; lip 1 x 8 x 1; hole R1.7. Four side/row placements.

R01 / top view and aperture schedule / 1 piece



APERTURES: centre X,Y; W x H; R

T rows: see triangle definition in JSON

B01 91.00,42.25 rr 15.00 x 15.00 R3.00
B02 111.00,42.25 rr 15.00 x 15.00 R3.00
B03 131.00,42.25 rr 15.00 x 15.00 R3.00
B04 111.00,66.25 rr 15.00 x 15.00 R3.00
B05 131.00,66.25 rr 15.00 x 15.00 R3.00

CLOSURE AXES X,Y (28 overall)

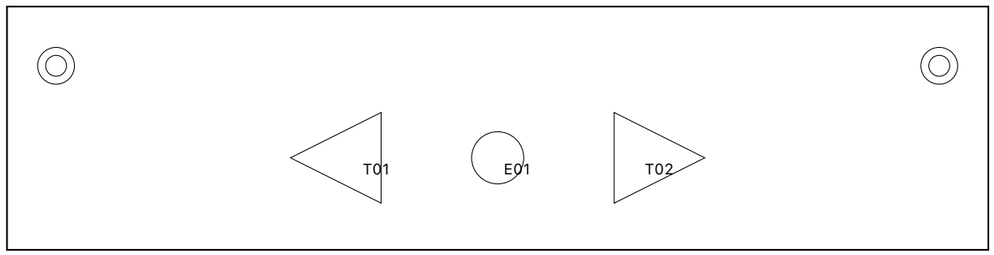
8.00, 8.00
110.00, 8.00
8.00, 132.00
160.00, 132.00
212.00, 132.00

Outline shown sharp: outside corners R3; top rim R0.8.

Jog circles clipped to panel outline. Exact CSG and ticks: JSON.

49 cap occurrences and nine exact faceted definitions: companion caps_poses / caps_mesh JSON.

R03 / top view and aperture schedule / 1 piece



APERTURES: centre X,Y; W x H; R

T rows: see triangle definition in JSON

E01 300.00,125.00 circle 8.50 x 8.50 R0.00
T01 277.00,125.00 rr 8.00 x 8.00 R1.40
T02 323.00,125.00 rr 8.00 x 8.00 R1.40

CLOSURE AXES X,Y (28 overall)

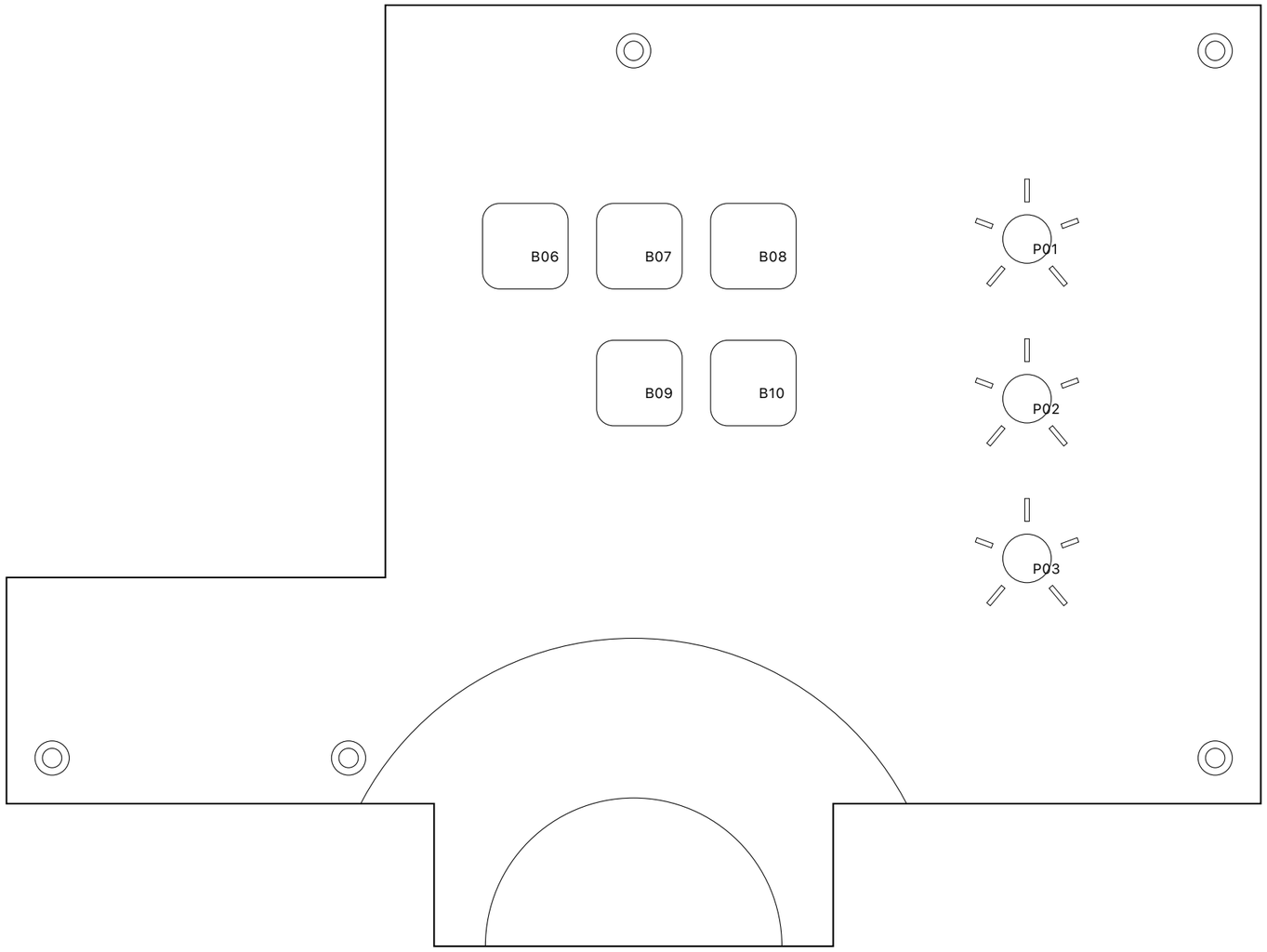
228.00, 110.00
372.00, 110.00

Outline shown sharp: outside corners R3; top rim R0.8.

Jog circles clipped to panel outline. Exact CSG and ticks: JSON.

49 cap occurrences and nine exact faceted definitions: companion caps_poses / caps_mesh JSON.

R04 / top view and aperture schedule / 1 piece



APERTURES: centre X,Y; W x H; R

T rows: see triangle definition in JSON

P01 559.00,41.00 circle 8.50 x 8.50 R0.00
P02 559.00,69.00 circle 8.50 x 8.50 R0.00
P03 559.00,97.00 circle 8.50 x 8.50 R0.00
B06 471.00,42.25 rr 15.00 x 15.00 R3.00
B07 491.00,42.25 rr 15.00 x 15.00 R3.00
B08 511.00,42.25 rr 15.00 x 15.00 R3.00
B09 491.00,66.25 rr 15.00 x 15.00 R3.00
B10 511.00,66.25 rr 15.00 x 15.00 R3.00

CLOSURE AXES X,Y (28 overall)

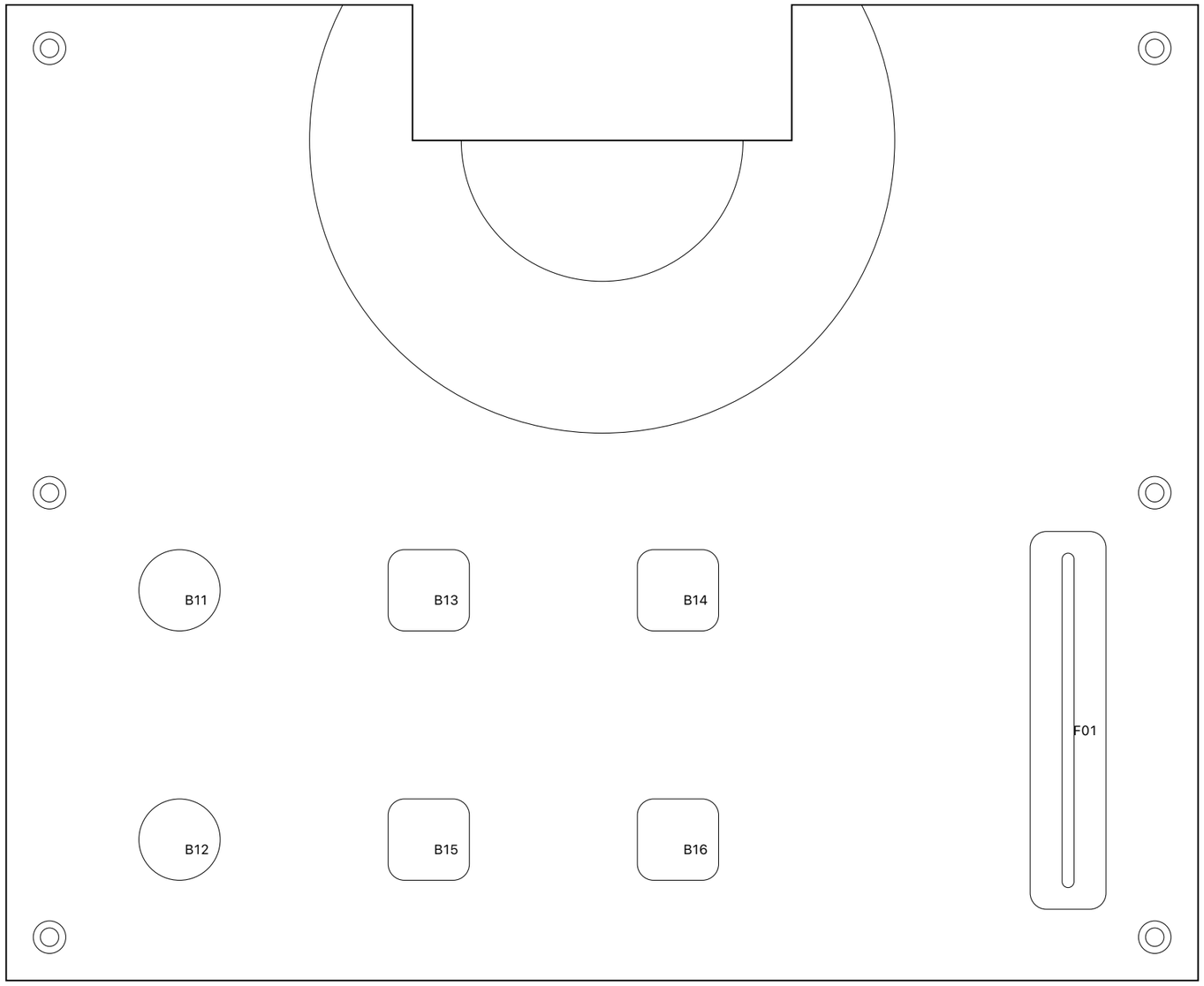
490.00, 8.00
592.00, 8.00
388.00, 132.00
440.00, 132.00
592.00, 132.00

Outline shown sharp: outside corners R3; top rim R0.8.

Jog circles clipped to panel outline. Exact CSG and ticks: JSON.

49 cap occurrences and nine exact faceted definitions: companion caps_poses / caps_mesh JSON.

R05 / top view and aperture schedule / 1 piece



APERTURES: centre X,Y; W x H; R

T rows: see triangle definition in JSON

B11 32.00,248.00 circle 15.00 x 15.00 R0.00
B12 32.00,294.00 circle 15.00 x 15.00 R0.00
B13 78.00,248.00 rr 15.00 x 15.00 R3.00
B14 124.00,248.00 rr 15.00 x 15.00 R3.00
B15 78.00,294.00 rr 15.00 x 15.00 R3.00
B16 124.00,294.00 rr 15.00 x 15.00 R3.00
F01 196.00,272.00 slot 2.20 x 61.70 R1.10

CLOSURE AXES X,Y (28 overall)

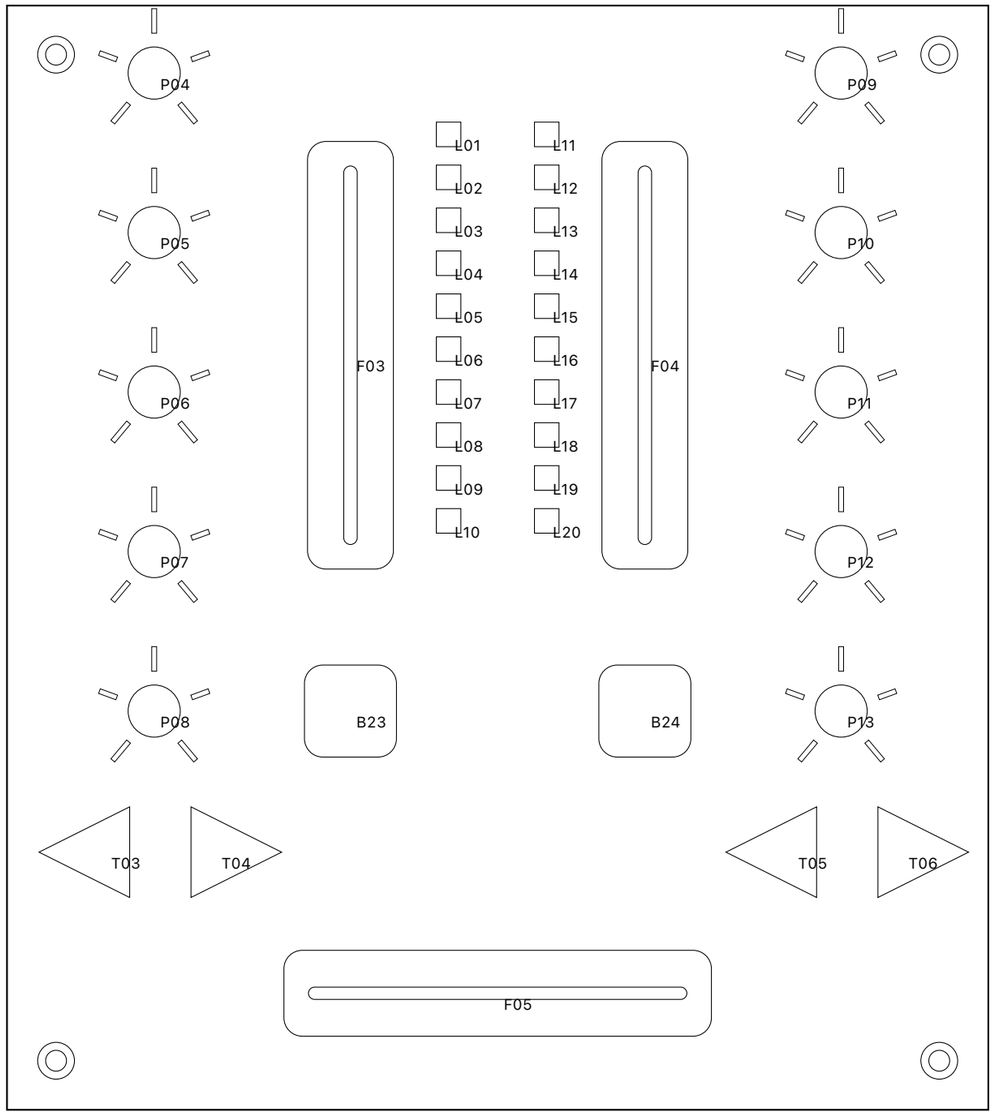
8.00, 148.00
212.00, 148.00
8.00, 230.00
212.00, 230.00
8.00, 312.00
212.00, 312.00

Outline shown sharp: outside corners R3; top rim R0.8.

Jog circles clipped to panel outline. Exact CSG and ticks: JSON.

49 cap occurrences and nine exact faceted definitions: companion caps_poses / caps_mesh JSON.

R06 / top view and aperture schedule / 1 piece



APERTURES: centre X,Y; W x H; R

T rows: see triangle definition in JSON

P04 244.00,151.00 circle 8.50 x 8.50 R0.00
P05 244.00,177.00 circle 8.50 x 8.50 R0.00
P06 244.00,203.00 circle 8.50 x 8.50 R0.00
P07 244.00,229.00 circle 8.50 x 8.50 R0.00
P08 244.00,255.00 circle 8.50 x 8.50 R0.00
P09 356.00,151.00 circle 8.50 x 8.50 R0.00
P10 356.00,177.00 circle 8.50 x 8.50 R0.00
P11 356.00,203.00 circle 8.50 x 8.50 R0.00
P12 356.00,229.00 circle 8.50 x 8.50 R0.00
P13 356.00,255.00 circle 8.50 x 8.50 R0.00
B23 276.00,255.00 rr 15.00 x 15.00 R3.00
B24 324.00,255.00 rr 15.00 x 15.00 R3.00
T03 236.00,278.00 rr 8.00 x 8.00 R1.40
T04 254.00,278.00 rr 8.00 x 8.00 R1.40
T05 348.00,278.00 rr 8.00 x 8.00 R1.40
T06 366.00,278.00 rr 8.00 x 8.00 R1.40
F03 276.00,197.00 slot 2.20 x 61.70 R1.10
F04 324.00,197.00 slot 2.20 x 61.70 R1.10
F05 300.00,301.00 slot 61.70 x 2.00 R1.00
L01 292.00,161.00 rect 4.00 x 4.00 R0.00
L02 292.00,168.00 rect 4.00 x 4.00 R0.00
L03 292.00,175.00 rect 4.00 x 4.00 R0.00
L04 292.00,182.00 rect 4.00 x 4.00 R0.00
L05 292.00,189.00 rect 4.00 x 4.00 R0.00
L06 292.00,196.00 rect 4.00 x 4.00 R0.00
L07 292.00,203.00 rect 4.00 x 4.00 R0.00
L08 292.00,210.00 rect 4.00 x 4.00 R0.00
L09 292.00,217.00 rect 4.00 x 4.00 R0.00
L10 292.00,224.00 rect 4.00 x 4.00 R0.00
L11 308.00,161.00 rect 4.00 x 4.00 R0.00
L12 308.00,168.00 rect 4.00 x 4.00 R0.00
L13 308.00,175.00 rect 4.00 x 4.00 R0.00
L14 308.00,182.00 rect 4.00 x 4.00 R0.00
L15 308.00,189.00 rect 4.00 x 4.00 R0.00
L16 308.00,196.00 rect 4.00 x 4.00 R0.00
L17 308.00,203.00 rect 4.00 x 4.00 R0.00
L18 308.00,210.00 rect 4.00 x 4.00 R0.00
L19 308.00,217.00 rect 4.00 x 4.00 R0.00
L20 308.00,224.00 rect 4.00 x 4.00 R0.00

CLOSURE AXES X,Y (28 overall)

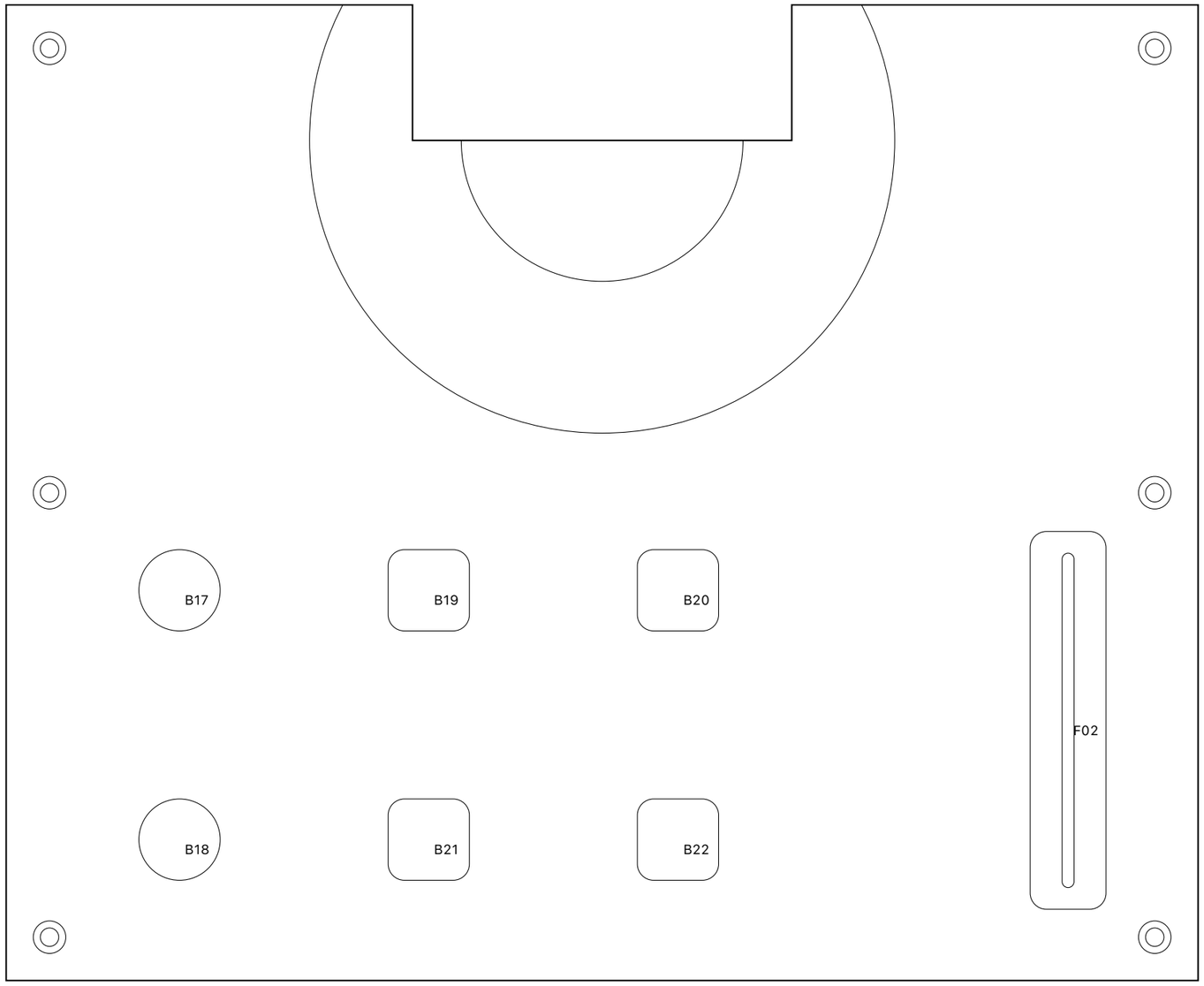
228.00, 148.00
372.00, 148.00
228.00, 312.00
372.00, 312.00

Outline shown sharp: outside corners R3; top rim R0.8.

Jog circles clipped to panel outline. Exact CSG and ticks: JSON.

49 cap occurrences and nine exact faceted definitions: companion caps_poses / caps_mesh JSON.

R07 / top view and aperture schedule / 1 piece



APERTURES: centre X,Y; W x H; R

T rows: see triangle definition in JSON

B17 412.00,248.00 circle 15.00 x 15.00 R0.00
B18 412.00,294.00 circle 15.00 x 15.00 R0.00
B19 458.00,248.00 rr 15.00 x 15.00 R3.00
B20 504.00,248.00 rr 15.00 x 15.00 R3.00
B21 458.00,294.00 rr 15.00 x 15.00 R3.00
B22 504.00,294.00 rr 15.00 x 15.00 R3.00
F02 576.00,272.00 slot 2.20 x 61.70 R1.10

CLOSURE AXES X,Y (28 overall)

388.00, 148.00
592.00, 148.00
388.00, 230.00
592.00, 230.00
388.00, 312.00
592.00, 312.00

Outline shown sharp: outside corners R3; top rim R0.8.

Jog circles clipped to panel outline. Exact CSG and ticks: JSON.

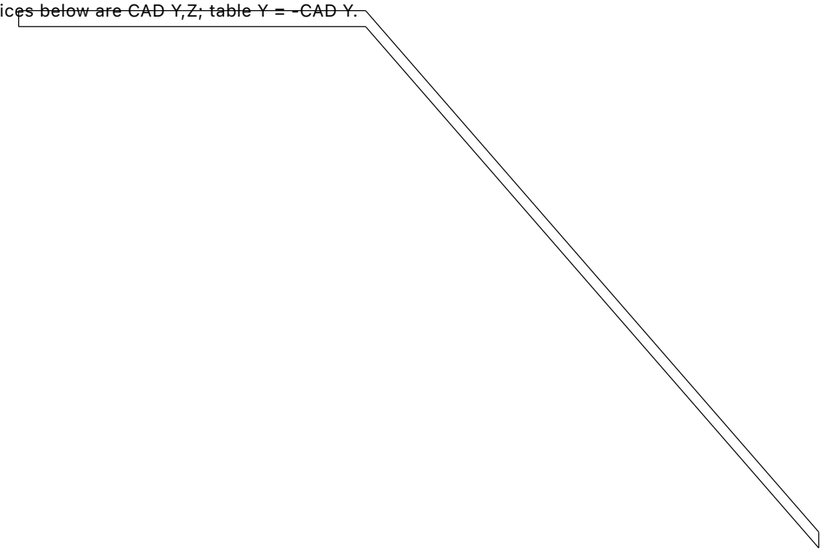
49 cap occurrences and nine exact faceted definitions: companion caps_poses / caps_mesh JSON.

D02L / D02R / roof interface section

YZ section looking along X; actual profile, scale 1.3:1

X intervals D02L 153.55..335; D02R 335..446.45.

Section vertices ~~below are CAD Y,Z; table Y = -CAD Y.~~



V1 0.00, 83.38

V2 -43.52, 83.38

V3 -100.35, 18.00

V4 -100.35, 20.00

V5 -43.52, 85.38

V6 0.00, 85.38

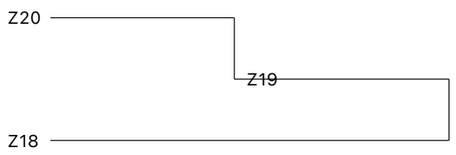
Slope 49 degrees; nominal vertical sheet thickness 2.

Display opening 272 x 72 is cut in the roof frame; roof end walls, support feet,
joint bosses, countersinks and transformed local support operations are defined in the JSON.
This section is not a complete orthographic depiction of those supports.

Sections / jog, fader and closure interfaces

Schematic orthographic sections, dimensions in mm; Z levels global; not to a common scale.

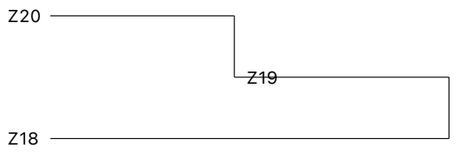
A-A jog recess



R54 recess; R26 through; top rim 0.8 x 45 deg

Centres (110,165), (490,165); split across paired panels.

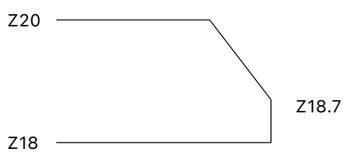
B-B fader well



Floor 14 x 69.7 R3; F05 rotated; upper edge chamfer 0.8.

Raised fader marks Z19..19.4; 22 per fader, source recipe in JSON.

C-C closure



Through diameter 3.4; conical seat diameter 6 at Z20, diameter 3.4 at Z18.7.

Exact profiles, chamfer selections, operation order, 65 raised P polygons and roof CSG
are in frontpanel_data.json. Sections are schematic; reconstruction uses the JSON.

Faceted caps are exact mesh references; no parametric history, fit, physical or production approval.

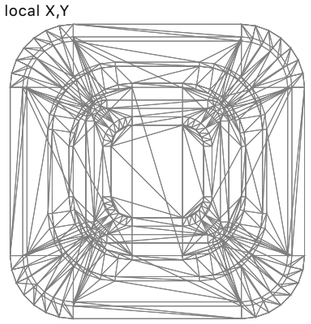
Faceted cap definitions / local views and occurrence schedule

Exact approved mesh data: B26_CONSOLIDATED_01_caps_mesh_DRAFT.json; placements: caps_poses JSON.

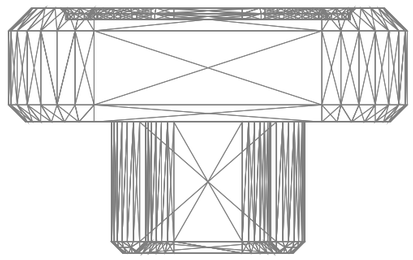
All facet edges are shown, including hidden edges; views are not analytic or parametric definitions.

CAP_B / quantity 12 / 1044 triangles

TOP / local X,Y



SIDE / local X,Z



Local X: -7.000 .. 7.000 mm

Local Y: -7.000 .. 7.000 mm

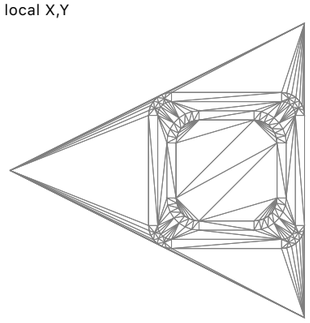
Local Z: -4.600 .. 4.000 mm

Installed owners: B01, B02, B03, B04, B05, B06, B07, B08, B09, B10, B23, B24

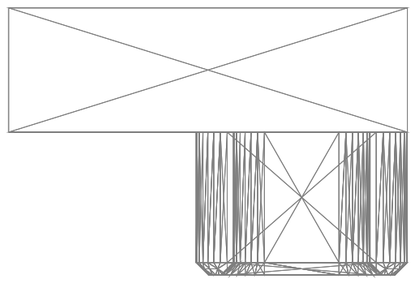
Bounds rounded to 0.001 mm; full facet coordinates and winding in JSON.

CAP_T / quantity 6 / 416 triangles

TOP / local X,Y



SIDE / local X,Z



Local X: -9.436 .. 3.400 mm

Local Y: -6.418 .. 6.418 mm

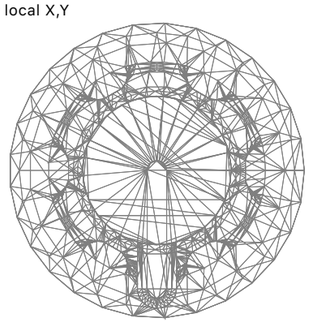
Local Z: -4.600 .. 4.000 mm

Installed owners: T01, T02, T03, T04, T05, T06

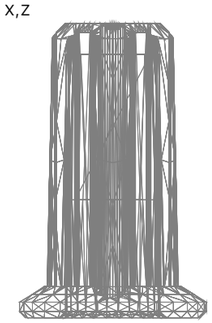
Bounds rounded to 0.001 mm; full facet coordinates and winding in JSON.

CAP_P / quantity 13 / 2054 triangles

TOP / local X,Y



SIDE / local X,Z



Local X: -6.015 .. 6.050 mm

Local Y: -6.041 .. 6.041 mm

Local Z: 0.000 .. 19.000 mm

Installed owners: P01, P02, P03, P04, P05, P06, P07, P08, P09, P10, P11, P12, P13

Bounds rounded to 0.001 mm; full facet coordinates and winding in JSON.

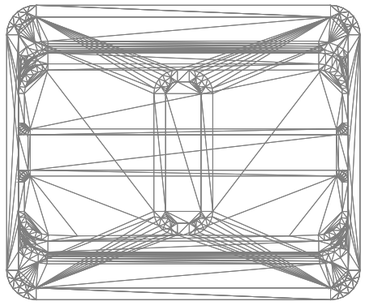
Faceted cap definitions / local views and occurrence schedule

Exact approved mesh data: B26_CONSOLIDATED_01_caps_mesh_DRAFT.json; placements: caps_poses JSON.

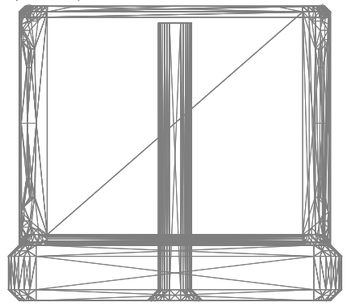
All facet edges are shown, including hidden edges; views are not analytic or parametric definitions.

CAP_F / quantity 4 / 1368 triangles

TOP / local X,Y



SIDE / local X,Z



Local X: -6.000 .. 6.000 mm

Local Y: -5.000 .. 5.000 mm

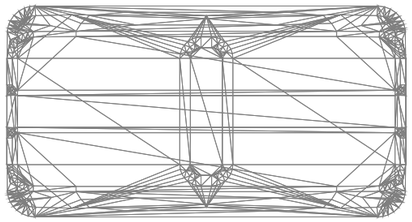
Local Z: 0.000 .. 10.600 mm

Installed owners: F01, F02, F03, F04

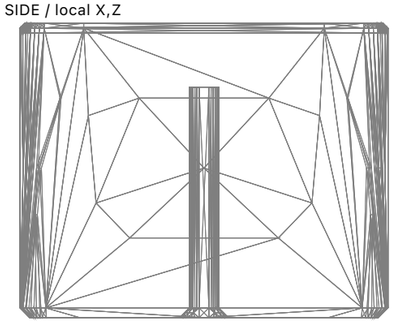
Bounds rounded to 0.001 mm; full facet coordinates and winding in JSON.

CAP_C / quantity 1 / 884 triangles

TOP / local X,Y



SIDE / local X,Z



Local X: -7.500 .. 7.500 mm

Local Y: -3.967 .. 3.967 mm

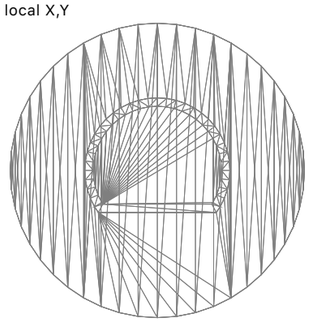
Local Z: 0.000 .. 12.000 mm

Installed owners: F05

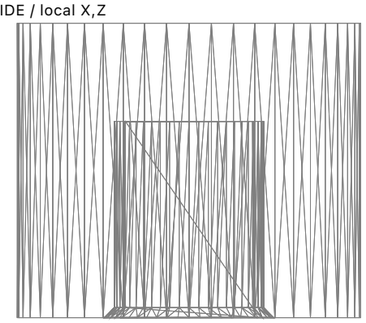
Bounds rounded to 0.001 mm; full facet coordinates and winding in JSON.

CAP_E / quantity 1 / 468 triangles

TOP / local X,Y



SIDE / local X,Z



Local X: -7.000 .. 7.000 mm

Local Y: -7.000 .. 7.000 mm

Local Z: 0.000 .. 12.000 mm

Installed owners: E01

Bounds rounded to 0.001 mm; full facet coordinates and winding in JSON.

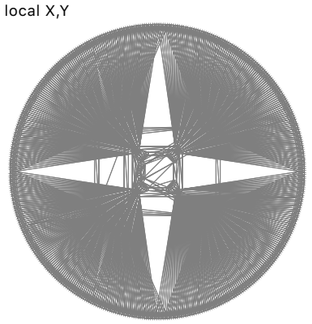
Faceted cap definitions / local views and occurrence schedule

Exact approved mesh data: B26_CONSOLIDATED_01_caps_mesh_DRAFT.json; placements: caps_poses JSON.

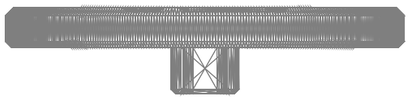
All facet edges are shown, including hidden edges; views are not analytic or parametric definitions.

CAP-CUE40 / quantity 2 / 4304 triangles

TOP / local X,Y



SIDE / local X,Z



Local X: -20.000 .. 20.000 mm

Local Y: -20.000 .. 20.000 mm

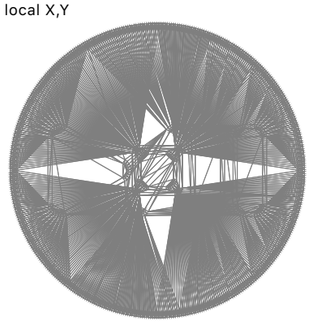
Local Z: -4.600 .. 4.000 mm

Installed owners: B11, B17

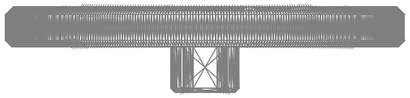
Bounds rounded to 0.001 mm; full facet coordinates and winding in JSON.

CAP-PP40 / quantity 2 / 4688 triangles

TOP / local X,Y



SIDE / local X,Z



Local X: -20.000 .. 20.000 mm

Local Y: -20.000 .. 20.000 mm

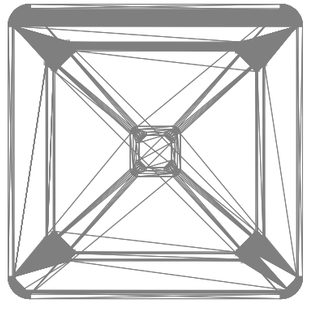
Local Z: -4.600 .. 4.000 mm

Installed owners: B12, B18

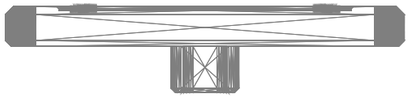
Bounds rounded to 0.001 mm; full facet coordinates and winding in JSON.

CAP-HC40 / quantity 8 / 9108 triangles

TOP / local X,Y



SIDE / local X,Z



Local X: -20.000 .. 20.000 mm

Local Y: -20.000 .. 20.000 mm

Local Z: -4.600 .. 4.000 mm

Installed owners: B13, B14, B15, B16, B19, B20, B21, B22

Bounds rounded to 0.001 mm; full facet coordinates and winding in JSON.

Exact panel, roof, mark and cap-placement schedule

PANEL CONSTRUCTION AND EXACT DIMENSIONAL PRECEDENCE

Numerical schedule display is rounded to 0.000000001 mm; the companion JSON retains the original source numbers. This is not a manufacturing tolerance.

FRONT = 96 + 5 / tan(49 degrees) = 100.34643368908114 mm; table datum and CAD transformation are on each drawing sheet.

All boxes below are table coordinates, Z18..20. R03: X220..380 YFRONT..140; R06 X220..380 Y140..320.

R01/R04: X0..220/X380..600 Y0..140 plus X75..145/X455..525 Y140..165, subtract X153.55..220/X380..446.45 Y0..FRONT.

R05/R07: X0..220/X380..600 Y140..320 subtract X75..145/X455..525 Y140..165.

Outside corners vertical fillet R3 then exterior top edges fillet R0.8. Aperture profile through Z17..22. T polygon: a=7.2+1.6 sqrt(5), sign + for odd T, - for even; CAD vertices (X+sign*u,-Y+sign*v) for (-a,0),(4,(a+4)/2),(4,-(a+4)/2). Other shapes circle width diameter; rect; slot overall max(width,height), width min; rr rounded rectangle.

Fader pocket rounded rectangle 14 x 69.7, radius 3; F05 69.7 x 14, Z19..22; chamfer all its top edges 0.8. Jog at (110,165) or (490,165): through radius 26 Z17..22, upper recess radius 54 Z19..22; chamfer top edge 0.8. Chamfer B and T aperture top edges 0.8.

P marks polygons listed here extruded Z20..20.4. Fader ticks: 11 positions u=-travel/2+j*travel/10, j=0..10, travel 28 F05 otherwise 41, each side sign +/-; rectangle local u±0.4, signed v=2.4..2.4+length, length 4 at j=0,5,10 else 2. F05 table=(x+u,y-v); others=(x+v,y+u). Extrude Z19..19.4.

Closure axes cylindrical through radius 1.7 Z17..22; countercone Z18.7..20 radius 1.7 bottom, 3 top. Operations applied in order as listed; fillets/chamfers selected by resulting upper edges.

ROOF CONSTRUCTION / CAD COORDINATES

CAD coords for this recipe, units mm. a=49 degrees; c=cos(a), s=sin(a), FRONT=96+5/tan(a), j=96-80*c, u=25+80*s, cy=96-40*c, h=23+40*s.

section(x,width,points) extrudes a polygon of CAD (Y,Z) along +X. D02L x0=153.55,x1=335; D02R x0=335,x1=446.45. Base section vertices (0,u-2),(-j,u-2),(-FRONT,18),(-FRONT,20),(-j,u),(0,u). Add section x=153.55 left or 444.45 right, width 2, vertices (-2,18),(-FRONT,18),(-FRONT,20),(-j,u),(-2,u).

Frame origin (300,-cy,h), x_dir=(side,0,0), z_dir=(0,-s,c); y_dir=z_dir cross x_dir. For place(shape,side,row), mirror shape across local XZ if side*row>0, then apply frame. Subtract place(box(-136,136,-36,36,-5,5),1,1).

For x in 210,230,300,370,390 strictly inside [x0,x1], subtract cylinder (x,-4,u-2.01), r1.7 height2.02 and cone (x,-4,u-1.3), bottom r1.7 top r3 height1.3.

Left add box X325..345 Y-29..-13 Z u-10..u-2; subtract cylinders at (340,-18/-24,u-9) r1.25 h7.01. Right subtract cylinders at (340,-18/-24,u-2.01) r1.7 h2.02 and cones bottom r1.7 top r3 at Z u-1.3 h1.3.

x=228 left /372 right: add section(x-5,10,[(-95,16),(-116,16),(-116,18),(-FRONT,18),(-95,23+tan(a))]); subtract cylinder(x,-110,15.99) r1.7 h2.02.

Exact panel, roof, mark and cap-placement schedule

For row=-1,+1: support union boxes (138.5,144.2,18,30,-21.2,0), (138.25,140,20,39.797470349551496,-21,-0.25), (130,140,38.297470349551496,39.797470349551496,-21,-0.25). Subtract box (136,146,19.9,28.1,-21.2,-18), cylinder (140.7,24,-18.1) r1.25 h7.1, cylinder (140.7,24,-60) r4 h39. Add place(support,side=-1 left / +1 right,row).

Fillet exterior top edges at Z=u radius 0.8, if present. box tuple means Xmin,Xmax,Ymin,Ymax,Zmin,Zmax.

FADER WELL AND TRAVEL REFERENCE ROWS

{"existing_slot_length": 61.7, "existing_slot_radius": 1.1, "existing_slot_width": 2.2, "existing_travel": 41, "id": "F01", "panel": "R05", "table_xy": [196, 272], "travel_axis": "+Y"}

{"existing_slot_length": 61.7, "existing_slot_radius": 1.1, "existing_slot_width": 2.2, "existing_travel": 41, "id": "F02", "panel": "R07", "table_xy": [576, 272], "travel_axis": "+Y"}

{"existing_slot_length": 61.7, "existing_slot_radius": 1.1, "existing_slot_width": 2.2, "existing_travel": 41, "id": "F03", "panel": "R06", "table_xy": [276, 197], "travel_axis": "+Y"}

{"existing_slot_length": 61.7, "existing_slot_radius": 1.1, "existing_slot_width": 2.2, "existing_travel": 41, "id": "F04", "panel": "R06", "table_xy": [324, 197], "travel_axis": "+Y"}

{"existing_slot_length": 61.7, "existing_slot_radius": 1, "existing_slot_width": 2, "existing_travel": 28, "id": "F05", "panel": "R06", "table_xy": [300, 301], "travel_axis": "+X"}

65 RAISED P-MARK POLYGONS / TABLE XY, MM

{"id": "P04", "panel": "R06", "polygon": [[247.871701686, 156.236403924], [250.442852124, 159.300581697], [251.055687679, 158.786351609], [248.48453724, 155.722173836]], "z": [20, 20.4]}

{"id": "P04", "panel": "R06", "polygon": [[250.244810092, 149.152746117], [253.063887955, 148.126685687], [252.79027184, 147.37493159], [249.971193978, 148.40099202]], "z": [20, 20.4]}

{"id": "P04", "panel": "R06", "polygon": [[244.4, 144.5], [244.4, 140.5], [243.6, 140.5], [243.6, 144.5]], "z": [20, 20.4]}

{"id": "P04", "panel": "R06", "polygon": [[238.028806022, 148.40099202], [235.20972816, 147.37493159], [234.936112045, 148.126685687], [237.755189908, 149.152746117]], "z": [20, 20.4]}

{"id": "P04", "panel": "R06", "polygon": [[239.51546276, 155.722173836], [236.944312321, 158.786351609], [237.557147876, 159.300581697], [240.128298314, 156.236403924]], "z": [20, 20.4]}

{"id": "P05", "panel": "R06", "polygon": [[247.871701686, 182.236403924], [250.442852124, 185.300581697], [251.055687679, 184.786351609], [248.48453724, 181.722173836]], "z": [20, 20.4]}

{"id": "P05", "panel": "R06", "polygon": [[250.244810092, 175.152746117], [253.063887955, 174.126685687], [252.79027184, 173.37493159], [249.971193978, 174.40099202]], "z": [20, 20.4]}

{"id": "P05", "panel": "R06", "polygon": [[244.4, 170.5], [244.4, 166.5], [243.6, 166.5], [243.6, 170.5]], "z": [20, 20.4]}

{"id": "P05", "panel": "R06", "polygon": [[238.028806022, 174.40099202], [235.20972816, 173.37493159], [234.936112045, 174.126685687], [237.755189908, 175.152746117]], "z": [20, 20.4]}

Exact panel, roof, mark and cap-placement schedule

{"id": "P05", "panel": "R06", "polygon": [[239.51546276, 181.722173836], [236.944312321, 184.786351609], [237.557147876, 185.300581697], [240.128298314, 182.236403924]], "z": [20, 20.4]}

{"id": "P06", "panel": "R06", "polygon": [[247.871701686, 208.236403924], [250.442852124, 211.300581697], [251.055687679, 210.786351609], [248.48453724, 207.722173836]], "z": [20, 20.4]}

{"id": "P06", "panel": "R06", "polygon": [[250.244810092, 201.152746117], [253.063887955, 200.126685687], [252.79027184, 199.37493159], [249.971193978, 200.40099202]], "z": [20, 20.4]}

{"id": "P06", "panel": "R06", "polygon": [[244.4, 196.5], [244.4, 192.5], [243.6, 192.5], [243.6, 196.5]], "z": [20, 20.4]}

{"id": "P06", "panel": "R06", "polygon": [[238.028806022, 200.40099202], [235.20972816, 199.37493159], [234.936112045, 200.126685687], [237.755189908, 201.152746117]], "z": [20, 20.4]}

{"id": "P06", "panel": "R06", "polygon": [[239.51546276, 207.722173836], [236.944312321, 210.786351609], [237.557147876, 211.300581697], [240.128298314, 208.236403924]], "z": [20, 20.4]}

{"id": "P07", "panel": "R06", "polygon": [[247.871701686, 234.236403924], [250.442852124, 237.300581697], [251.055687679, 236.786351609], [248.48453724, 233.722173836]], "z": [20, 20.4]}

{"id": "P07", "panel": "R06", "polygon": [[250.244810092, 227.152746117], [253.063887955, 226.126685687], [252.79027184, 225.37493159], [249.971193978, 226.40099202]], "z": [20, 20.4]}

{"id": "P07", "panel": "R06", "polygon": [[244.4, 222.5], [244.4, 218.5], [243.6, 218.5], [243.6, 222.5]], "z": [20, 20.4]}

{"id": "P07", "panel": "R06", "polygon": [[238.028806022, 226.40099202], [235.20972816, 225.37493159], [234.936112045, 226.126685687], [237.755189908, 227.152746117]], "z": [20, 20.4]}

{"id": "P07", "panel": "R06", "polygon": [[239.51546276, 233.722173836], [236.944312321, 236.786351609], [237.557147876, 237.300581697], [240.128298314, 234.236403924]], "z": [20, 20.4]}

{"id": "P08", "panel": "R06", "polygon": [[247.871701686, 260.236403924], [250.442852124, 263.300581697], [251.055687679, 262.786351609], [248.48453724, 259.722173836]], "z": [20, 20.4]}

{"id": "P08", "panel": "R06", "polygon": [[250.244810092, 253.152746117], [253.063887955, 252.126685687], [252.79027184, 251.37493159], [249.971193978, 252.40099202]], "z": [20, 20.4]}

{"id": "P08", "panel": "R06", "polygon": [[244.4, 248.5], [244.4, 244.5], [243.6, 244.5], [243.6, 248.5]], "z": [20, 20.4]}

{"id": "P08", "panel": "R06", "polygon": [[238.028806022, 252.40099202], [235.20972816, 251.37493159], [234.936112045, 252.126685687], [237.755189908, 253.152746117]], "z": [20, 20.4]}

{"id": "P08", "panel": "R06", "polygon": [[239.51546276, 259.722173836], [236.944312321, 262.786351609], [237.557147876, 263.300581697], [240.128298314, 260.236403924]], "z": [20, 20.4]}

{"id": "P09", "panel": "R06", "polygon": [[359.871701686, 156.236403924], [362.442852124, 159.300581697], [363.055687679, 158.786351609], [360.48453724, 155.722173836]], "z": [20, 20.4]}

Exact panel, roof, mark and cap-placement schedule

{"id": "P09", "panel": "R06", "polygon": [[362.244810092, 149.152746117], [365.063887955, 148.126685687], [364.79027184, 147.37493159], [361.971193978, 148.40099202]], "z": [20, 20.4]}

{"id": "P09", "panel": "R06", "polygon": [[356.4, 144.5], [356.4, 140.5], [355.6, 140.5], [355.6, 144.5]], "z": [20, 20.4]}

{"id": "P09", "panel": "R06", "polygon": [[350.028806022, 148.40099202], [347.20972816, 147.37493159], [346.936112045, 148.126685687], [349.755189908, 149.152746117]], "z": [20, 20.4]}

{"id": "P09", "panel": "R06", "polygon": [[351.51546276, 155.722173836], [348.944312321, 158.786351609], [349.557147876, 159.300581697], [352.128298314, 156.236403924]], "z": [20, 20.4]}

{"id": "P10", "panel": "R06", "polygon": [[359.871701686, 182.236403924], [362.442852124, 185.300581697], [363.055687679, 184.786351609], [360.48453724, 181.722173836]], "z": [20, 20.4]}

{"id": "P10", "panel": "R06", "polygon": [[362.244810092, 175.152746117], [365.063887955, 174.126685687], [364.79027184, 173.37493159], [361.971193978, 174.40099202]], "z": [20, 20.4]}

{"id": "P10", "panel": "R06", "polygon": [[356.4, 170.5], [356.4, 166.5], [355.6, 166.5], [355.6, 170.5]], "z": [20, 20.4]}

{"id": "P10", "panel": "R06", "polygon": [[350.028806022, 174.40099202], [347.20972816, 173.37493159], [346.936112045, 174.126685687], [349.755189908, 175.152746117]], "z": [20, 20.4]}

{"id": "P10", "panel": "R06", "polygon": [[351.51546276, 181.722173836], [348.944312321, 184.786351609], [349.557147876, 185.300581697], [352.128298314, 182.236403924]], "z": [20, 20.4]}

{"id": "P11", "panel": "R06", "polygon": [[359.871701686, 208.236403924], [362.442852124, 211.300581697], [363.055687679, 210.786351609], [360.48453724, 207.722173836]], "z": [20, 20.4]}

{"id": "P11", "panel": "R06", "polygon": [[362.244810092, 201.152746117], [365.063887955, 200.126685687], [364.79027184, 199.37493159], [361.971193978, 200.40099202]], "z": [20, 20.4]}

{"id": "P11", "panel": "R06", "polygon": [[356.4, 196.5], [356.4, 192.5], [355.6, 192.5], [355.6, 196.5]], "z": [20, 20.4]}

{"id": "P11", "panel": "R06", "polygon": [[350.028806022, 200.40099202], [347.20972816, 199.37493159], [346.936112045, 200.126685687], [349.755189908, 201.152746117]], "z": [20, 20.4]}

{"id": "P11", "panel": "R06", "polygon": [[351.51546276, 207.722173836], [348.944312321, 210.786351609], [349.557147876, 211.300581697], [352.128298314, 208.236403924]], "z": [20, 20.4]}

{"id": "P12", "panel": "R06", "polygon": [[359.871701686, 234.236403924], [362.442852124, 237.300581697], [363.055687679, 236.786351609], [360.48453724, 233.722173836]], "z": [20, 20.4]}

{"id": "P12", "panel": "R06", "polygon": [[362.244810092, 227.152746117], [365.063887955, 226.126685687], [364.79027184, 225.37493159], [361.971193978, 226.40099202]], "z": [20, 20.4]}

{"id": "P12", "panel": "R06", "polygon": [[356.4, 222.5], [356.4, 218.5], [355.6, 218.5], [355.6, 222.5]], "z": [20, 20.4]}

{"id": "P12", "panel": "R06", "polygon": [[350.028806022, 226.40099202], [347.20972816, 225.37493159], [346.936112045, 226.126685687], [349.755189908, 227.152746117]], "z": [20, 20.4]}

Exact panel, roof, mark and cap-placement schedule

{"id": "P12", "panel": "R06", "polygon": [[351.51546276, 233.722173836], [348.944312321, 236.786351609], [349.557147876, 237.300581697], [352.128298314, 234.236403924]], "z": [20, 20.4]}

{"id": "P13", "panel": "R06", "polygon": [[359.871701686, 260.236403924], [362.442852124, 263.300581697], [363.055687679, 262.786351609], [360.48453724, 259.722173836]], "z": [20, 20.4]}

{"id": "P13", "panel": "R06", "polygon": [[362.244810092, 253.152746117], [365.063887955, 252.126685687], [364.79027184, 251.37493159], [361.971193978, 252.40099202]], "z": [20, 20.4]}

{"id": "P13", "panel": "R06", "polygon": [[356.4, 248.5], [356.4, 244.5], [355.6, 244.5], [355.6, 248.5]], "z": [20, 20.4]}

{"id": "P13", "panel": "R06", "polygon": [[350.028806022, 252.40099202], [347.20972816, 251.37493159], [346.936112045, 252.126685687], [349.755189908, 253.152746117]], "z": [20, 20.4]}

{"id": "P13", "panel": "R06", "polygon": [[351.51546276, 259.722173836], [348.944312321, 262.786351609], [349.557147876, 263.300581697], [352.128298314, 260.236403924]], "z": [20, 20.4]}

{"id": "P04-copy", "panel": "R04", "polygon": [[562.871701686, 46.236403924], [565.442852124, 49.300581697], [566.055687679, 48.786351609], [563.48453724, 45.722173836]], "z": [20, 20.4]}

{"id": "P04-copy", "panel": "R04", "polygon": [[565.244810092, 39.152746117], [568.063887955, 38.126685687], [567.79027184, 37.37493159], [564.971193978, 38.40099202]], "z": [20, 20.4]}

{"id": "P04-copy", "panel": "R04", "polygon": [[559.4, 34.5], [559.4, 30.5], [558.6, 30.5], [558.6, 34.5]], "z": [20, 20.4]}

{"id": "P04-copy", "panel": "R04", "polygon": [[553.028806022, 38.40099202], [550.20972816, 37.37493159], [549.936112045, 38.126685687], [552.755189908, 39.152746117]], "z": [20, 20.4]}

{"id": "P04-copy", "panel": "R04", "polygon": [[554.51546276, 45.722173836], [551.944312321, 48.786351609], [552.557147876, 49.300581697], [555.128298314, 46.236403924]], "z": [20, 20.4]}

{"id": "P04-copy", "panel": "R04", "polygon": [[562.871701686, 74.236403924], [565.442852124, 77.300581697], [566.055687679, 76.786351609], [563.48453724, 73.722173836]], "z": [20, 20.4]}

{"id": "P04-copy", "panel": "R04", "polygon": [[565.244810092, 67.152746117], [568.063887955, 66.126685687], [567.79027184, 65.37493159], [564.971193978, 66.40099202]], "z": [20, 20.4]}

{"id": "P04-copy", "panel": "R04", "polygon": [[559.4, 62.5], [559.4, 58.5], [558.6, 58.5], [558.6, 62.5]], "z": [20, 20.4]}

{"id": "P04-copy", "panel": "R04", "polygon": [[553.028806022, 66.40099202], [550.20972816, 65.37493159], [549.936112045, 66.126685687], [552.755189908, 67.152746117]], "z": [20, 20.4]}

{"id": "P04-copy", "panel": "R04", "polygon": [[554.51546276, 73.722173836], [551.944312321, 76.786351609], [552.557147876, 77.300581697], [555.128298314, 74.236403924]], "z": [20, 20.4]}

{"id": "P04-copy", "panel": "R04", "polygon": [[562.871701686, 102.236403924], [565.442852124, 105.300581697], [566.055687679, 104.786351609], [563.48453724, 101.722173836]], "z": [20, 20.4]}

Exact panel, roof, mark and cap-placement schedule

{"id": "P04-copy", "panel": "R04", "polygon": [[565.244810092, 95.152746117], [568.063887955, 94.126685687], [567.79027184, 93.37493159], [564.971193978, 94.40099202]], "z": [20, 20.4]}

{"id": "P04-copy", "panel": "R04", "polygon": [[559.4, 90.5], [559.4, 86.5], [558.6, 86.5], [558.6, 90.5]], "z": [20, 20.4]}

{"id": "P04-copy", "panel": "R04", "polygon": [[553.028806022, 94.40099202], [550.20972816, 93.37493159], [549.936112045, 94.126685687], [552.755189908, 95.152746117]], "z": [20, 20.4]}

{"id": "P04-copy", "panel": "R04", "polygon": [[554.51546276, 101.722173836], [551.944312321, 104.786351609], [552.557147876, 105.300581697], [555.128298314, 102.236403924]], "z": [20, 20.4]}

49 CAP OCCURRENCES / LOCAL-TO-CAD FRAMES

Each frame uses CAD origin in mm and right-handed x_dir, z_dir. Exact faceted definitions are in B26_CONSOLIDATED_01_caps_mesh_DRAFT.json.

B01 / CAP_B: origin [91, -42.25, 21.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

B02 / CAP_B: origin [111, -42.25, 21.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

B03 / CAP_B: origin [131, -42.25, 21.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

B04 / CAP_B: origin [111, -66.25, 21.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

B05 / CAP_B: origin [131, -66.25, 21.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

B06 / CAP_B: origin [471, -42.25, 21.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

B07 / CAP_B: origin [491, -42.25, 21.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

B08 / CAP_B: origin [511, -42.25, 21.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

B09 / CAP_B: origin [491, -66.25, 21.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

B10 / CAP_B: origin [511, -66.25, 21.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

B11 / CAP-CUE40: origin [32, -248, 21.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

B12 / CAP-PP40: origin [32, -294, 21.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

Exact panel, roof, mark and cap-placement schedule

B13 / CAP-HC40: origin [78, -248, 21.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

B14 / CAP-HC40: origin [124, -248, 21.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

B15 / CAP-HC40: origin [78, -294, 21.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

B16 / CAP-HC40: origin [124, -294, 21.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

B17 / CAP-CUE40: origin [412, -248, 21.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

B18 / CAP-PP40: origin [412, -294, 21.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

B19 / CAP-HC40: origin [458, -248, 21.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

B20 / CAP-HC40: origin [504, -248, 21.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

B21 / CAP-HC40: origin [458, -294, 21.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

B22 / CAP-HC40: origin [504, -294, 21.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

B23 / CAP_B: origin [276, -255, 21.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

B24 / CAP_B: origin [324, -255, 21.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

E01 / CAP_E: origin [300, -125, 23.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

F01 / CAP_F: origin [196, -272, 20.8]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

F02 / CAP_F: origin [576, -272, 20.8]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

F03 / CAP_F: origin [276, -197, 20.8]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

F04 / CAP_F: origin [324, -197, 20.8]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

F05 / CAP_C: origin [300, -301, 20.8]; x_dir [0, -1, 0]; z_dir [0, 0, 1]

Exact panel, roof, mark and cap-placement schedule

P01 / CAP_P: origin [559, -41, 20.8]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

P02 / CAP_P: origin [559, -69, 20.8]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

P03 / CAP_P: origin [559, -97, 20.8]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

P04 / CAP_P: origin [244, -151, 20.8]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

P05 / CAP_P: origin [244, -177, 20.8]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

P06 / CAP_P: origin [244, -203, 20.8]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

P07 / CAP_P: origin [244, -229, 20.8]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

P08 / CAP_P: origin [244, -255, 20.8]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

P09 / CAP_P: origin [356, -151, 20.8]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

P10 / CAP_P: origin [356, -177, 20.8]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

P11 / CAP_P: origin [356, -203, 20.8]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

P12 / CAP_P: origin [356, -229, 20.8]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

P13 / CAP_P: origin [356, -255, 20.8]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

T01 / CAP_T: origin [277, -125, 21.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

T02 / CAP_T: origin [323, -125, 21.5]; x_dir [-1, 0, 0]; z_dir [0, 0, 1]

T03 / CAP_T: origin [236, -278, 21.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

T04 / CAP_T: origin [254, -278, 21.5]; x_dir [-1, 0, 0]; z_dir [0, 0, 1]

Exact panel, roof, mark and cap-placement schedule

T05 / CAP_T: origin [348, -278, 21.5]; x_dir [1, 0, 0]; z_dir [0, 0, 1]

T06 / CAP_T: origin [366, -278, 21.5]; x_dir [-1, 0, 0]; z_dir [0, 0, 1]

Dimensioned profile and cap-data schedule / continued

DATUM AND COMPANION DATA

units: mm

drawing: table X right, table Y toward front/operator, Z up

cad: (X, -Y, Z); all operation bounds and centres below use table coordinates unless CAD is explicitly stated

precision: Decimal coordinates are numerical transcriptions/approximations of the source, not manufacturing tolerances. FRONT is exactly 96 + 5/tan(49 degrees), approximately 100.34643368908114.

B26_CONSOLIDATED_01_caps_mesh_DRAFT.json: Verbatim engineer-selected nine legacy mesh definitions, not analytic or editable BRep feature histories.

B26_CONSOLIDATED_01_caps_poses_DRAFT.json: Verbatim 49 occurrence identities, mesh keys and global CAD frames after B26 translation. Original 01 revision/status labels describe provenance, not approval of revision 02.

frontpanel_data.json: Verbatim B26-CONSOLIDATED-01 source-derived dataset: 69 aperture records with panel identity, 28 closure axes, five fader centres, 65 raised P-mark polygons, panel operation summary and roof information. Panel operations below clarify the source order and edge cases; the roof fields are outside this addendum.

PANEL PROFILES; Z18..20

R01: Union X0..220 Y0..140 and X75..145 Y140..165; subtract X153.55..220 Y0..FRONT.

R03: X220..380 YFRONT..140.

R04: Union X380..600 Y0..140 and X455..525 Y140..165; subtract X380..446.45 Y0..FRONT.

R05: X0..220 Y140..320; subtract X75..145 Y140..165.

R06: X220..380 Y140..320.

R07: X380..600 Y140..320; subtract X455..525 Y140..165.

ORDERED PANEL OPERATIONS

1. Subtract the 69 through apertures assigned to their panel in frontpanel_data.json, from Z17 through Z22. A circle has diameter W; a rect is W by H; rr is W by H with corner radius R; a slot has overall length max(W,H), width min(W,H), long axis Y when H > W and X otherwise. T IDs override their listed rr metadata: let a=7.2+1.6*sqrt(5), s=+1 for odd numeric T ID and -1 for even; CAD XY vertices are (X+s*u,-Y+s*v) for (u,v)=(-a,0),(4,(a+4)/2),(4,-(a+4)/2).

2. Round the vertical outside corner edge R3 on R01 at table (0,0), R04 at (600,0), R05 at (0,320), R07 at (600,320). Then fillet R0.8 the Z20 edges on the external 600 by 320 outline including its R3 corner arcs; do not round internal seams or aperture edges at this step.

Dimensioned profile and cap-data schedule / continued

3. For each assigned fader centre subtract a rounded rectangle from Z19 through Z22: F01-F04 14 X by 69.7 Y; F05 69.7 X by 14 Y; corner radius 3. Chamfer its Z20 perimeter 0.8. The already-cut central slot extends through the remaining panel; the pocket floor is Z19 and is not the panel underside.

4. On each of R01 and R05 cut at table (110,165), and on each of R04 and R07 cut at (490,165): radius 26 from Z17 through Z22 and radius 54 from Z19 through Z22. Chamfer the resulting Z20 R54 edge 0.8. The centres lie on the split: cut each panel separately; do not add a full unsplit ring.

5. Chamfer Z20 edges of B and T apertures 0.8. P, F, L and E aperture edges do not receive this cap-edge operation.

6. Add the 65 P-mark polygons in frontpanel_data.json at their listed panel, coordinates and Z20..20.4. They are individually placed polygons, not a font or text approximation. For provenance, original P04 polygons are also translated to P01/P02/P03 at X559, Y41/69/97 by the released source. Preserve the transcribed final coordinates; do not translate them a second time.

7. Add 22 fader tick rectangles per fader, at Z19..19.4, i.e. on the recessed floor. For j=0..10 let u=-travel/2+j*travel/10, travel=28 for F05 and 41 otherwise. For sign=-1,+1 and length=4 when j is 0,5,10, otherwise 2, use local corners (u-0.4,sign*2.4), (u+0.4,sign*2.4), (u+0.4,sign*(2.4+length)), (u-0.4,sign*(2.4+length)). If local coordinates are (uu,vv), table XY is (x+uu,y-vv) for F05, otherwise (x+vv,y+uu). The fader centres (x,y) are in frontpanel_data.json.

8. Finally subtract the 28 closure holes at their panel-specific centres: cylinder radius 1.7 from Z17 through Z22, followed by a coaxial conical countersink from Z18.7 to Z20, bottom radius 1.7 and top radius 3. Dimensions are nominal; actual fastener fit is unverified.

The released code tests five sampled points on candidate edges for the specified Z20 perimeter predicates with 1e-5 mm tolerance. This is an operation-selection tolerance, not a part tolerance. Corner edges use 1e-6 mm coordinate comparisons. Boolean kernel outcomes and fillet continuity are not independently established by this document.

EXACT FACETED CAPS AND PLACEMENTS

The triangles' vertices_mm are local mesh Cartesian coordinates in millimetres, with no scaling on import. Preserve each triangle normal, vertex order, and 16-bit attribute. To recover each selected binary STL byte-for-byte, decode its 80-byte header_base64, append triangle_count as unsigned little-endian uint32, then each normal and its three vertices as 12 little-endian IEEE-754 binary32 values followed by the unsigned little-endian uint16 attribute. Source SHA-256 is recorded per definition in the mesh dataset; these are faceted surfaces, not inferred parametric features.

For each occurrence use its cad_origin_mm as O, cad_x_direction as ex and cad_z_direction as ez. Set ey=ez cross ex as the right-handed build123d Plane convention. A local mesh vertex (u,v,w) maps to global CAD O + u*ex + v*ey + w*ez; apply no further drawing-Y reflection to this CAD result. To express a resulting position in table coordinates, negate its global CAD Y. The source B24 inventory pose translation was in cm and is multiplied by 10; local STL vertices are already mm. The poses dataset already includes B26 X/Y owner deltas: do not apply them again.

Each occurrence is distinguished by owner and full inventory_id; mesh_definition is one of nine keys. A shared definition does not make the 49 occurrences interchangeable. These are retained control-cap references; purchased mating geometry and physical fit are TBD.

LIMITATIONS

No revision-02 drawing, geometry, STEP/STL, physical fit, manufacture or electrical safety is approved by this addendum. Bought-part and electrical interfaces are incomplete schematic references; unspecified geometry, fasteners, mounting fit,

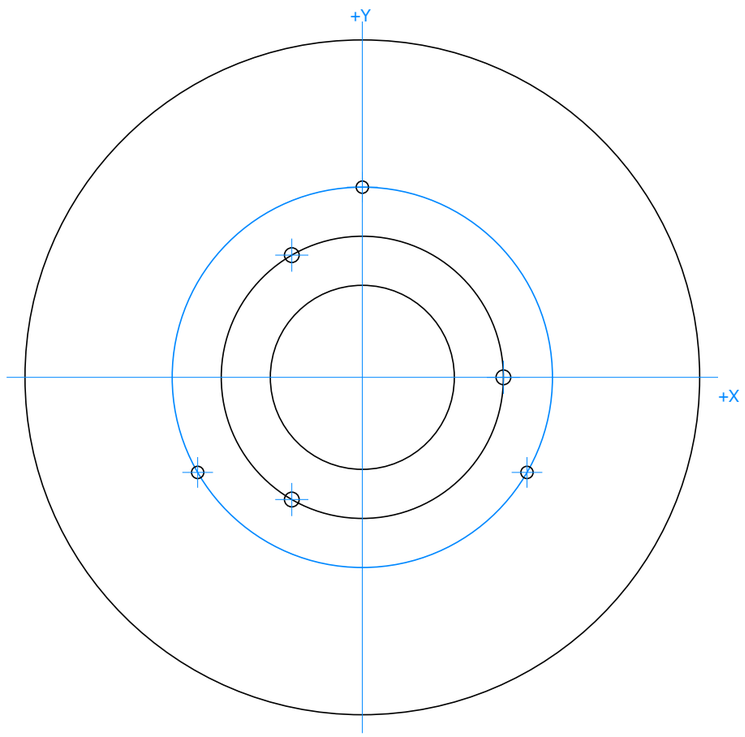
Dimensioned profile and cap-data schedule / continued

materials and tolerances remain TBD by the engineer.

The 01 dataset and this addendum have not been independently reconstructed or compared against the released source Boolean surfaces. Prior 01 count and mesh byte checks are reported in its validation JSON, not new checks. Display roofs and control retainers require their own documentation.

P01 here is a panel control identifier and must not be confused with jog P01 rotor. The released STEP strict default-volume check for each jog P01 rotor failed by 0.641091522 mm3 against 0.0001 mm3; the engineer accepted only a bounded unpowered-trial exception. Adaptive difference 0.000001609 mm3 and zero directed cut volumes do not establish exact surface equivalence. See models/release/enclosure/B26/B26-build123d03_RELEASE/B26-build123d03_handoff.md, STEP exception section.

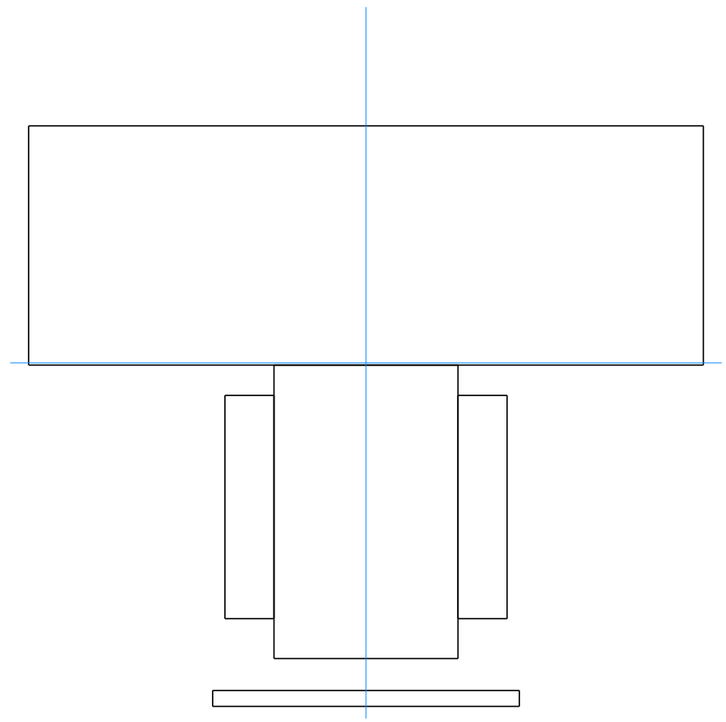
1 / Assembly datum, occurrence IDs and mounting — top + axial stack



Top: rotor D110; mounting PCD62; covers PCD46

3 floor axes 90/210/330; 3 cover axes 0/120/240

Stack / axial Z, schematic; boundaries and heights in schedule



P01: Z3..36.4; D110 / D105

P02: Z1..19.5, R31 mounting

Bearings: Z5.5..12.5 and 12.5..19.5

P05 Z19.5..21; P06 Z4..5.5

P04 Z4..5.5; P07 Z3..4

Sensor plate Z0..3; P08 toes Z3..4.3

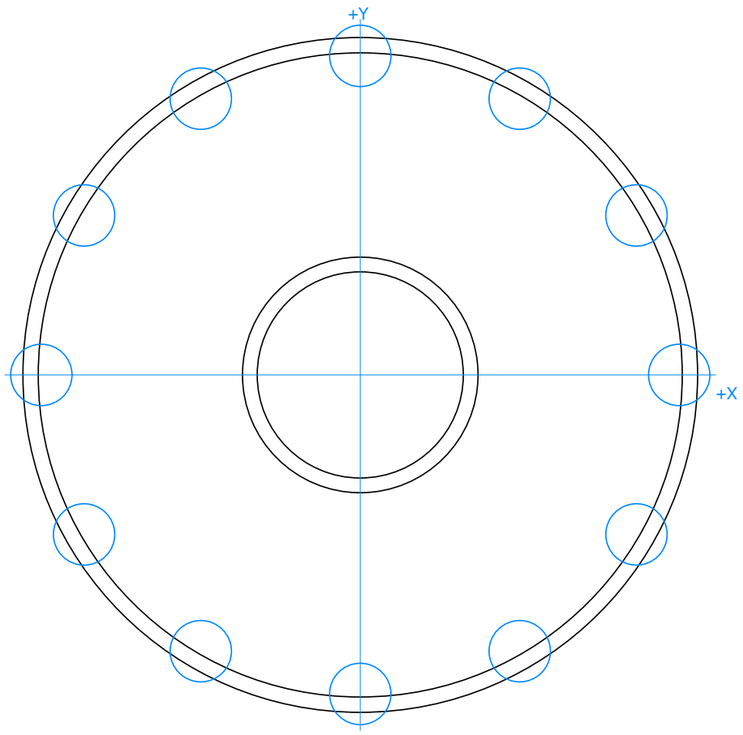
Per jog: P01, P02, P04, P07, P05, P06, sensor plate each x1; P08 right/left x1 each = 9 printed occurrences.

Two identical jogs: local-to-case map and exact R31 mounting coordinates are in the schedule. Local +X is rotated +60 deg in CAD.

Bearing, magnet, board and fasteners are incomplete nominal reference envelopes; no hardware detail or tolerances inferred.

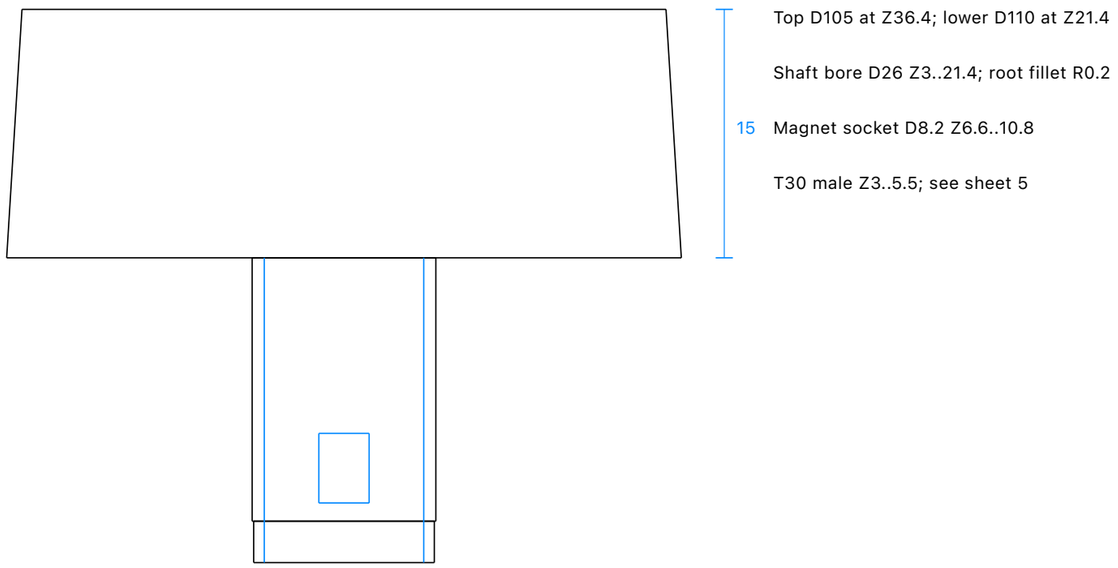
Section is explanatory, not a Boolean projection. For all hidden/recessed boundaries use the ordered feature schedule.

2 / P01 rotor — nominal plan, axial section and groove



TOP / 12 radial R5 tools, pitch 30 deg

C-groove rings only indicate radii; exact opening and loft in schedule



AXIAL SECTION / schematic

Top D105 at Z36.4; lower D110 at Z21.4

Shaft bore D26 Z3..21.4; root fillet R0.2

Magnet socket D8.2 Z6.6..10.8

T30 male Z3..5.5; see sheet 5

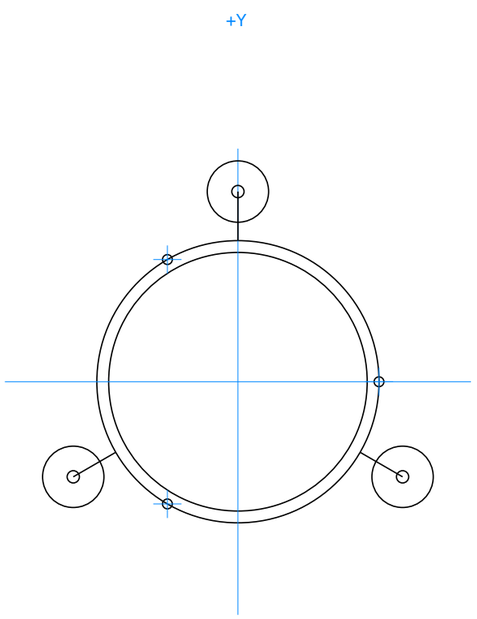
12 cutter axes begin R50.25 at Z28.9, radial outward length 9.75; R5 with R1 start-edge fillet.

C-groove ruled loft: Z35.6 offset 0, Z36.4 offset 0.8; eight exact arc/line segments per profile, see feature schedule.

The section deliberately omits projected hidden edges and is not an exact silhouette of the cut groove or helical ridge.

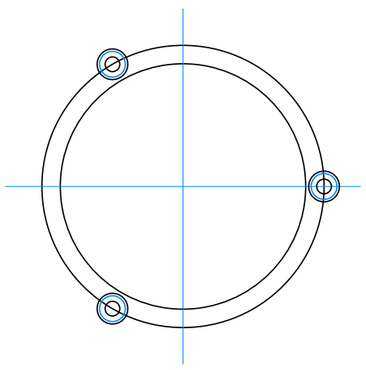
P01 STEP strict volume comparison failed; bounded numerical trial exception is not proof of BRep or surface identity.

3 / P02 carrier and P05/P06 covers — plan and sections



P02 TOP / D46 outside, D42.15 bore

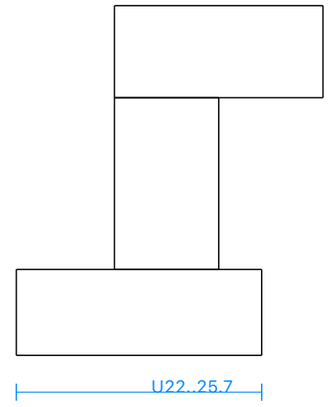
R31 ears D10, pilot D2; 3 at 120 deg



P05/P06 TOP / D46 / D40

3 lobes R2.5 on R23, D2.4 holes

RADIAL EAR SECTION, schematic



P02 annulus Z5.5..19.5

Ear bridge U22..31, V +/-5, Z6..9

Foot U22..25.7, V +/-2, Z1..3.8

Web U24..25.7, V +/-2, Z3.8..9

P05 Z19.5..21; P06 Z4..5.5

D4.2 countersink, depth 0.9

P05 from top; P06 from bottom

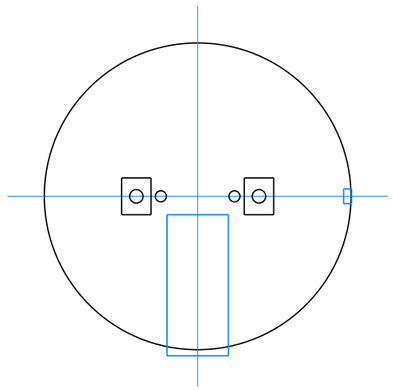
P02 cover bosses: R2.5 at R23 and 0/120/240 deg; lower Z5.5..9.5 and upper Z15.5..19.5.

Their D1.6 pilot cuts are Z5.5..8.5 and Z16.5..19.5 respectively. Ear pilots are D2 Z6..9.

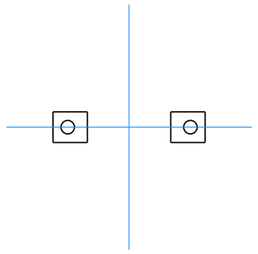
Drawn section is schematic; exact rotated rectangular union and bore recut order are in the feature schedule.

4 / Sensor plate and P08 toes — plans, section and datum

+Y



+X

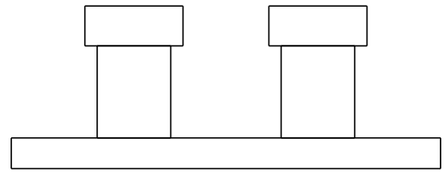


P08 LEFT and RIGHT / TOP

5.6 x 5; D2.2 at X +/-10, Y0

LEFT is RIGHT rotated 180 deg

SENSOR STACK / section schematic



SENSOR PLATE TOP / D50 base Z0..1

Base notch and exit shown in blue

Base Z0..1; support blocks Z1..3

Pins D1.8 at X +/-6: Z1..2.5

Blocks X +/-7.6..12.4, Y +/-3

P08 Z3..4.3, countersink D2.6

D2.2 plate holes start at Z0.2

Hex AF4.1 pocket Z0.6..2.3

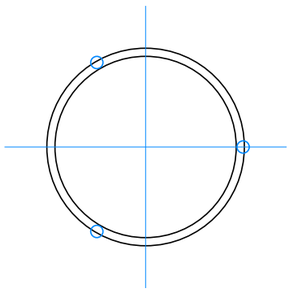
Slot toward +Y: Y0..8

Plate exit is B(-5,5,-26,-3,0,1); edge notch B(23.8,25.1,-1.2,1.2,0,1).

P08 right X6.8..12.4, Y-2.5..2.5; left is rotated 180 deg. Top countersink only Z4.1..4.3.

Sensor board, nuts and screws shown by locations in the BOM schedule; components, connector and wire geometry unknown.

5 / P04 and P07 collars; T30 thread and P01 groove details

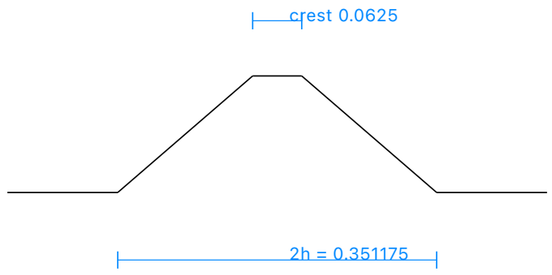


COLLAR PLAN / D32.2 outer

D29.55 bore before thread tool

3 slots: U15.6..16.2 V +/-1

T30 AXIAL PROFILE / enlarged schematic



Pitch 0.5 RH; root R14.725, crest R14.975

Female cutting ridge offset +0.05 radial

P04 Z4..5.5; P07 Z3..4

Same absolute Z phase, not reset at collar bottom

Curves and CSG in schedule define exact source construction

GROOVE / TOP PROFILES

At Z35.6: outer R19.2, inner R16.8

At Z36.4: outer R20, inner R16

Profile centres and arc directions:

see P01 schedule, no offset substitution.

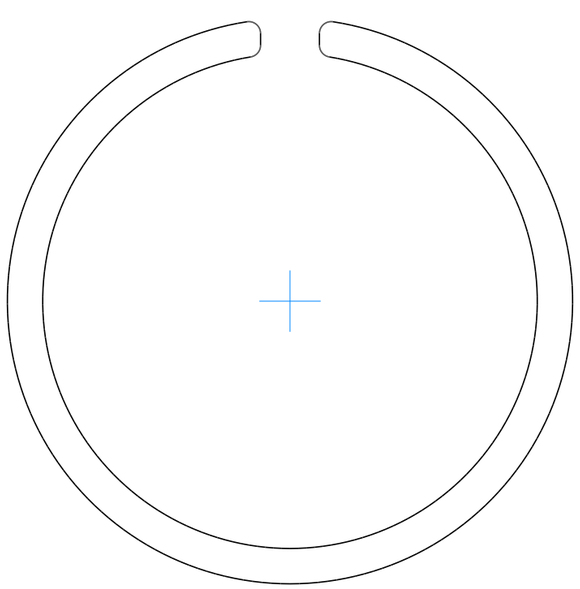
T30 is a custom modeled profile, NOT a specified ISO tolerance or fit. Its source sweep and same-Z phase require independent verification.

Each of six helical edges makes seven turns from its own axial-offset start; adjacent edges form ruled faces, then end caps and Z trim.

The source's shell validity checks are not independent proof of nominal phase, fit, surfaces or printable mating geometry.

P01 STEP was accepted under a bounded numerical exception after the strict P01 volume comparison failed. No surface equivalence claimed.

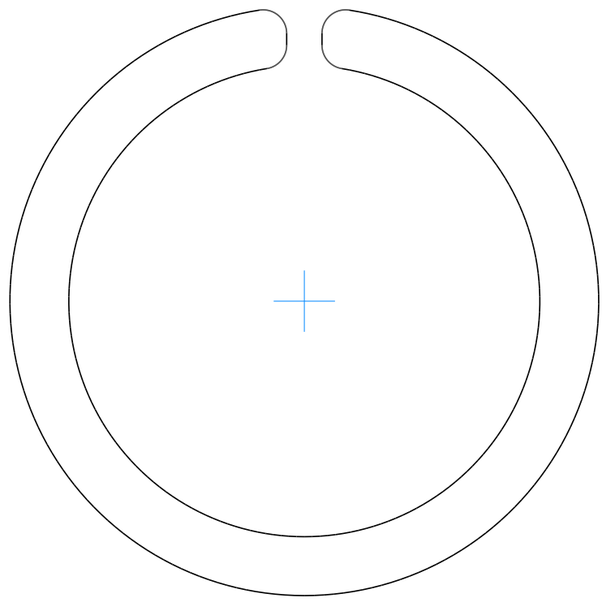
6 / P01 groove: two analytic planar tool profiles, not a model projection



Z35.6 / d=0

Outer R19.2; inner R16.8

Opening endpoints X=+/-2



Z36.4 / d=0.8

Outer R20; inner R16

Opening endpoints X=+/-1.2

Top is local +Y. The material removed is the ruled loft between these profiles.

Profiles are sampled 2D polylines for illustration; exact arc centres, endpoints and directions are in the schedule.

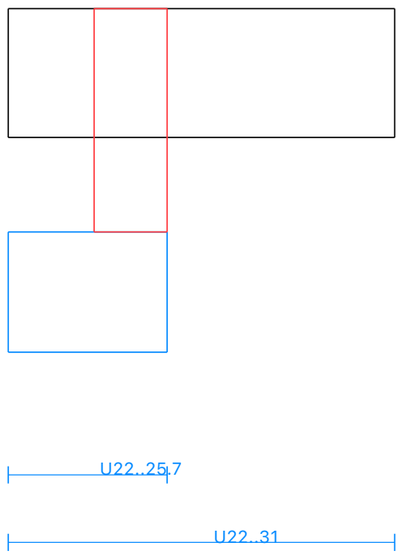
yo=sqrt(18.4^2-2.8^2); yi=sqrt(17.6^2-2.8^2). Ends centred at (+/-2.8, yo/yi).

Outer arc is the long CCW arc; inner arc is the long clockwise arc. Do not substitute complete rings.

Platter finger tools: 12 x R5, radial at 30 deg; axis Z28.9 begins R50.25, tool length 9.75; start-edge fillet R1.

7 / P02 ear, cover and P08 / dimensioned operation details

P02 one radial ear / U-Z operation overlay; U radial, V tangential



Bridge Z6..9, V +/-5

Foot Z1..3.8, V +/-2

Web Z3.8..9, U24..25.7, V +/-2

End cylinder R5 at U31 Z6..9

Pilot R1 at U31 Z6..9

Replicate at 90 / 210 / 330 deg

Covers / section through each fixing axis

P05: Z19.5..21; P06: Z4..5.5

R23 lobe axes 0 / 120 / 240 deg

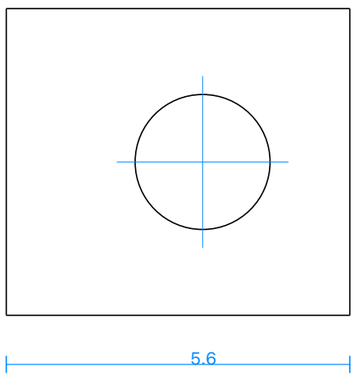
R2.5 lobes; R1.2 fixing bore

Upper cone R1.2 to R2.1 Z20.1..21

Lower cone R2.1 to R1.2 Z4..4.9

Main annulus R20..23; 1.5 thick

P08 right / plan; left is Rz(180) copy



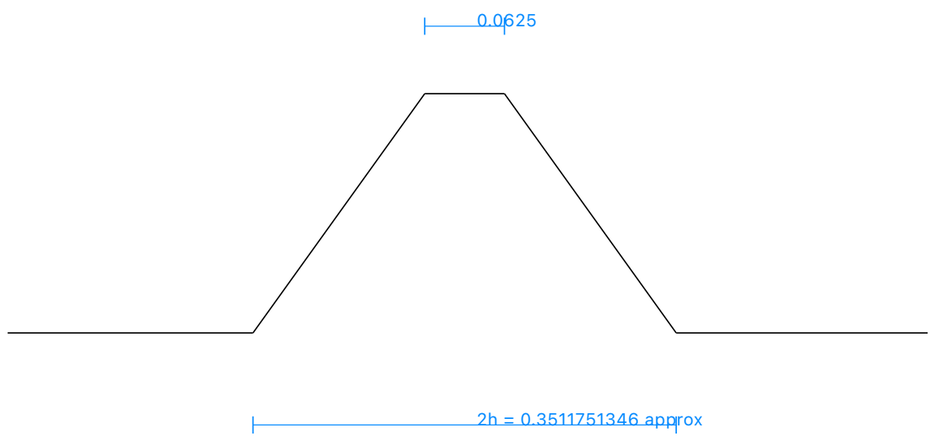
Scaled 10:1; X6.8..12.4, Y-2.5..2.5; Z3..4.3

Hole D2.2 at (10,0); countersink D2.6 top, Z4.1..4.3.

Ear outlines are operation overlays, not resulting Boolean sections.

8 / T30 phase and sensor plate / additional operation details

T30 ideal axial section at fixed angle / diagram not to scale



Male root R14.725; crest R14.975; radial height 0.25.

Female cutting tool has the same profile at radii +0.05; base bore R14.775.

Pitch 0.5 RH, seven turns before trim. At +X, centres Z=2.5+0.5n.

At -X, centres Z=2.75+0.5n; collars share the same absolute phase.

Ridge is clipped to Z3..5.5; ruled surfaces and ends require independent geometric verification.

Sensor plate / support and pocket axial operations

Base R25, Z0..1. Pins R0.9 at X+/-6 Z1..2.5. Blocks X+/-7.6..12.4, Y+/-3, Z1..3.

Cut exit X-5..5 Y-26..-3 Z0..1; key notch X23.8..25.1 Y-1.2..1.2 Z0..1.

At X+/-10: hex AF4.1, Z0.6..2.3 and slot X centre+/-2.1, Y0..8, Z0.6..2.3.

At X+/-10: R1.1 cutter Z0.2..3. Source cutting ranges, not inferred through-holes.

Sensor parts, connector, wiring, tolerances, material and physical fits remain undefined.

P01 STEP strict default-volume FAIL: 0.641091522 mm3 per rotor, limit 0.0001 mm3.

Bounded accepted trial exception only; no strict pass, surface equality or new drawing approval.

Exact nominal construction and reference schedule / continued

AUTHORITY AND VERIFICATION

SOURCE-TRANSCRIPTION CROSS-CHECK: independently reread f13_jog.py lines 22-278 against the entire 01 jog_data.json ordered schedule. This checks transcription, not BRep reconstruction, Boolean equivalence, or manufacturing fitness.

Nominal trial-model construction authority: models/release/enclosure/B26/B26-build123d03_RELEASE/SOURCE/models/enclosure/B26/BUILD123D03/f13_jog.py, lines 22-278; source snapshot selected by SOURCE/source-manifest.json.

F13-STATIC-01 sheets 2-3, F13-STATIC-02 sheets 1-2, F13-STATIC-04 sheet 1 and F12-D05 sheets 6 and 10 are historical visual references only: the feature definitions below supersede their incompatible text and features.

The approved trial model is not a manufacturing approval; this new consolidated drawing has no approval. No physical, electrical, safety, fit or strength approval is implied.

P01 STEP strict default-volume comparison failed for BOTH rotors: 0.641091522 mm3 each versus 0.0001 mm3 threshold. Accepted bounded exception applies to the unpowered trial representation only; not a strict pass, STL waiver, surface identity or approval of this drawing. Release handoff records adaptive integration difference 0.000001609 mm3, directed zero-volume cuts and sampled comparisons, not exact equality.

Purchased parts are nominal envelopes only, not complete measured hardware. PCB mounts elsewhere in the enclosure are intentionally omitted. No general tolerance is specified.

DATUM AND TRANSFORM

All lengths mm; third-angle projection. Local jog O is the axis at floor top; +X/+Y as source, +Z up; angles counterclockwise from local +X.

Enclosure drawing datum rear-left, X right, Y toward front, Z up. CAD coordinates are (X,-Y,Z). Local point p maps to CAD (Xj,-165,2) + Rz(+60 degrees)p; Xj=110 left and 490 right.

Thus case point is (Xj + x cos60 - y sin60, 165 - x sin60 - y cos60, 2+z). Both jogs have identical local geometry and phase.

The R31 ear centres, in enclosure drawing XY, are left (83.153212,149.5), (110,196), (136.846788,149.5); add 380 to X for right. Correspondence to local angles 90,210,330 is respectively first, second, third. These decimal values are rounded to 6 places; exact coordinates use the stated trigonometric transform.

BOOLEAN NOTATION AND ORDER

C(r,z0,z1,x,y): solid cylinder radius r, axis +Z at (x,y), from z0 through z1; omit x,y for (0,0). K(r0,r1,z0,z1,x,y): coaxial cone frustum bottom/top radii r0/r1.

B(x0,x1,y0,y1,z0,z1): axis-aligned closed box. RB(a,u0,u1,v0,v1,z0,z1): B(u0,u1,v0,v1,z0,z1) rotated +a about Z. XY(r,a)=(r cos a,r sin a).

All union, subtraction, intersection, rotation and fillet operations are ordered; the subordinate constructions are built independently before insertion. Only the explicitly identified fillets occur in f13_jog.py. The source is nominal construction authority; text and diagrams are not evidence of BRep equivalence.

PRINTED P01 rotor — quantity per jog: 1

Platter K(55,52.5,21.4,36.4), D110 lower / D105 upper; 12 recesses at a=0+30k, k=0..11. Subtract horizontal cylinder R5, axis radial +X, start (50.25,0,28.9), length 9.75; fillet its start circular edge R1 before rotating each tool about Z.

Exact nominal construction and reference schedule / continued

Top C-groove is ruled loft of the two exact planar profiles below at Z35.6 and Z36.4, subtracted from platter. No generic offset approximation: use the following arc-centre construction.

For profile offset d=0 at Z35.6, d=0.8 at Z36.4: O=19.2+d; I=16.8-d; xc=2.8; yo=sqrt(18.4^2-xc^2); yi=sqrt(17.6^2-xc^2). OL=(-xc,yo), OR=(xc,yo), IL=(-xc,yi), IR=(xc,yi). rad(P,r)=P*r/|P|.

Boundary in order: CCW arc centre (0,0) rad(OL,O) to rad(OR,O); CCW arc centre OR to (2-d,yo); line to (2-d,yi); CCW arc centre IR to rad(IR,I); clockwise arc centre (0,0) to rad(IL,I); CCW arc centre IL to (-2+d,yi); line to (-2+d,yo); CCW arc centre OL to rad(OL,O). Arc sweep is modulo 360 in indicated direction. Each arc radius is distance from its centre to its start. All four arc destinations are explicitly listed in the original sequence: arc OR ends at right_up=(2-d,yo); arc IR starts at right_low=(2-d,yi) and ends at rad(IR,I); arc IL starts at rad(IL,I) and ends at left_low=(-2+d,yi); arc OL starts at left_up=(-2+d,yo) and ends at rad(OL,O). The first centre-O arc is the long CCW arc, not the short arc across the opening.

Shaft: union C(14.725,3,5.5), T30 ridge offset 0 (see T30), C(14.975,5.5,21.4), C(16.1,19.5,21.4); subtract C(13,3,21.4). Fillet R0.2 the circular shaft/shoulder root at Z19.5, circumference 2*pi*14.975.

Union C(6,6.6,21.4) and platter; subtract magnet socket C(4.1,6.6,10.8). Magnet D8x4 is a reference, not part of P01.

PRINTED P02 fixed carrier — quantity per jog: 1

Union C(23,5.5,19.5); for a=0,120,240 union C(2.5,5.5,9.5,XY(23,a)) and C(2.5,15.5,19.5,XY(23,a)).

For a=90,210,330 union C(5,6,9,XY(31,a)), RB(a,22,31,-5,5,6,9), RB(a,22,25.7,-2,2,1,3.8), RB(a,24,25.7,-2,2,3.8,9).

Subtract C(21.075,5.5,19.5); for a=0,120,240 subtract C(0.8,5.5,8.5,XY(23,a)) and C(0.8,16.5,19.5,XY(23,a)); for a=90,210,330 subtract C(1,6,9,XY(31,a)).

PRINTED P04 / P07 collars — quantity per jog: 1 each

P04 z0=4,z1=5.5; P07 z0=3,z1=4. For each: C(16.1,z0,z1) subtract C(14.775,z0,z1), subtract T30 ridge offset 0.05, subtract RB(a,15.6,16.2,-1,1,z0,z1) for a=0,120,240.

Same global local-Z T30 phase as P01; no phase reset to collar bottom. They are custom printed threads, not a claimed ISO fit.

PRINTED P05 / P06 bearing covers — quantity per jog: 1 each

P05 z0=19.5,z1=21; P06 z0=4,z1=5.5. For each union C(23,z0,z1) and C(2.5,z0,z1,XY(23,a)) for a=0,120,240; subtract C(20,z0,z1), then C(1.2,z0,z1,XY(23,a)).

P05 countersinks: subtract K(1.2,2.1,20.1,21,XY(23,a)); P06 countersinks: subtract K(2.1,1.2,4,4.9,XY(23,a)), same three axes. No F12 spring reliefs.

PRINTED Sensor plate — quantity per jog: 1

Union C(25,0,1); for s=-1,+1 union C(0.9,1,2.5,6s,0), and B(-12.4,-7.6,-3,3,1,3) for s=-1 or B(7.6,12.4,-3,3,1,3) for s=+1.

Subtract B(23.8,25.1,-1.2,1.2,0,1) and B(-5,5,-26,-3,0,1). The cut is limited to Z0..1; ignore F12's taller exit.

For x=-10,+10 subtract regular hexagonal prism AF4.1 centered (x,0), vertices at angles 30+60i, i=0..5, Z0.6..2.3; subtract B(x-2.1,x+2.1,0,8,0.6,2.3); subtract C(1.1,0.2,3,x,0).

PRINTED P08 opposed toe — quantity per jog: 2

Right: B(6.8,12.4,-2.5,2.5,3,4.3) subtract C(1.1,3,4.3,10,0) and K(1.1,1.3,4.1,4.3,10,0). Left is the same shape rotated 180 degrees about local Z.

SOURCE CROSS-CHECK AND PROVENANCE

f13_jog.py:22-47 primitives, transforms and AF hex; 49-96 P01 groove and platter; 99-118 six helices, ruled faces, end caps and cylinder intersection; 121-156 rotor and collars; 159-180 carrier; 183-198 covers; 201-222 plate and toes; 225-235 reference board and bearing; 238-278 build, quantities, reference fasteners and placement.

Groove boundary consists of eight oriented elements, with major CCW outer and major clockwise inner circular arcs; 01 circles are not the groove. Segment endpoint definitions and d=0/d=0.8 profiles are in P01 feature schedule.

P01 source guard checks shaft before final boss and platter; collar guard checks groove after slot cuts. These are source checks only and were not executed in this documentation task.

The 01 source-manifest recorded source SHA-256 8fa9d5a7f6bb4ae8c6cde7079d26bf9a2b58bec395b94c31ae3c97e89cb0d65e; 01 jog_data SHA-256 25fb51f4c89d606da5440d86cb76dc22663ec153232b01fd81755f2a3100e388; 01 jog PDF SHA-256 722cfad1f2ab7597c9f0a7667963a1de202ba465b336969c3394403dc6f4c6ac, as documented in 01 validation.

T30 HELICAL RIDGE CONSTRUCTION

Exact nominal source ridge before trim: radial offset d=0 for male and d=0.05 for female cutting tool. root=14.725+d; crest=14.975+d; h=0.03125+0.25/sqrt(3). Ordered (radius, axial offset) vertices: (root-0.025,-h), (root,-h), (crest,-0.03125), (crest,+0.03125), (root,+h), (root-0.025,+h).

For every vertex construct RH helix of pitch 0.5, axial height 3.5, start (r,0,2.5+axial offset), around +Z. Join corresponding adjacent helices, including last to first, by ruled faces; cap both ends at Z offset 0 and 3.5 with polygonal faces; form closed solid and intersect with C(16,3,5.5).

The common ridge centre phase is Zc=2.5+0.5*(theta/360+n), n integer, for unwrapped theta on the +Z right-handed helix; +X at Z=2.5 is the start, and at theta=180 degrees ridge centres occur Z=2.75+0.5n. Wrap t=Z-2.5-0.5*theta/360 to [-0.25,0.25); ideal profile crest |t|<=0.03125 and root at |t|=h=0.03125+0.25/sqrt(3) (~0.1755875673). This is an ideal interior analytic interpretation, not proof of ruled surface parameterization, finite ends or boolean output. Female tool offset is +0.05 radial and both collars use the identical absolute phase; no ISO fit is implied.

Helix hand and positive winding are sourced to installed build123d topology/one_d.py make_helix lines 2546-2607: normal +Z, lefthand=False, positive 2*pi per positive pitch. Source P01 check probes (-14.97,0,4.25) and collar check probes (-15,0,z0+0.25); these internal checks do not independently validate geometry.

REFERENCE BOM AND LOCATIONS PER JOG

Printed: P01 x1, P02 x1, P04 x1, P07 x1, P05 x1, P06 x1, sensor plate x1, P08 x2 (9 printed occurrences). Two jogs total 18.

Bearing SKF 61806-2RS1 x2: simplified annular envelopes R15..21, Z5.5..12.5 and 12.5..19.5; actual internal geometry, corners and seals absent.

Diametric magnet D8 x4 x1: C(4,6.6,10.6); AS5048A PCB x1: C(9,1,3) intersect B(-7.5,7.5,-8.5,8.5,1,3), subtract D2.2 holes at x=+/-6,y=0. Components, connector, wiring absent.

Exact nominal construction and reference schedule / continued

M2 nuts x2: nominal threadless hex AF4 at x=+/-10, z0.65..2.25 minus R1 bore. M2x4 cap screws x2: R1 shank z0.3..4.3 plus R1.9 head z4.3..6.3; no drive/thread modeled.

ETTINGER M2.5x8 floor screws x3 at R31/90,210,330: R1.25 shank z0.5..8.5 plus R2.25 head z-2..0.5. M2x4 DIN7991 screws x6 at R23/0,120,240: upper shank R1 z17..20 and cone R1 to R2 z20..21; lower shank R1 z5..8 and cone R2 to R1 z4..5. All screws are threadless nominal envelopes, not complete hardware.

LIMITS AND UNVERIFIED FEATURES

Rotor, collar and thread geometry describe nominal trial-source construction, not physical fit or manufacturing tolerances. Bounded STEP volume acceptance is not surface identity; no geometric equivalence claim.

No proof of tool access, wire clearance, bearing fit, magnet retention, preload, printability, strength, motion or electrical performance. Actual sensor components and connector are unknown.

Drawing 01 sheet 2 rings and R5 circles, sheet 3 ear section, sheet 4 sensor section, and sheet 5 collar circles/thread profile are explanatory approximations, not true projected Boolean contours. The sheet 5 2h and crest bars are schematic, not scaled profile geometry. Exact operations, not those sketches, govern reconstruction.

Electrical schematic references for actual sensor PCB components, connector, wiring and interfacing remain incomplete; do not infer them from the nominal contour. Unknown materials, fits and tolerances remain TBD; no new 3D, STEP or STL made.

Nominal equipment envelopes and reference planes

Table (X,Y,Z) maps to CAD (X,-Y,Z). X right, Y toward operator, Z up. Reference boxes below are axis-aligned table-coordinate intervals; all quantities are one unless specified.

PI-envelope: table XY centre (217.5, 57.95), size (88.99, 58.4), Z 3.9..23.8.

DAC-envelope: table XY centre (305.5, 57.75), size (65.0, 58.0), Z 3.9..23.8.

PSU-envelope: table XY centre (388.0, 54.25), size (78.0, 51.0), Z 3.0..31.0.

U1-envelope: table XY centre (28.75, 170.0), size (25.4, 62.74), Z 3.0..14.0.

U2-envelope: table XY centre (181.0, 185.0), size (32.0, 21.0), Z 3.0..14.0.

U3-envelope: table XY centre (419.0, 170.0), size (32.0, 21.0), Z 3.0..14.0.

U4-envelope: table XY centre (571.25, 170.0), size (18.0, 41.0), Z 8.0..11.6.

J01-body: union CAD cylinders along +Y axis at X101 Z10: radius 4.075 from Y=-318 to -306.2 and radius 3 from Y=-322.5 to -318. J01-nut: CAD cylinder radius 4, Y=-321.8..-320; subtract coaxial radius 3, Y=-321.81..-319.99. These are round nominal envelopes, not complete hardware.

D01-front-plane: set c=cos(49 degrees), s=sin(49 degrees), CY=96-40*c, H=23+40*s. Four CAD vertices (300+u, -CY-c*v, H-s*v), in order (u,v)=(-138,-38),(-138,38),(138,38),(138,-38); face only, no thickness.

RCA-front-plane: four CAD vertices (28.85,0,0.05), (28.85,0,19.95), (80.55,0,19.95), (80.55,0,0.05); face only, no depth.

Nominal control bodies and limitations

For each of the 49 distinct owners, use the local-to-table 4x4 transform in the corresponding FIXED group of shell/shell_dataset_DRAFT.json; for E01 use E01-fixed_cage. Evaluate the following local primitives in (u,v,w), transform to table and negate resulting table Y for CAD. These are incomplete purchased references, not printed retainer geometry.

B01-B24 and T01-T06: box u[-6,6], v[-6,6], w[0,4.3]. A separate zero-thickness nominal tip face has vertices (-1.9,-1.9,7.3),(-1.9,1.9,7.3),(1.9,1.9,7.3),(1.9,-1.9,7.3); apply the same transform.

P01-P13: box u[-5.7,5.7], v[-5.5,7], w[0,6.8].

F01-F05: union box u[-30,30], v[-4.5,4.5], w[0,6] and box u[-2,2], v[-0.5,0.5], w[6,21].

E01: union box u[-5.9,5.9], v[-7,7.5], w[0,6.5], cylinder radius 3.5 w[6.5,13.5] and cylinder radius 3 w[13.5,21.5]; subtract box u[-5,5], v[1.5,5], w[14.5,21.51]. Cylinders on u=v=0.

Not complete geometry of purchased controls, bearings, boards, display, jack, connectors or cables. U4 is body-only; U1-U4 mounts are intentionally omitted.

Numerical data describe model placeholders, not permission to infer actual dimensions, mating fits, electrical clearance or safety.

P01 rotor STEP strict default volume comparison failed; the accepted bounded exception applies only to the released unpowered mechanical trial.